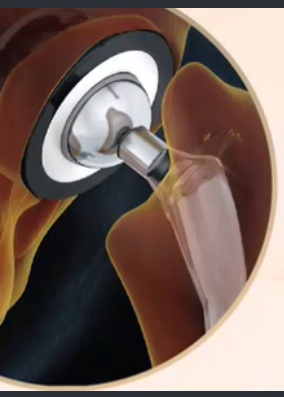
COURSE

# Hip Arthroplasty

*31 lectures*



## Kasr Al Ainy
## Hip Arthroplasty Course

*From Basics to Practice*

Watch course

15 October 2026

TRANS PRO

Developed by

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- "(see Fig.)" jumps to the figure; "↑ back to text" under the figure brings you back to where it was cited.
- "Ask AI" beside each heading opens a page that turns that section into a ready question for an AI: simplify it, clarify what the recording left unclear, or check for newer guidelines.
- Orange notes such as [C1] and [Q1] are AI corrections and doubts - mostly where the recording was hard to transcribe. Each lecture lists them after its index of figures.

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# Biomechanics of THA

*(Forces, Offset, Version, Range of Motion, Stability and Templating)*

**Prof Dr Mohamed Gobba**

**Watch Original Lecture**

*Lecture transcript processed with AI assistance.*

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## Why Biomechanics of THA Matters Ask AI

- Question: importance of **biomechanics of the total hip**
- Last 60–70 years → formal **analysis** of every defect and of performance
- Science behind any craft → very large difference
- Residents implant joints before understanding them
- Implanting without knowing what fits into what, and why
- Every unexplained clinical finding → a **biomechanical explanation**
- Pre-op: discuss head options with patient
- Intra-op: choose which set of heads to use
- Typical problems
  - **Head size:** rep has no 28 mm head, only 32 or 36 → implant or postpone?
  - Post-op leg turns inward
  - New limp absent before surgery
  - **Lateral pain** → cannot sleep on that side
  - Urgent 2 a.m. WhatsApp call from patient
- Question: could anything have been done?
- **Six biomechanical points** → introduction only; detail in textbooks
  - Forces around the hip
  - Offset
  - Version
  - Range of motion
  - Stability
  - Templating

## Forces Around the Hip Ask AI

- Two principles
  - Principle 1 — door and hinge
    - Push at point A vs point B
    - Farther from hinge → less force needed
  - Principle 2 — hip forces behave like a **seesaw**
    - Hip = fulcrum in the middle; two sides move
    - Light loads on both sides → hinge survives
    - Heavy loads → problem
- All loads summed at hip → opposed from below by **joint reaction force**
- **Newton's third law:** every force → equal and opposite force
- Heavier loads → joint reaction force increases considerably
- Weight on both sides ↑ → **failure**

# Single-Leg Stance and Abductor Deficiency Ask AI

## Normal Single-Leg Stance Ask AI

- Double stance → body weight distributed over both legs
- **Single stance** → different loading
- Single stance = half of the gait cycle
- Body weight → acts downward
- **Abductors** → counteract body weight → level the pelvis
- Joint reaction force from below → opposes both, like the seesaw
- Normal single stance: body weight = one force
- Abductors act from **insertion to origin**
  - Level opposite side of pelvis → prevent pelvic drop
- Lever arms balance forces: **A × B = C × D**

## Abductor Damage Ask AI

Abductors

- Scenario: THA via **lateral approach**, inexperienced hand → abductors completely damaged
- Equation disturbed → A × B ≠ C × D
- Must reduce C and D → re-balance → level the pelvis
- Weight loss by tomorrow → not possible
- Patient reduces **D** instead
- Mechanism: leans instead of standing upright
- Lean → **center of mass** moves inward toward joint → D ↓ → equation balanced
- Result: equilibrium, but forced to lean
- Complaint: "I limp whenever I walk" → **Trendelenburg gait**

## Correction with a Cane Ask AI

Cane

- **Cane** in the **contralateral hand**
- Cane → redistributes the load
- Cane balances equation → no need to lean
- Cane → opposite hand, not same side as operated hip

**High Yield**

- Abductor loss → A × B ≠ C × D → lean → Trendelenburg gait
- Lean → center of mass toward joint → D ↓
- Cane → contralateral hand

# Femoral Offset Ask AI

- **Offset:** distance between **center of the head** and femoral **axis**
- Adjustment → requires **modular stems**
  - Most → no comparative data → almost nobody uses them

| Offset | Mechanism | Result |
|---|---|---|
| Too small | No offset → **greater trochanter** strikes | **Bony impingement;** hip instability |
| Too large | Friction ↑ between **IT band** and **trochanteric bursa** → constant pressure | **Trochanteric bursitis;** continuous inflammation |

- Complaint of constant lateral pain → offset chosen incorrectly
- Prevention → correct choice through **templating**

## Femoral Version Ask AI

- Lateral approach → view with **patella facing the ceiling**
- Main fear → dislocation

| Approach | Dislocation direction | Stem version | Aim |
|---|---|---|---|
| **Lateral** | Anterior | More **retroverted** | Head travels longer distance before dislocating |
| **Posterior** | Posterior (opposite) | More **anteverted** | Head travels longer distance before dislocating |

### Excess Stem Anteversion Ask AI

Version

- Stem excessively anteverted → head recentered in cup only with leg internally rotated
- Pediatric analogue: **increased femoral anteversion** (**in-toeing**)
- Anteversion ↑ intra-op → leg comes out **internally rotated**
- In-toeing not corrected after two months → stem excessively anteverted

**Clinical Pearl**

- Persistent post-op in-toeing → excessively anteverted stem

## Range of Motion Ask AI

- ROM in THA → not for sport/activity
- ROM in THA → for **total hip survival**
- Needed for daily movements → range not small
- **Extremes of range → impingement**
  - → **polyethylene wear** on impingement side
  - → or on opposite side
  - → or **complete failure**
- ROM determinants
  - **Head–neck ratio** factors
  - **Cup factors**

## Head–Neck Ratio Ask AI

- **Head–neck ratio** = head diameter : neck diameter
- Larger head → greater ROM
- Same neck, small vs large head → large head → higher ROM
- Step `1` → very large head
- Same head, thick vs thin neck → **thin neck** → higher ROM
- Apparent solution → large head + thin neck
  - Manufactured widely → after a year → prosthesis **fracturing**
  - Thin neck cannot carry large head

## Trapezoidal Neck Ask AI

Neck

- **Trapezoidal neck**: wide in one plane → no impingement in that area
- Impingement area removed → neck becomes trapezoidal
- Modern prosthetic neck → not completely round → slightly flattened

## Skirted vs Unskirted Heads Ask AI

Heads

- Options: **skirted** and **unskirted** heads
- Scenario: stem grips well → trial shows need for **long head** → needs **skirt**
- Skirt → thickens **neck diameter**
- Thicker neck → head–neck ratio ↓ → ROM ↓

> **Pitfall**
> - Long skirted head → neck diameter ↑ → head–neck ratio ↓ → ROM ↓

## Cup Factors: Range vs Stability Ask AI

Cup

- **Spectrum**: range at one end, **stability** at the other
  - Toward range → ROM ↑, stability ↓
  - Toward stability → stability ↑, ROM ↓ considerably

| Cup option | Design | ROM | Stability | Consequence |
|---|---|---|---|---|
| **Liner augmentation** | Added piece on liner | ↓ | ↑ | — |
| **Captive cup** | Augmentation above and below liner | Very small (e.g. `60°`) | Most stable component known | High **failure rate**; young patient → **loosening** after a year |

- Moving beyond head and neck → **liner offset**
- **Liner offset**
  - Normal: head **center of rotation** moves with **center of rotation of the cup**
  - Liner offset → head center pushed slightly outward
  - → head travels greater distance
  - Reasonable idea, with its own issues

# Stability Ask AI

## Head Size and Jumping Distance Ask AI

Stability

- Stability → depends on **head size**
- Head size → **jumping distance**
  - Ball exits completely → half its diameter must pass over rim
  - **Jumping distance** = head **radius**
- Larger head → greater distance to travel → more stable
- Video: normal range → continued motion → dislocation = jumping distance
- **3–4 mm** of jumping distance → major role in stability

## Soft Tissue Repair Ask AI

- **Posterior capsule** repair after **posterior approach** → matters for jumping distance
  - Head reaches jumping distance + capsule closed → stays
  - Capsule not closed → dislocates

## Component Alignment Ask AI

- Cup cannot be placed anyhow
- Patient's native cup position → range patient is accustomed to
- Fear of dislocation → cup over-anteverted, further with **broach** → ROM altered
- Departure from native ROM → disturbed
- Goal → reproduce patient's **native anatomy**
- **Primary arc of range** of total hip
  - Cup within patient's native range → 100%
  - Excessive anteversion or retroversion → **cup impingement** → implant lifespan ↓

**High Yield**

- Larger head → larger jumping distance → more stable
- Posterior capsule repair → retains head at jumping distance
- Cup outside native range → cup impingement → lifespan ↓

# Templating Ask AI

## Principle Ask AI

- **Templating**: pre-op planning together in the office
- Example: retained plates → **hardware removal** → template, see design, plan
- Whole operation performed in the office
- Best operation = one performed in the office
  - Choose and request everything needed, including templates
- Templating **X-ray** → specific requirements
  - Covered in the course by **Dr. Mahmoud Abdel-Karim**

### Leg Length Ask AI

Templating

- Cup center not coinciding with head center → leg lengthened; offset ↑ or ↓
- **Stem 9 mm higher** → leg lengthened 9 mm on reduction into cup
- **Neck cut** calculated relative to **lesser trochanter**
- Neck cut 15 mm above lesser trochanter → stem seated
  - → stem **center of rotation** higher than cup center of rotation
- Result: normal length restored → lengthening known exactly in millimeters
  - → via the template

### Offset and Implant Design Ask AI

Templating

- Case: patient with **arthritic hip**
- Arthroplasty performed neatly → check **AP** view
- Finding → **offset increased**
- Avoidable? Not if no templating and simply using prosthesis supplied to hospital
- Template: offset slightly ↑ → lengthening; measured offset segment slightly smaller
- Matching **centers of rotation** after implantation → offset ↑
- Second case: templating pre-op would have shown the problem
- Template offset vs **actual** prosthesis → actual jumps further → offset ↑ considerably
- → that **stem** unsuitable → different design needed
- Templating → not only for **size**
- Templating → choose **design**: this company's set suits the patient or not

**Must-Know**

- Templating → size, leg length, offset and implant design
- Neck cut level relative to lesser trochanter → predicts lengthening in mm

## Closing Principles Ask AI

- Most THA problems → **scientific basis**
- Complaining patient → do not blame patient or nature → investigate cause
- Every complaining patient → find out why

# Tribology

*(Friction, Lubrication and Wear in Hip Arthroplasty Bearings)*

**Prof Dr Mohamed Abo El Soud**

**Watch Original Lecture**

*Lecture transcript processed with AI assistance.*

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## Tribology: Definition and Scope Ask AI

- Lecture first given in a **Basic Science** course, about 5 years ago
- Slide picture: time needed just to recall what **tribology** covers
- **Tribology** = science of friction
  - **Tribos** = friction (rubbing)
  - Not "three"; not the science of camera tripods
  - English began theorising → **David Tabor** first to speak about it
- Three components
  - **Friction**
  - **Lubrication**
  - **Wear**

## The Tribosystem Ask AI

- **Tribosystem** = system discussed; all theories applied to it
  - Body + another body
  - **Lubricant** between them = interfacial material → opposes friction
  - Environment: temperature + humidity
  - Pressure on system = **load**
- Outputs of the system
  - **Wear particles**
  - Surface shows the effect of wear

| Element | Tribosystem role |
|---|---|
| Two bodies | Rubbing surfaces |
| Lubricant | Interfacial material, opposes friction |
| Temperature, humidity | Environment |
| Load | Pressure applied to system |
| Wear particles, worn surface | Outputs |

## Friction Ask AI

- **Friction** = rubbing of two bodies
  - Resistance to motion between two bodies moving on each other
- Friction equation: $F = \mu \times W$
  - F = **friction force**
  - $\mu$ = **coefficient of friction**
  - W = **load** applied
- Equation has no term for **surface area** or contact area
- **Apparent contact area** → the two surfaces seen touching
- **Real contact area** → surfaces never touch completely, however polished

  - Cause → surface is rough
- Polished surface stays rough → high spot + low spot
- **Surface roughness** of hip materials

| Material | Surface roughness |
|---|---|
| Polyethylene | 1 micrometre |
| Polished metals | 0.05 micrometre (most cases) |

- Surface roughness → leads into lubrication

**High Yield**
- F = μ × W → no contact-area term
- Apparent contact area ≠ real contact area
- Polyethylene roughness 1 micrometre; polished metal 0.05 micrometre

# Lubrication Regimes and the Lambda Ratio Ask AI

## Regimes Ask AI

Lubrication

- **Asperities** = surface roughness
- Asperities completely touching → **boundary lubrication**
  - Lubricant = substance added to smooth motion
  - Present only as a very thin layer
  - Layer still allows all asperities to touch
- More lubricant → asperities moved apart
- Asperities separated → **fluid film** forms between the two surfaces
- Halfway between → **mixed lubrication**
- Surfaces completely separated → fluid film; surfaces touching → boundary
  - Expressed by an equation → lambda ratio

## Lambda Ratio Ask AI

Lambda

- **Lambda ratio (λ)** = minimum film thickness (h-minimum) / surface roughness of the two surfaces
  - h-minimum = minimum distance between the two surfaces
- Bigger fluid film → h-minimum exceeds surface roughness → λ rises → better lubrication
- λ versus **coefficient of friction**
  - λ rising → boundary → mixed → fluid film lubrication
  - Coefficient of friction falls → lubrication improving
  - Shortly after λ passes 3 → coefficient of friction high again
- Reason for rise at λ greater than 3
  - Film that large needs a very **viscous** (sticky) lubricant
  - Lubricant itself resists motion
- λ greater than 3 → lubrication decreases again → coefficient of friction rises again
- Review of the two extremes
  - **Boundary lubrication** → surfaces touching, total asperity contact
    - Lubrication at the level of molecules

    - Single lubricant molecule **adsorbed** on the two surfaces → smooths motion
  - **Fluid film** → no asperity contact at all, complete separation of surfaces

| Regime | Asperity contact | Lubricant | Coefficient of friction |
|---|---|---|---|
| Boundary | Total | Very thin layer; single molecule adsorbed on surfaces | Highest |
| Mixed | Partial (halfway) | Intermediate | Falling |
| Fluid film | None; complete separation | Film thicker than roughness | Lowest |
| λ greater than 3 | None | Very viscous, resists motion | High again |

**Exam Pearl**

- λ = h-minimum / surface roughness
- Rising λ → boundary → mixed → fluid film
- λ just past 3 → friction rises again (viscous lubricant resists motion)

## Fluid Film Lubrication: Hydrostatic versus Hydrodynamic Ask AI

- **Fluid film lubrication** → two types
- **Hydrostatic lubrication** → factories
  - Fluid film pumped between two surfaces by a piston → keeps them apart
  - Not present in the body
- **Hydrodynamic lubrication** → present in the body
  - Most important part of the lecture

| Type | Mechanism | Setting |
|---|---|---|
| Hydrostatic | Film pumped by piston, surfaces kept apart | Factories; not in body |
| Hydrodynamic | Film generated by motion | The body |

### Water-Skier Model Ask AI

Hydrodyn.

- Man gliding on water on skis, boat in front (unseen)
- Factors making him glide better
  - **Speed** of boat → higher speed, better gliding
  - **Load** → tall, fat man sinks into water
- Lighter man → better gliding
- **Angle** between ski and water
  - Larger angle → acts as resistance
  - Smaller angle → faster glide
  - Zero or negative angle → sinks into water
  - Needed: angle present, as small as possible

- Rough, uneven ski undersurface → acts as a brake → **surface roughness** matters
- Four factors: speed, load (applied by patient), surface roughness, angle

| Factor | Better gliding | Worse gliding |
|---|---|---|
| Speed | High | Low |
| Load | Light | Heavy → sinks |
| Angle | Present, as small as possible | Large → resists; zero or negative → sinks |
| Surface roughness | Smooth undersurface | Rough, uneven → brakes |

## Hydrodynamic Lubrication in the Hip Ask AI

### Wettability and Surface Finish Ask AI

Hip

- **Ceramic head** "loves water" → water spreads over its surface
  - High **wettability**
  - Surface finish completely smooth
  - Both help fluid film formation
- Next factor → angle between the two surfaces

### Radial Clearance Ask AI

Hip

- Large angle between surfaces → polar contact, no fluid film
  - Fluid escapes out entirely
  - Two surfaces touch each other
- **Radial clearance** = radius of liner − radius of head
  - Must be present
  - Must be small
  - Result → polar contact with high conformity

### Entrainment Ask AI

Hip

- Polar contact + high conformity → fluid film via **entrainment**
- At rest → polar contact
- At rest → fluid film only slight, at the sides
- Motion begins with boundary lubrication
- Head moves → head itself spreads fluid film over the other surface = **entrainment**
- Achieved by small radial clearance

**Must-Know**

- Radial clearance: present but small
- Large clearance angle → polar contact, fluid escapes, no film
- Entrainment = moving head spreads fluid film over opposing surface
- Sequence: rest (polar contact) → boundary lubrication → fluid film

### Head Size and Velocity Ask AI

Hip

- Does size matter? Bigger head →
- → more lubrication
- **Velocity** = Δd / Δt
  - Δd = distance moved
  - Thigh flexed 90 degrees: 28 head versus 36 head
  - Δt constant
- Distance moved differs → 36 head moves more
- 36 head → greater distance → higher velocity → better lubrication
- Remaining factors
  - Patient's weight
  - **Synovial fluid** present

| Skier factor | Hip equivalent |
|---|---|
| Speed | Head velocity → larger head (36 versus 28) faster |
| Load | Patient's weight |
| Surface roughness | Surface finish, wettability (ceramic head) |
| Angle | Radial clearance → small |

## Wear Mechanisms Ask AI

- **Wear** = third component of tribology
- Many types; three best known
  - Abrasive wear
  - Adhesive wear
  - Fatigue wear

### Abrasive Wear Ask AI

Wear

- **Abrasive wear** = ploughing
  - Analogy: kid with a nail scratching cars as he walks
- Occurs in metal-on-poly
  - Steel, however polished, has asperities
  - Asperities scratch the **polyethylene** with walking
- Also the type produced by **third body wear**
  - Cement piece embedded in polyethylene → scratches femoral head
  - Or scratches femoral component of the knee

### Adhesive Wear Ask AI

Wear

- **Adhesive wear** → bonds form between the two surfaces
  - Intermolecular bonds resist motion → produce friction
  - Responsible for **spot welding**, e.g. head–trunnion
- Spot welding at head–trunnion → as in **trunnionosis**

## Fatigue Wear Ask AI

Wear

- **Fatigue wear** → repeated stress on polyethylene surface
  - Microscopic cracks form on the surface
  - Cracks coalesce → large piece
  - Piece eventually breaks off

| Type | Mechanism | Example |
|---|---|---|
| Abrasive | Ploughing; asperities scratch softer surface | Metal-on-poly; third body wear (cement in polyethylene) |
| Adhesive | Intermolecular bonds between surfaces → spot welding | Head–trunnion, trunnionosis |
| Fatigue | Microscopic cracks coalesce → large piece breaks off | Polyethylene surface |

# Linear versus Volumetric Wear and Head Size Ask AI

- **Linear wear** versus **volumetric wear;** small head versus large head
- **Pressure** = load (patient's weight) / surface area
  - Small head → smaller surface area → higher pressure
  - Linear wear → small head sinks into polyethylene faster
- **Volume** involves $\pi r^2$ → radius to the power of 2
  - Large head → higher volume of particles generated
  - Large head → more wear particles
- Trade-off of the large head
- Stability + lubrication on one side
- Linear or volumetric wear on the other

| Feature | Small head | Large head |
|---|---|---|
| Surface area | Smaller | Larger |
| Pressure | Higher | Lower |
| Linear wear | Faster penetration into polyethylene | Slower |
| Volumetric wear | Lower | Higher → more wear particles |
| Stability, lubrication | Less | Better |

**High Yield**

- Pressure = load / surface area
- Small head → high pressure → more linear wear
- Large head → $\pi r^2$ → more volumetric wear, more particles
- Large head trade-off: stability and lubrication versus wear

## Wear Rate Ask AI

- **Wear rate** → last point
- Joint implanted → wearing in occurs
- Asperities begin to bed onto each other
  - Analogy: running in a new car → parts still bedding in
  - Continues until asperities worn down and flattened
- Beginning → very high wear rate = **wearing in** (running in)
- Then → steady
- Finally → approaches end of life of the prosthesis
- Wear rate rises a second time → **catastrophic failure** of the implant

| Phase | Wear rate | Event |
|---|---|---|
| Wearing in (running in) | Very high | Asperities bed in, worn down, flattened |
| Steady state | Steady | — |
| End of prosthesis life | Rises again | Catastrophic failure of implant |

# Biomaterials in hip arthroplasty

*(Material Properties, Metals, Ceramics and Polyethylene)*

**Prof Dr Mahmoud Fahmy**

**Watch Original Lecture**

*Lecture transcript processed with AI assistance.*

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## Biomaterials: Definition and Classes Ask AI

- **Biomaterial** → any substance or combination, synthetic or natural
  - Used for any period of time
  - As a whole or as part of a system
  - Treats, augments or replaces any tissue, organ or body function
- Materials in **orthopaedic implants**
  - **Metallic alloys** → stainless steel, titanium, cobalt-chromium
  - **Non-metallic materials** → ceramics, polymers (polyethylene)

## Stress–Strain Curve Ask AI

- Basis of material behaviour → **stress–strain curve**
  - Relationship between stress and strain
- Applied stress → **strain** → change in shape and dimensions
- Two key zones
  - Elastic zone
  - Plastic zone of deformation

### Elastic and Plastic Zones Ask AI

- **Elastic zone** → load removed → reversible recovery → original shape
- **Plastic deformation** → irreversible recovery → no return to original shape

### Young's Modulus Ask AI

- Elastic zone slope
  - Steeper slope → **stiffer** material
  - Flatter slope → less stiff, more flexible material
  - Slope = **Young's modulus of elasticity**
    - Specific to each material
- **Young's modulus** ranking, highest → lowest
  - Ceramic → metal alloys → cortical bone → polyethylene, bone cement

**High Yield**

- Young's modulus = slope of elastic zone
- Steeper slope → stiffer material
- Ceramic → metal alloys → cortical bone → polyethylene, bone cement

## Material Behaviour Terms Ask AI

### Brittle and Ductile Ask AI

- **Plastic zone** → internal breakdown without gross fracture
  - **Gross fracture** → at failure point
- Very little plastic deformation → **brittle**
  - Example → glass; in orthopaedics → ceramic

- Large plastic deformation → **ductile**
  - Example → stainless steel

### Notch Sensitivity Ask AI

- **Notch sensitivity** → susceptibility to fracture with crack or scratch
  - Glass or ceramic cutter → notch first → tap → easy break

### Hardness Ask AI

- **Hardness** → resistance to surface scratching and indentation
  - Unrelated to stress–strain curve
  - Hardest material overall → diamond
  - Hardest in orthopaedics → **ceramic**

### Toughness Ask AI

- **Toughness** → area under the stress–strain curve
  - = energy absorbed before failure
- More precise than **ultimate tensile strength**
  - Ultimate tensile strength → highest stress point on curve
  - Same ultimate tensile strength, larger area → ductile vs brittle material

## Metal Alloys Ask AI

- **Pure metals** → soft, ductile → alloys developed
- **Alloy** → metal + one or more other elements
  - Stronger, harder
  - Corrosion resistant
  - Better surface finish
- Key alloys → stainless steel, cobalt-chromium, titanium

### Stainless Steel Ask AI

- Composition → on slide
- Advantages
  - **Very stiff**, fracture resistant
    - Use → **cemented stems** → less bending stress on cement → cement protected from failure
- Problems
  - **Stress shielding**
  - Susceptible to **corrosion**

### Cobalt-Chromium Ask AI

- Composition → on slide
- Better corrosion resistance, lower **wear rate**
  - → bearing surface, e.g. **femoral head**
- **Highest modulus of elasticity** among metals, least ductility
  - → also used in **cemented stems**

### Titanium Ask AI

- Composition → on slide
- **Very biocompatible**

  - No tissue irritation
- Corrosion resistant
- **Low modulus of elasticity** → very close to bone
  - Directly above cortical bone in Young's modulus ranking
  - No stress shielding, bone loss or resorption → **uncemented systems**
    - Uncemented stems, uncemented cups
- Problem → **poor wear** → metal debris
  - → not used in **bearing surfaces**

### Tantalum Ask AI

- Newest metal
- **Low modulus of elasticity** → closer to bone
- **Extremely porous** → bone ingrowth
- Promising → **acetabular reconstruction,** revision of acetabular cups

**Clinical Pearl**

- High modulus → stainless steel, cobalt-chromium → cemented stems
- Low modulus → titanium, tantalum → uncemented stems and cups
- Titanium → poor wear → not a bearing surface

## Modes of Material Failure Ask AI

### Wear and Fatigue Ask AI

- All materials fail by **wear** and fatigue failure
  - **Fatigue failure** → failure below ultimate tensile stress  repetitive loading
- Metals → additional mode: **corrosion** (chemical reaction)

### Corrosion Types Ask AI

| Type | Mechanism | Key point |
|---|---|---|
| **Galvanic** | Two different metals coupled → cathode + anode → electrochemical reaction | Best-known pair → stainless steel + cobalt-chromium |
| **Crevice** | Crack or scratch on surface | Stainless steel most susceptible; avoid surface cracks at insertion |
| **Fretting** | Micromotion between two materials | Always at head–neck junction |

## Metal Debris and Its Effects Ask AI

- Wear or corrosion → metal ions, particulates, **metal debris**
- Debris deposited in tissues → **dose-related allergic response**
  - Local or systemic
- Local → **ALVAL** syndrome, pseudotumours, bone and tissue destruction

- Systemic → metal ion elevation, organ toxicity, **carcinogenesis**
- Consequence → **metal-on-metal** bearings now much less used

## Ceramics Ask AI

- **Ceramics** → solid inorganic compounds, metallic + non-metallic elements
  - Bound by ionic and covalent bonds

### Advanced Ceramics Ask AI

- Current **advanced ceramics** → alumina, zirconia, alumina matrix composite

| Ceramic | Wear | Fracture risk |
|---|---|---|
| **Alumina** | Very low wear rate | Brittle, susceptible to fracture |
| **Zirconia** | Higher wear rate | Less risk of fracture |
| **Alumina matrix composite** | No (minimal) wear | Very low fracture risk |

### Generations Ask AI

- Generations → 1974 to 2003 → current **BIOLOX delta** = alumina matrix composite
  - Aim → lower fracture risk via smaller **grain size**

### Material Characteristics Ask AI

Ceramic

- **Stiff** → first in Young's modulus ranking
  - No deformation under high load
- **Brittle** → very little plastic deformation → higher fracture risk
- **Hard** → scratch resistant → resists **third-body wear**
- Strong in compression, poor in bending
  - → used as **bearing surface**
  - → not used in stems → would fracture
- **Biologically inert**
  - Particles → no allergic, immunologic or inflammatory response
  - No tissue response → **bioinert**
- **Hydrophilic** → wettability
  - Uniform synovial fluid distribution → lower **coefficient of friction**

### Advantages Ask AI

Ceramic

- **Very low wear rate**
  - → less osteolysis, especially less **aseptic loosening**
- Very small, bioinert particles → no tissue response
- Longer **implant longevity**
  - → still recommended bearing surface in **young adults**

### Disadvantages Ask AI

Ceramic

- **Squeaking** → noisy sounds, up to 30% of patients
  - Exact cause unknown; suggested causes
    1. Insufficient lubrication
    2. Neck impingement on metallic rim

        - Localized stripe wear
- **Fracture** → very rare in new generations
    - Increased by poor surgical technique
        - Cup **retroversion** → markedly accelerated fracture
- **Stripe wear** → see figure
    - Head–liner separation on loading
    - Taper or neck impingement on acetabulum
- **Not forgiving**
    - Requires learning curve
    - Requires proper surgical technique and component insertion
- **Very expensive**

### Revision Drawbacks Ask AI

Ceramic

- Ceramic liner removal difficult → specific **suction instrument**
- Head removal → **taper damage**
- **"Once a ceramic, always a ceramic"**
    - Ceramic particles remain in tissues, not easily removed
    - → revision only to ceramic, not to other bearing surfaces

**Pitfall**

- Cup retroversion → markedly accelerated ceramic fracture
- Ceramic not forgiving → proper technique and component insertion
- Once a ceramic, always a ceramic

## Polyethylene: Structure and Manufacture Ask AI

- **Polyethylene** → polymer of ethylene molecule
- Types depend on → molecular weight, cross-linking

### Molecular Weight and Structure Ask AI

- **Molecular weight** → number of monomers (ethylene molecules) per polymer molecule
    - Low, medium, high, ultra-high molecular weight
    - **UHMWPE** → more than `200,000` monomers per molecule
- Structure
    - Parallel lines, no connections between monomers → **linear homopolymer**
- **Cross-linking** → connections between monomers
    - Resists **intermolecular mobility**
    - → more resistant to wear and deformation

### Manufacture of UHMWPE Ask AI

- **Ethylene** ($C_2H_4$) → polymerization → polyethylene
    - = chain of carbon atoms + bonds between them
- Processing → one of two methods

| Method | Process | Wear |
|---|---|---|
| **Ram extrusion** | Rods → cut and machined into cup shape | — |
| **Direct compression moulding** | Mould in exact cup shape → cup produced directly | Better wear characteristics |

# Sterilization and Cross-Linking of Polyethylene Ask AI

## Sterilization Methods Ask AI

- **Sterilization** → key step after processing
- Ethylene oxide or gas plasma → no cross-linking → **inferior wear** properties
- **Irradiation** → dose approx. 2–4 Mrad

## Irradiation and Oxygen Ask AI

- Irradiation outcome → depends on **oxygen availability**

| Environment | Result | Effect |
|---|---|---|
| Vacuum or inert gas (no oxygen) | **Cross-linking** | Chains connected to each other |
| Oxygen present | Chain scission + oxidation → **oxidative degradation** | All properties spoiled; ↑ wear, ↑ fracture risk |

## Free-Radical Mechanism Ask AI

- Irradiation → breaks **covalent bond** between carbon atoms
  - → unpaired electron → **free radical**
- Free radical → strong **affinity to oxygen**
  - Oxygen present → binds oxygen → polyethylene degradation
  - Oxygen absent → **cross-linking** between carbon atoms → resists intermolecular mobility

## Higher-Dose Cross-Linking Ask AI

- More cross-linking
  - → higher radiation dose, up to 8–10 Mrad
- Benefits → resists intermolecular mobility
  - More resistant to deformation, wear, **osteolysis**
  - Improved implant longevity
- Cost → worse mechanical properties
  - Lower **fatigue** and fracture resistance

## Free Radical Removal Ask AI

- Oxygen re-entry after sterilization still possible
  - **Shelf storage**
  - **In vivo** environment
- Free radical removal → two options

| | Remelting | Annealing |
|---|---|---|
| Process | Plastic melted again | — |
| Free radicals | All removed | Not all removed |
| Crystalline part | Affected | Not affected |
| Fatigue cracking | Prone | Not prone |

- **Crystalline part** → responsible for mechanical strength of polyethylene
- Alternative strategies
  - Irradiation + **antioxidant**, e.g. vitamin E
  - Sequential irradiation and annealing

**Must-Know**

- Irradiation without oxygen → cross-linking
- Irradiation with oxygen → oxidative degradation
- Oxygen re-entry → shelf storage, in vivo
- Remelting → all free radicals removed, prone to fatigue cracking
- Annealing → crystalline part spared, free radicals remain

### HXLPE vs UHMWPE Ask AI

| | UHMWPE | HXLPE |
|---|---|---|
| Wear | Reference | ↓ approx. 55–90%, metal or ceramic head |
| Larger **head diameter** | ↑ volumetric wear | No increase in volumetric wear |

## Polyethylene: Advantages and Disadvantages Ask AI

### Advantages Ask AI

PE

- **Biocompatible** → not toxic or carcinogenic like metal ions
- No **squeaking**, unlike ceramic
- **High wear resistance** → especially HXLPE
- Less **fatigue failure**
- **Easy revision**
- **Familiar** → no specific learning curve
  - Same learning curve for HXLPE and UHMWPE
- **Adaptability** → liner options
  - Extended liner, enhanced posterior wall, offset liner, constrained liner
- **Forgiving** → two key situations
  - **Impingement** → stem neck vs polyethylene edge → wear
    - Catastrophic in metal and ceramic; forgiving in polyethylene
  - Increased **cup abduction** → edge loading, head vs polyethylene → wear
    - Catastrophic in metal and ceramic

    - Forgiving in polyethylene
- **Lower cost**

### Disadvantages Ask AI

PE

- **Wear** → osteolysis → aseptic loosening
    - Vicious circle, nothing to stop it

## Bearing Surface Comparison Ask AI

| Bearing | Wear |
|---|---|
| Cobalt-chromium on polyethylene | Approx. 200 µm per year |
| Ceramic head on polyethylene | Half |
| On cross-linked polyethylene | Approx. 10% |
| Metal-on-metal, ceramic-on-ceramic | Lower still |

- Metal-on-metal, ceramic-on-ceramic → better wear than polyethylene (agreed)
- Polyethylene still favoured for
    - Intra-operative liner options → **adaptability**
    - Forgiveness
    - Easy revision
    - Cost
    - Familiarity
    - Avoids other bearings' specific disadvantages
        - Metal-on-metal → **ALVAL** syndrome → now rarely used
        - Ceramic → squeaking, fracture, other problems
- **Metal-on-polyethylene** → gold standard, preferred by most surgeons worldwide
    - National joint registry, America, 2003–2013 → most used
    - Swedish Hip Arthroplasty Register, 2005–2013 → lion's share

**Exam Pearl**

- Metal-on-polyethylene → gold standard bearing
- Cobalt-chromium on polyethylene → approx. 200 µm per year
- Cross-linked polyethylene → wear approx. 10%

# Summary Ask AI

## Metals Ask AI

- **Alloys** → stronger, harder than pure metals

| Alloy | Modulus of elasticity | Key point | Use |
|---|---|---|---|
| **Stainless steel** | Very high | Less bending stress on cement → cement protected | Cemented stems |
| **Cobalt-chromium** | Very high | Very good wear properties | Cemented stems; femoral head |
| **Titanium** | Low, closer to bone | No stress shielding or bone resorption | Uncemented stems and cups |
| **Tantalum** | Low, closer to bone | No stress shielding or bone resorption | Uncemented stems and cups; acetabular reconstruction (very promising) |

- Metal ion effects, local and systemic → **metal-on-metal** use much decreased

## Ceramics Ask AI

- Superior in **wear** and osteolysis
- **Fracture** → much decreased in new generations, still present
  - → needs precise surgical technique

## Polyethylene Ask AI

- **UHMWPE** → more than 200,000 monomers per molecule
- Irradiation in oxygen-free environment → **cross-linking**
  - Higher dose → **highly cross-linked polyethylene** (HXLPE)
- Up to 90% decrease in **wear**
- Allows **large heads** → greater stability
- Still biocompatible, familiar, adaptable, forgiving, low cost

# How to choose your Bearing

*(Wear, Lubrication and Bearing Surface Selection)*

**Prof Dr Khaled Fawzy**

**Watch Original Lecture**

*Lecture transcript processed with AI assistance.*

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# AI detected corrections (Mostly bad transcripts)

### Corrected - the text was changed; the transcript said

**C1** (p. 40) “activation of the osteoblasts” → “osteoclast activation”

### Queried - kept as spoken, but looks wrong

**Q1** (p. 40) as spoken; force × coefficient of friction gives frictional force, not wear (wear depends on load and sliding distance)

# Case: The Painful Hip After Arthroplasty Ask AI

## Initial Presentation Ask AI

Case

- Patient with **arthritis** → previous technique, options, balance between them
- Relatively young, newly presenting arthritis
- Decision → **osteotomy** vs arthroplasty
- Next question → which bearing?
- At this age → **metal-on-polyethylene**
- Fixation → cemented vs uncemented; cemented proposed, rationale questioned
- Key question → bone **osteoporotic**? → yes
- Treated with this implant

## Symptomatic Years Later Ask AI

Case

- Symptomatic 3 years later, at age 75 → **prosthesis type**?
- Second scenario → symptomatic 10 years after replacement → prosthesis type?
- Question → which **bearing** was used?
- Cause of pain → suggested: **loosening**
- Beyond loosening → other obvious findings?
- Suggested cause of loosening → **cement**
- Component **not cemented** → still osteolysis
  - Osteolysis → several zones; then inspect other component
- Question → osteolysis from cement, or another cause?
- Never-cemented component + **osteolysis** → cause?
- **Instability** proposed → questioned as cause of this change

## Eccentric Head Ask AI

Case

- **Head not centred** in cup
- Position unchanged → good position at implantation 10 years earlier
- Normal → head sits **concentrically** within cup
- Eccentric head → not shortening → process inside joint
- Cause → **polyethylene wear**

**Clinical Pearl**

- Eccentric head in cup on X-ray → polyethylene wear
- Osteolysis around uncemented component → not cement-related

# Polyethylene Wear and Osteolysis Ask AI

## Particle Disease Ask AI

PE wear

- Polyethylene → **wear particles** → osteolysis
- Mechanical ball-and-socket → rusts
- Prosthetic joint → no rust → **polyethylene wear** instead
- **Polyethylene wear particles** → very small

  - → engulfed by **macrophages**
  - → **cytokines** released
  - → **osteoclast** activation → osteolysis *[C1 · transcript: "activation of the osteoblasts"]*

### Wear Generation Ask AI

PE wear

- Head smooth to eye; electron microscope → **asperities**
  - Asperities scratch polyethylene → shed **polyethylene particles**
- After wear → cup cavity **larger**
- Case X-ray → worn polyethylene → particles → **osteolysis**
- Weakest link → **polyethylene** → reason for alternative bearings

**High Yield**

- Wear particles → macrophages → cytokines → osteoclasts → osteolysis
- Head asperities → scratch polyethylene → particles
- Weakest link of the bearing → polyethylene

## Friction and Wear Mechanisms Ask AI

### Coefficient of Friction Ask AI

- Wear → force × **coefficient of friction** *[Q1 · as spoken; force × coefficient of friction gives frictional force, not wear (wear depends on load and sliding distance)]*
- **Metal-on-polyethylene** → very low coefficient of friction
  - **Charnley** → low friction arthroplasty
  - Precisely → **low frictional torque arthroplasty**

### Mechanisms of Wear Ask AI

- Four mechanisms:
  - **Abrasion**
  - **Adhesion**
  - **Third-body wear**
  - Polyethylene **fatigue**
- Microscopic asperities rub → mutual abrasion → **abrasive wear**
- **Ceramic** → **surface pits** → rub on bearing surfaces → wear

### Joint Load Ask AI

- Hip stresses → up to 5× **body weight**
  - 80 kg patient, walking up/down hill → ~400 kg across hip

## Lubrication Ask AI

### Types of Lubrication Ask AI

- Requirements → low coefficient of friction, low force, good **lubrication**
- Three types:
  - **Fluid film lubrication** → occurs with movement
  - **Boundary lubrication** → **lubricin** molecules in synovial fluid

  - **Mixed lubrication**

### Head Size and Lubrication Ask AI

| Head size | Dominant lubrication |
|---|---|
| Large head | Fluid film lubrication |
| Small head (e.g. 28 mm) | Boundary lubrication |

- Boundary → **lubricin** lines surface asperities; contact persists
- Fluid film → **fluid layer** separates the two surfaces
  - Analogy → car oil between two metal surfaces
- Car engine → additive applied to metal → **boundary lubrication** also needed
- Oil build-up between surfaces → **fluid film** → minimal friction
- Basis → mechanics and biomechanics

## Improving Ultra-High Molecular Weight Polyethylene Ask AI

### Oxidation Ask AI

UHMWPE

- **UHMWPE** particles → osteolysis → need for improvement
- Cut section, cup left long on shelf → **oxidation**
- Oxidation → **accelerated wear**
- Unrecognised when low friction arthroplasty evolved → discovered later
- Oxygen + **irradiation** → oxidation
- Irreversible → oxygen not removable afterwards → act before oxidation

### Processing Ask AI

UHMWPE

- **Sterilisation** → required
- **Annealing** → part of processing sequence
- Sterilise in **inert gas** or vacuum
- **Cross-linking** → high-dose radiation
  - After radiation → **anneal** or melt
- Most up-to-date → **vitamin E**
- Results:
  - Improved **wear rates**
  - Long-term **longevity**
  - Allows **large head**

### Conventional vs Cross-Linked Ask AI

UHMWPE

| Feature | Conventional PE | Cross-linked PE |
|---|---|---|
| Large head → volumetric wear | Increased (possibly ↑ linear wear) | Not increased |
| Fatigue strength, fracture toughness | — | Decreased |
| Risk–benefit | — | Benefit outweighs risk |

**Exam Pearl**

- Oxygen + irradiation → oxidation → accelerated wear
- Sterilise in inert gas or vacuum
- High-dose radiation → anneal or melt → cross-linked PE
- Cross-linked PE → ↓ fatigue strength and fracture toughness
- Latest → vitamin E

## Resurfacing Ask AI

- Operation → **resurfacing**
- Bearing → **metal-on-metal**
- Questions → osteolysis with MoM? common? what **fractured**?
- Fracture predisposition → small **neck** size, not stem size
- **Females** → contraindicated; no longer used
  - Reason → **smaller necks** and heads → fracture; not only childbearing
- **Notching the neck** during insertion → fracture
- **Technique-sensitive** → training required

## Metal-on-Metal Bearings Ask AI

### Advantages and Toxicity Ask AI

MoM

- Advantage → femoral **bone stock** likely preserved
- Problem identified later:
  - **Toxicity** → metal ions pass into blood

### Run-In Phase Ask AI

MoM

- Engine/car analogy → joint **runs in**
- First year → irregularities wear each other off → **smoother surfaces**
- Run-in → ↑ blood **cobalt** and chromium; then decrease
- Graph → very high **cobalt and chromium levels** at implantation → decline after first year
- Levels **rising** on analysis → something wrong

### Contraindications Ask AI

MoM

- **Renal impairment** → cannot be used; harmful
- **Women of childbearing age** → teratogenic

### The Painful Metal-on-Metal Hip Ask AI

MoM

- Painful, no **fracture** → design fault?
- Option → **aspiration**
- Painful, nothing wrong, **no gap**
- Imaging → **MARS MRI** or other imaging
  - MARS → Metal Artifact Reduction Sequence
- Finding → large collection → **ALVAL**
  - ALVAL → Aseptic Lymphocyte-dominated Vasculitis-Associated Lesion
- **Metal debris** clearly visible
- Metal ions + metal hypersensitivity → tissue destruction → **pseudotumour**
- Concerns:
  - Metal ions in **blood**
  - **Renal impairment**
  - Possible **carcinogenic** effects
  - **Pseudotumour**
- Pseudotumour → majority of MoM implants **withdrawn from market**

**Must-Know**

- Resurfacing fracture → small neck, female sex, neck notching
- MoM contraindications → renal impairment, women of childbearing age
- Rising cobalt/chromium after first year → something wrong
- Painful MoM, normal X-ray → MARS MRI → ALVAL / pseudotumour
- Pseudotumour → most MoM withdrawn from market

## Ceramic-on-Ceramic: Squeaking and Fracture Ask AI

### Squeaking Ask AI

CoC

- **Squeaking** → fracture of something?
- Prosthesis → **ceramic-on-ceramic**
- CoC → well known for **squeaking**
  - Very noisy, distressing → especially going **upstairs**
- Another CoC → **fracture**
- Evolving problems:
  - **Squeaking**
  - **Fractures**

### Identifying the Bearing on Radiographs Ask AI

| Radiograph | Bearing surface |
|---|---|
| Left side | Metal-on-polyethylene |
| Other side | Ceramic-on-polyethylene |
| This side | Ceramic-on-ceramic |

- Advanced countries → **implant records** document the prosthesis

- Cup appearance → probably **polyethylene liner**, not ceramic; head visible
- **Half ball, broken** → not metal (very hard, does not break in half) → ceramic

### Ceramic Fracture Ask AI

Ceramic

- Question → why does ceramic break? (glass analogy)
- **Micro-cracks** → first, second, third generations → propagate
  - → **fracture**
- **Crack propagation** + trauma → fracture
- If using ceramic → fourth generation **BIOLOX delta**
  - **Pinkish** colour
  - Very low **fracture risk**
- Diagnosis by **X-ray**?
- Often not visible on **X-ray**
- Diagnosis:
  - **Pain** complaint
  - **Squeaking**
  - **CT**
  - Possibly **synovial fluid** analysis
- Revision timing → **as soon as possible**

**Exam Pearl**

- Fourth generation BIOLOX delta → pinkish → very low fracture risk
- Ceramic fracture → often invisible on X-ray → pain, squeaking, CT
- Fractured ceramic → revise as soon as possible

## Revision of a Fractured Ceramic Ask AI

### Choice of Revision Bearing Ask AI

Revision

- Revise to → **ceramic**
- Revision to **polyethylene-on-metal:**
  - Broken ceramic debris → **third body** → wear
  - Debris scratches **metal head** → rougher → accelerated wear
- Real option → **ceramic-on-ceramic**

### Taper and Stem Ask AI

Revision

| Finding | Action |
|---|---|
| Taper intact, stem stable | Metal sleeve |
| Taper damaged | Remove and revise stem → damaged taper damages ceramic → fracture |

**Pitfall**

- Fractured ceramic → metal head on polyethylene → third-body wear
- Ceramic debris scratches metal head → accelerated wear
- Damaged taper retained → ceramic damage → fracture

## Ceramic-on-Ceramic: Advantages and Concerns Ask AI

### Advantages Ask AI

CoC

- **Hard-on-hard** bearing
- Low **surface roughness**
- Low **coefficient of friction**
- Small, inert **wear particles**
- **Least volumetric wear**
- **Wettable, hydrophilic** → encourages fluid film lubrication
- Low **scratch profile**

### Scratch Behaviour and Wettability Ask AI

| Property | Metal | Ceramic |
|---|---|---|
| Accidental scratch | Asperities → more polyethylene wear | V-shaped pit → less wear |
| Water contact angle | 86° | 45° |

- Smaller **contact angle** → more wettable surface

### Concerns Ask AI

CoC

- **Fracture**
- **Squeaking**
- No **lipped inserts**
- **Limited modularity** → head sizes, neck lengths
- Needs **accurate positioning**
- **Expensive**
- **Stripe wear**

# Choosing the Bearing Ask AI

## Options and Wear Ask AI

Bearings

- Options:
  - **Metal-on-polyethylene**
  - **Ceramic-on-polyethylene**
  - Metal or ceramic on **highly cross-linked polyethylene**
  - **Metal-on-metal**
  - **Ceramic-on-ceramic**
- Metal-on-polyethylene → **largest volumetric wear**
- Ceramic-on-ceramic, metal-on-metal → **least volumetric wear**

## Decision Factors Ask AI

Bearings

- **Sensible choice**, not absolute → based on:
  - Research **evidence**
  - **Benefit–risk ratio**
  - **Price**
  - Patient age, activity, comorbidities, **life expectancy**

## Choice by Age Ask AI

Bearings

| Patient group | Preferred bearing | Alternative |
|---|---|---|
| Young, active, below 60 | Ceramic-on-ceramic, large head (32 or 36 mm) | Ceramic-on-cross-linked PE |
| 60–65, active | Ceramic-on-ceramic | Ceramic-on-cross-linked PE |
| 65–75 | Ceramic-on-cross-linked PE | — |
| Above 75 | Cobalt-chrome head on cross-linked PE | — |

## Resurfacing Indications Ask AI

Resurfacing

- **Limited indications:**
  - First → **males**
  - **Large head size** → 56–64 mm; small sizes not used
  - **McMinn hip** → Birmingham hip; other hips withdrawn from market
  - Needs **expertise** → otherwise conventional hip replacement

## Caution with Ceramic-on-Ceramic Ask AI

CoC

- Choice → depends on many **factors**
- **Ligamentous laxity** → CoC with care
- ↑ Preoperative **range of motion** → CoC with care
- **Anatomical deformities** → CoC with care

## Summary: Classification of Bearing Surfaces Ask AI

| Category | Bearing | Key point |
|---|---|---|
| Hard-on-soft | Metal-on-polyethylene | — |
| Hard-on-soft | Metal-on-cross-linked polyethylene | — |
| Hard-on-soft | Ceramic-on-polyethylene | — |
| Hard-on-hard | Metal-on-metal | Major concerns; withdrawn from market; limited indications |
| Hard-on-hard | Ceramic-on-ceramic | Better wear rates and longevity vs other bearings |

**High Yield**

- Below 60, active → ceramic-on-ceramic, large head (32 or 36 mm)
- 65–75 → ceramic-on-cross-linked PE
- Above 75 → cobalt-chrome head on cross-linked PE
- Resurfacing → males, 56–64 mm head, Birmingham hip, expert surgeon
- Laxity, ↑ preop ROM, deformity → ceramic-on-ceramic with care

# Implant fixation

*(Cemented and Cementless Fixation)*

**Prof Dr Mahmoud Fahmy**

**Watch Original Lecture**

*Lecture transcript processed with AI assistance.*

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# AI detected corrections (Mostly bad transcripts)

## Corrected - the text was changed; the transcript said

**C1** (p. 56) “grade C radiolucency 50% to 90%” → “grade C radiolucency 50–99%”

**C2** (p. 56) “grade D: absence of the stem” → “absent cement below stem tip”

**C3** (p. 58) “extensive coating loads proximal diaphysis” → “extensive coating loads diaphysis”

# Implant Fixation: Definition and Types Ask AI

- Topic → **types of implant fixation**
- **Implant fixation** → coupling of prosthetic component to musculoskeletal system
  - Prosthetic + natural elements → act together harmoniously
- Goal → eliminate motion at implant–bone interface
- Two types
  - **Cemented fixation**
  - **Cementless fixation** → biologic fixation
    - Bone growth into or onto porous surfaces

# Indications: Cemented vs Cementless Ask AI

## Schools of Thought Ask AI

- Many schools on indications
  - Some surgeons → cement in every patient
  - Others → cementless in every patient
- Cup → **porous-coated cementless cup** preferred by most surgeons
  - Reason → reliable long-term results
- Cemented → still acceptable in elderly, **poor bone stock**
  - Both cemented and cementless → acceptable options

## Dorr Classification Ask AI

Dorr

- **Dorr classification** → major guide to fixation choice
- Dorr type → largely decides cemented vs cementless stem

| Factor | Cementless stem | Cemented stem |
|---|---|---|
| Dorr type | A and B | C (stovepipe femur) |
| Age | Young | Elderly |
| Activity | High-activity | — |
| Other | — | Irradiated bone (summary) |

- Any patient → either fixation type remains possible

**High Yield**

- Dorr A and B, young, high-activity → **cementless stem**
- Dorr C (stovepipe femur), elderly → **cemented stem**
- Cup → porous-coated cementless preferred

# Bone Cement: Composition and Polymerization Ask AI

## Composition Ask AI

PMMA

- Cement → polymer: **polymethyl methacrylate (PMMA)**
- Supplied to theatre → powder + liquid
- **Powder**
  - **Polymer** → two carbon atoms
    - First carbon → two hydrogen atoms
    - Second carbon → methyl group + methacrylate group
  - **Initiator** of polymerization → benzoyl peroxide
  - **Radio-opacifier**
    - Gives cement radio-opacity on X-ray
  - Antibiotics
- **Liquid**
  - **Monomer** → methyl methacrylate
  - **Accelerator** of polymerization → dimethyl-p-toluidine
  - **Stabilizer** (inhibitor) → prevents monomer polymerization inside liquid
  - Additives
- Why not liquid alone
  - Very long setting time intra-operatively
  - Large energy (heat) release
  - **Shrinkage**
- Compensation → **pre-polymerized powder**

## Polymerization Ask AI

- Initiator + activator → free radicals → break carbon double bond
- Large energy release → **exothermic reaction**
- In vitro → 80–85 °C
- In vivo → much lower
  - Thin cement layer → heat dissipated via stem + bone blood circulation
  - Cement **imbibes water** → further cooling

**Exam Pearl**

- Powder initiator → **benzoyl peroxide**
- Liquid accelerator → **dimethyl-p-toluidine**
- Liquid alone → slow setting, high heat, shrinkage
- Polymerization exothermic → in vitro 80–85 °C

## Cementing Generations Ask AI

- Successive generations → better canal preparation, pressurization, mixing

| Generation | Features |
|---|---|
| First | Hand mixing; no bone preparation, no plug, no pressurization → blood/void inclusion, ↓ penetration |
| Second | Plug (cement restrictor); retrograde filling with **cement gun**; pressurization → ↑ mechanical interlock |
| Third | Improved pressurization with **rubber seal**; **vacuum mixing** |
| Recent advances | Aim → ↓ mixing time |

## Mechanical Properties of Cement Ask AI

- Two groups → short-term, long-term properties

### Short-Term Properties Ask AI

- **Brittle**
  - No significant plastic deformation before fracture
  - Brittle → **notch-sensitive**
    - Notch → **stress riser** → easy breakage
  - Revision → crack cement first → then remove

| Load | Cement strength |
|---|---|
| Compression | Strong |
| Tension | Poor |
| Shear | Moderate |

### Long-Term Properties Ask AI

- Susceptible to **creep**
- **Stress relaxation**
- **Fatigue failure**

### Creep Ask AI

Cement

- **Creep** → ↑ plastic deformation under constant load over time
- In vitro experiment
  - Cement inside **serrated metal cone**
  - One specimen → loaded long period; other → unloaded
  - After set period
    - Unloaded → no serration imprint
    - Loaded → serrations imprinted
  - → cement undergoes creep with time

### Fatigue Failure Ask AI

Cement

- Last long-term property
- **Fatigue failure** → failure below ultimate tensile strength under cyclic loading
- Site → points of high stress concentration
- Protection → **stress relaxation**
- Creep + stress relaxation → key clinical role (below)

## Cement Function and Stem Design Philosophies Ask AI

### Cement as Grout Ask AI

- Function → macro- and micro-interlock + **interdigitation** with trabecular bone
- Cement → **grout, not glue** (mechanical grout)
  - Fills space → forms **mantle** → load transfer to bone
  - More load to bone; likely more even stress distribution
- Function requires preparation of both surfaces
  - Bone surface → cleaned of blood + debris → ↑ penetration → interdigitation
  - Stem surface → one of two design philosophies → different stem behavior

### Stem Design Philosophies Ask AI

- Two cemented stem philosophies

| Feature | Composite beam (shape-closed) | Loaded taper |
|---|---|---|
| Surface | Rough | Polished |
| Geometry | — | Tapered in two planes |
| Cement interface | Strong bond with cement | — |
| Subsidence | Not allowed; subsidence → failure | Designed to subside |

- Loaded taper subsidence → controlled by **air-filled (hollow) centralizer**
- Axial loading → subsidence
  - → **radial compressive forces** in cement
  - → transferred to bone as **tensile hoop stresses**

### Clinical Significance of Long-Term Properties Ask AI

Taper

- Long-term properties → clinical relevance
  - Axial load → stem subsides → radial compressive forces
  - **Creep** → ↑ interdigitation of cement within medullary canal
  - Radial compression → transmitted as tension (hoop stress) in bone → harmful
- Relief of bone tension
  - Sleep/rest → no load change; stress persists
  - → **stress relaxation**
    - Likely → bone stress dissipates; compressive force holding stem remains

**Must-Know**

- Cement → **grout, not glue**
- Composite beam → rough, no subsidence; subsidence → failure
- Loaded taper → polished, double-tapered, designed to subside
- Subsidence → radial compression in cement → hoop stress in bone
- Stress relaxation → protects against fatigue failure

# Cement Mantle Ask AI

## Mantle Stresses Ask AI

- Load transfer → through **cement mantle**
- Mantle → weakened by voids (porosity); located at high-stress site
- Forces
  - Compression
  - Shear + tension at **cement–metal interface**

## Avoiding Mantle Failure Ask AI

- **Mantle thickness** > 2 mm → ↓ fracture risk
- **Uniform** mantle
- ↓ **Porosity** → fewer stress points
- Good **pressurization** → ↑ interdigitation
- **Clean, dry bone** bed → ↑ interdigitation
- **Stiff stem** (biomaterials lecture) → ↓ bending stress on mantle
- No **sharp edges** on stem
  - → avoids stress concentration

**High Yield**

- Mantle > 2 mm, uniform
- ↓ Porosity, good pressurization
- Clean, dry bone bed
- Stiff stem, no sharp edges

# Radiographic Grading of Cementing Ask AI

## Barrack & Harris Grading Ask AI

Barrack

- **Barrack & Harris grading** → grades A, B, C, D

| Grade | Finding |
|---|---|
| A | Complete filling → "white-out" |
| B | Radiolucency up to 50% |
| C | Radiolucency 50–99% (stated as 50–90%) *[C1 · transcript: "grade C radiolucency 50% to 90%"]* |
| D | 100% radiolucency, or absent cement below stem tip *[C2 · transcript: "grade D: absence of the stem"]* |

## Zones Ask AI

- **Cup** → **DeLee and Charnley zones**
  - Numbered lateral → medial
- **Stem** → **Gruen zones**
  - Lateral → medial: zones 1–7
  - Anterior → posterior: zones 8–14

# Cementless Fixation and Bone Ingrowth Ask AI

## Ingrowth vs Ongrowth Ask AI

Cementless

- Cementless → **biologic fixation**
  - Bone growth into porous surfaces → secures implant
- Two ways
  - **Bone ingrowth** → into porous surface
  - **Bone ongrowth** → onto rough surface
- Image → ingrowth into porous surface vs ongrowth onto rough surface

## Bone Ingrowth Ask AI

- **Porous-coated surface** → cellular/physiological response
  - Resembles healing cascade of cancellous defect
  - → similar to **primary fracture healing**
- Porous coating types
  - **Sintered beads** (most popular)
    - Titanium or cobalt-chrome
    - Sintered to substrate at high temperature
  - **Titanium fiber mesh**
    - Cut titanium wires, bonded (probably diffusion bonding)
  - Likely **plasma-sprayed titanium alloy**
  - Probably **porous tantalum** (trabecular metal; biomaterials lecture) → highly porous

- Factors facilitating ingrowth
  - Implant → likely optimal pore size + porosity
  - Surgeon
    - Viable bone (probable)
    - Good contact; likely minimal gaps
    - ↓ **Micromotion** → via initial rigid fixation
- **Initial rigid fixation** → critical for ingrowth

### Initial Rigid Fixation Ask AI

- Two methods → **press-fit, line-to-line**

| Feature | Press-fit | Line-to-line |
|---|---|---|
| Principle | Minimal motion; rigid fixation via compression | Bone contour = implant size |
| Acetabular cup | No screws needed | Screws required |
| Femoral stem | — | Extensively porous coated → initial frictional fit |

- Press-fit → achieved by **under-reaming** of bone

> **Clinical Pearl**
> - Press-fit cup → compression alone, no screws
> - Line-to-line cup → screw fixation required
> - Line-to-line stem → extensive porous coating for frictional fit

## Bone Ongrowth and Hydroxyapatite Coating Ask AI

### Bone Ongrowth Ask AI

- **Bone ongrowth** → onto roughened, non-porous surface
- Microscopic picture → shows rough-surface production
- Production methods
  - **Plasma spray** → probably molten metal particles sprayed onto substrate
  - **Grit blasting**
    - Particles propelled onto surface → roughened
    - **Aluminum oxide** particles → rough surface + **microdivots**
- Ongrowth requirements
  - Rough surface with microdivots
  - Stem design → **double-wedge taper**
  - Initial rigid fixation → must be **press-fit**

### Hydroxyapatite Coating Ask AI

HA

- **Hydroxyapatite (HA) coating** → osteoconductive
- Adjuvant → porous-coated + grit-blasted surfaces
- Function → ↓ component **migration**
  - Mechanism → bidirectional closure of bone–prosthesis gap

- Osteoblasts adhere to HA → grow toward bone → bone ingrowth

## Femoral Stem Loading and Fixation Signs Ask AI

### Stem Loading by Coating Ask AI

- Load transfer → depends on coating: proximal vs distal (extensive)
- **Proximal coating** → load to metaphysis + proximal diaphysis
- **Extensive coating**
  - Load → diaphysis *[C3 · transcript: "extensive coating loads proximal diaphysis"]*
  - **Stress shielding** → proximal femur
  - **Spot welds** (endosteal consolidation) → near stem end

### Signs of Well-Fixed Cementless Components Ask AI

- Radiographic signs of well-fixed cementless components

| Femoral stem | Acetabular cup |
|---|---|
| Spot welds | Intact acetabular screws |
| Proximal stress shielding | No progressive radiolucent lines |
| No radiolucent lines around stem | No migration on serial radiographs |
| No subsidence on serial radiographs | — |

**High Yield**

- Extensive coating → proximal stress shielding + distal spot welds
- Spot welds → sign of well-fixed cementless stem
- Serial radiographs → no subsidence (stem), no migration (cup)

## Summary Ask AI

- Two types → cemented, cementless
- **Cemented**
  - Cement → polymer; grout, not glue
  - Brittle, **notch-sensitive**
  - Stress relaxation + creep
  - New cementing generations → better function
  - Stem mechanisms → loaded taper, composite beam
  - Indications → elderly, Dorr C, **irradiated bone**
  - Factors improving stem fixation in cement
- **Cementless**
  - Ingrowth (porous surface) + ongrowth (rough surface)
  - Indications
    1. Acetabular cups
    2. Dorr A and B

    - Young, high-activity
- Stem loading → depends on stem coating

# Stem designs

*(Composite Beam and Taper-Slip Concepts in Femoral Stems)*

**Prof Dr Ahmed Hazem**

**Watch Original Lecture**

*Lecture transcript processed with AI assistance.*

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**C1** (p. 68) “collum type cuts the whole neck” → “collum type retains whole neck”

## Queried - kept as spoken, but looks wrong

**Q1** (p. 66) As spoken; contradicts earlier statement that Types 1 and 2 are designed for Type A femur; double wedge suits Dorr A/B.

## Lecture Scope Ask AI

- Continuation of Dr. Mahmoud's lecture → focus on stem specifications
- Key topic: choosing a stem design within the available **portfolio**
  - Whole portfolio reviewed → how to choose a design
  - Stem designs reviewed → recognise the design at patient follow-up
  - Order: **cemented** designs → cementless designs

## Cemented Stems: Engineering Basis Ask AI

### Two Philosophies Ask AI

- **Two philosophies** for cemented stems → from engineering, not chosen by surgeons
  - Rod inside another material (cement) → two modes of interaction
    1. **Composite beam**
    2. **Loaded taper**

### Material Properties and Composite Action Ask AI

- **Cement** → strong in compression; compressive forces make it stronger
  - Shear, tension → voids → cement failure
- **Steel** → better in tension; no elastic stretch when pulled
- Same applies to shear
- **Composite action** → principle used in building construction
  - Concrete (cement) → resists compressive forces
  - Steel (iron) rods within → antagonise tension / shear forces
- Result → structure combining both forces
  - Compression antagonised alongside tension / shear
- Same philosophy → applied to cemented stems

## Composite Beam Ask AI

- Cement + stem components → bonded together in this region
- Stem and cement bonded firmly → **primary (initial) stability**
  - Requirement: no micromotion at all
- **Micromotion** → cement cracks
  - Same mechanism used deliberately in revision → cement cracked until failure → extracted
  - Any micromotion → harmful
  - Stem must stay fixed, stable in this zone

## Loaded Taper Ask AI

### Rationale: Secondary Stability Ask AI

- Problem: cement settles during the **first year**
  - Some micromotion → construct settles into final position

  - Engineers → use this micromotion → beyond initial stability → **secondary stability**
- Origin of the **loaded taper concept**
- Stem loading → forces converted into **compression**
  - → construct progressively tighter, stronger
  - → different kind of stability

### Barrel Analogy Ask AI

- Source: **barrels** (e.g., wine storage)
  - Good wooden blocks
  - Surrounded by **steel hoops**
- Barrel filled → more pressure on wooden blocks
  - Stainless steel wires (hoops) → compressive force converted into **hoop stress** (radial force)
  - → components tighter, more stable
- Stem equivalent: subsidence → compressive force converted into **radial force** within cement
  - → construct progressively tighter

### Design Requirements Ask AI

LT

- **Tapered** → able to subside
- **Polished (smooth)** → otherwise no subsidence
- **Void at the tip** → space beneath tip in cement mantle for subsidence
- Inserted with a **centraliser** → must be able to subside

### Controlled Subsidence Ask AI

- Aim: **secondary stability** after one year
- Expected stem subsidence: ~1 mm
  - Stem at insertion vs at one year → subsided inside cement mantle
- ~1 mm subsidence → termed **controlled subsidence**
  - Rate: ~1 mm per year (first year)

**Exam Pearl**

- Loaded taper requirements → tapered, polished, void at tip, centraliser
- Controlled subsidence → ~1 mm in first year
- Endpoint → secondary stability after one year

## Composite Beam vs Loaded Taper Ask AI

### Stability and Load Transfer Ask AI

- Composite beam → stable from start; locks and stays
- Loaded taper → locks, slides, subsides → **secondary stability** after one year
- Composite beam → load shared by stem, bone and cement together
  - Stresses delivered distally
- Loaded taper → **load transfer** to surrounding structures
  - Stem converts shear forces → compressive (radial) forces
    - Transmitted top → bottom, approximately along the stem

### Design Comparison Ask AI

| Feature | Composite beam | Loaded taper |
|---|---|---|
| Surface | Rough → cement interdigitates with stem | Polished |
| Shape | Anatomical | Straight, tapered |
| Aim | Stick to cement | Controlled subsidence within one year |
| Collar | Favoured → transmits stresses, reduces proximal stress shielding | Harmful → prevents subsidence |
| Void | Harmful → allows subsidence → cracks | Must (distal to tip) → centraliser required |
| Stability | Initial | Secondary (after one year) |
| Forgiveness | Not forgiving | More forgiving |

- **Centraliser** → allows stem to subside inside cement mantle
- Both concepts → work effectively

**Pitfall**
- No mixing between the two concepts
- Use one concept fully to its end, or the other fully to its end

## Cementless Stems: Fixation Principles Ask AI

### Biological Fixation Ask AI

- Goal: **biological fixation** (per Dr. Mahmoud)
  - Requirements:
    - **Immediate mechanical stability**
    - **Intimate bone contact** → as in primary bone healing
  - → stem designed to fit bone shape properly
    - Femur classified → stem fitted correctly to its shape

### Dorr Femoral Types Ask AI

- Three femoral types → **Dorr classification**
  - Type A, B, C

| Dorr type | Metaphysis | Zone of fixation |
|---|---|---|
| A | Very good zone of fixation | Metaphysis |
| B | Partly lost | Metaphysis (metaphyseal–diaphyseal) |

| Dorr type | Metaphysis | Zone of fixation |
|---|---|---|
| C | Entirely lost; good cortical support only | Diaphysis |

- Three fixation groups → metaphyseal / metaphyseal–diaphyseal / diaphyseal
- Analogy: primary stability in **revision surgery** when the classification was devised

## Khanuja Classification Ask AI

- **Khanuja classification** (2011) → still used today
  - Basis: zone of fixation + stem geometry
  - Method: all market stem geometries collected → where each manufacturer designed its stem to fix
  - Result: 4 main types + 2 additional types

| Khanuja type | Zone of fixation |
|---|---|
| 1, 2 | Metaphyseal |
| 3 (3A, 3B, 3C) | Metaphyseal + diaphyseal |
| 4 | Diaphysis |
| 5 (additional) | Modular |
| 6 (additional) | Anatomical |

## Types 1 and 2: Metaphyseal Fixation Ask AI

- Types 1 and 2 → **metaphyseal fixation** stems
  - Designed for the **Type A femur**

### Type 1: Single Wedge Ask AI

- Broad mediolaterally above; thin below (other plane)
  - → enters within the narrow cortex, fits the metaphysis
- Broad in one plane, thin in the other
- Depends on **correct positioning**

### Type 2: Double Wedge Ask AI

- Opposite of Type 1 → broad in both AP and lateral views
- Termed **metaphyseal filling** → fills metaphysis in AP and lateral
- Major problem: whole metaphysis filled → large amount of good bone removed
  - Likely not feasible through a **minimally invasive** approach
- Impossible in the **Type A femur** *[Q1 · As spoken; contradicts earlier statement that Types 1 and 2 are designed for Type A femur; double wedge suits Dorr A/B.]*
- Works well in the **Type C femur** → broad distally

## Type 3: Metaphyseal–Diaphyseal Fixation Ask AI

- Category **most surgeons** use; different devices
- All grip at the **metaphyseal–diaphyseal junction**
  - Initially designed for Type B femur → in practice also work well in Type C femur
- Difference from Type 1 → **straight**, continuous taper
  - No wide-above / narrow-below sudden change
  - Continuous taper top → bottom in AP and lateral views
- Three categories → 3A, 3B, 3C

### Type 3A Ask AI

- Simplest in operation
- **Conical**, not cylindrical → straight taper top → bottom
- **Three-point fixation** in AP and lateral views
- Most of today's market

### Type 3B: Wagner Design Ask AI

- **Splined**, long stems
- Grip mainly in **proximal diaphysis**, not metaphysis
- Long splines (flutes) → **three-point** resistance
- Splines cut into bone like a knife → **rotational stability**
- **Conical** → any position (version) possible
  - → use reportedly favoured in difficult / deformed femora

### Type 3C Ask AI

- Used by **most surgeons**
- Grips **metaphyseal–diaphyseal junction** + proximal diaphysis
  - → also works well in Type C femur
- **Rectangular cross-section**; fully coated
- Fixation in AP view:
  - Rectangular shape → **four-point** rotational stability
  - Flat surface → fits lateral position with **three-point** fixation
  - Rectangular stem → grips metaphysis; fixes approximately in metaphysis in **lateral view**

**Clinical Pearl**

- Type 3 → category most surgeons use
- 3A → most of today's market
- 3B (Wagner) → splines give rotational stability; any version
- 3C → also works well in Type C femur

## Type 4: Diaphyseal Fixation Ask AI

- Grip in the **diaphysis** → metaphysis lost
- Very tough prostheses
  - **Cylindrical columns** → diaphyseal fit
  - Broad enough to fill diaphysis → femur exposed to cracks / fractures

  - Enter and grip diaphysis → **diaphyseal scratch fit**
- **Wedge (lock)** inside diaphysis → diaphyseal problems, cracks
  - → most have declined in market use

## Types 5 and 6: Modular and Anatomical Stems Ask AI

### Type 5: Modular Stems Ask AI

- Used for **diaphyseal fixation**
- Fixed distally + proximally → two different parts
  - **Distal part** → fixed distally with screws or coating
  - **Proximal part** → grips metaphysis, possibly weak in that patient

### Type 6: Anatomical Stems Ask AI

- Back on the market now
- Broad laterally + broad posteriorly → fills **metaphysis**
- **Anatomical contour** → fits proximal anteversion
  - → right and left stems

## Short Stems Ask AI

- Newer design type
- Length: standard stem ~14 cm → **short stem** ~half
- **Metaphyseal engagement** only
- Very many types on the market
- Classified by amount of **calcar / neck** removed:
  - **Collum** type → retains whole neck *[C1 · transcript: "collum type cuts the whole neck"]*
  - **Partial collum** type
  - **Trochanteric-harming** type
- All fit in the metaphysis

## Summary Ask AI

### Cemented Stems Ask AI

Summary

**High Yield**

- Cemented stems → composite beam or loaded taper
- Composite beam → initial stability; rough, collar, anatomical; not forgiving
- Loaded taper → polished, collar incompatible, straight, tapered
- Loaded taper → more forgiving; secondary stability after initial insertion

### Cementless Stems Ask AI

Summary

- Grip metaphysis, metaphyseal–diaphyseal junction, or diaphysis → **four types**
  - Most surgeons use Type 3
- Two additional types → **modular, anatomical**

# Cup designs

*(Cemented and Cementless Acetabular Components)*

**Prof Dr Fady Kamal**

**Watch Original Lecture**

*Lecture transcript processed with AI assistance.*

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# AI detected corrections (Mostly bad transcripts)

## Queried - kept as spoken, but looks wrong

**Q1** (p. 77) As spoken; safe zone is posterosuperior quadrant only; anterosuperior screws endanger external iliac vessels.

# Overview of Acetabular Cup Designs Ask AI

- Topic → **acetabular cup design**
- Two groups:
  - **Cemented cups**
  - **Cementless cups**

# Cemented Cup Ask AI

Cemented

## Material Ask AI

- Made of **polyethylene** → two types (materials lecture):
  - **Ultra-high molecular weight polyethylene (UHMWPE)**
  - **Highly cross-linked polyethylene (HXLPE)**
- Very large difference in **longevity** between the two
- Features discussed:
  - **Cement spacers**
  - Cup **profile** → thick vs thin polyethylene
  - **Internal diameter**

## Cement Spacers Ask AI

- **PMMA spacers (cement spacers)** → adequate cement mantle all around cup
  - Cement mantle surrounds cup circumferentially
- **Cement mantle:** 2 to 4 mm
- Approximate value quoted: 2.5 mm

## Marker Wire Ask AI

- **Marker wire** → placed in desired direction:
  - Posterior
  - Posterosuperior
  - Superior
- Runs around cup → two functions:
  - Shows **position of the cup**
  - Shows probable **wear** on radiographs
- Possibly also estimates cup **thickness**
  - Ellipse of cup measured along cup diameter
- Ensures **thickness of the cement** around cup

### Outer Surface Grooves Ask AI

- **Grooves on the outer surface** → macro-interlock between cement and cup

### Profile Ask AI

Cemented

- High vs low profile → adjusts **lateralization** of cup

| Goal | Choice |
|---|---|
| Lateralize cup slightly | **High profile** |
| Medialize cup slightly | **Low profile** |

### Internal Diameter Ask AI

- Head sizes: 22 up to 40 mm
- Larger heads not usable with **UHMWPE** → wear
- UHMWPE concern → later **osteolysis**
- **HXLPE** → wear reduced by 90%
- 32, 36 and 40 mm heads → **HXLPE** only, not conventional polyethylene

| | UHMWPE (conventional) | HXLPE |
|---|---|---|
| Wear | Limits larger heads | Reduced by 90% |
| Late concern | **Osteolysis** | — |
| 32, 36, 40 mm heads | Not usable | Usable |

**High Yield**

- Cement mantle → 2 to 4 mm
- PMMA spacers → circumferential cement mantle
- High profile → lateralize; low profile → medialize
- **HXLPE** → wear reduced by 90%
- 32, 36, 40 mm heads → HXLPE only

## Cementless Cup Material Ask AI

- Cementless features discussed:
  - **Cup material**
  - **Cup shape**
  - **Surface coating** of the material
  - **Liner** and modularity (liner inside cup)
  - **Fixation**
- Options → cobalt-chrome / titanium / tantalum
- Most widely used now → **titanium**
- Titanium advantages:
  - Porous → **bone ingrowth**
  - **Modulus of elasticity** → less stress shielding on surrounding bone
  - Easier design and manufacture

  - Easier insertion → **press-fit** technique
- **Tantalum** → probably higher porosity than other materials used
- Tantalum drawback → somewhat **expensive**

## Cementless Cup Shape Ask AI

- Older classification (dated 1983 in lecture) → five types:
  - **Cylindrical**
  - **Square**
  - **Conical**
  - **Ellipsoid**
  - **Hemispherical**
- Now → only **hemispherical** cup used

## Generations of Cementless Cups Ask AI

- Cup in general use → **third generation**

### First Generation Ask AI

- Before **World War II** (before ~1945–1950)
- **Stainless steel**
- Aim → **mechanical fixation** only
  - Biological fixation concept not yet emerged
- Fixation → **screws**

### Second Generation Ask AI

- Appeared → **early 1980s**
- Problems:
  - **Difficult insertion**
  - **Vertical cups** → edge loading on polyethylene → ↑ polyethylene wear
  - **Locking mechanism** problems

### Third Generation Ask AI

- Used from early **2000s**
- Improvements:
  - Shape → hemispherical / **hemi-ellipsoid**
  - Polyethylene → **gamma-irradiated not in air** → ↑ longevity
  - **Locking mechanism** problems addressed
  - **Periphery** trimmed (possibly to about one third of cup)
    - → no impingement of periphery with neck of the femur
  - Inner surface highly polished → ↓ **backside wear**
    - Second generation: unpolished inner surface → backside wear between cup (shell) and polyethylene
  - Better engagement between **liner** and cup
- Net result:
  - ↓ **Dissociation**
  - ↓ **Wear**

  - ↓ **Failure of the locking mechanism**

**Exam Pearl**

- Third generation → cup in general use
- First generation → stainless steel, screws, mechanical fixation only
- Second generation → **vertical cups** → edge loading → polyethylene wear
- Third generation → highly polished inner surface → ↓ backside wear
- Third generation → trimmed periphery → no impingement with femoral neck

## Hemispherical Cup Geometry Ask AI

- Hemispherical cup → the one all use now → three geometries

| Geometry | Description | Note |
|---|---|---|
| **Single radius** | One diameter, periphery → dome | Obsolete |
| **Dual radius** | Smaller internal diameter at dome; much larger at periphery | Obsolete; can fracture the acetabulum |
| **Hemi-ellipsoid** | Periphery 2 mm wider than dome | Third-generation shape |

- Hemi-ellipsoid → **maximizes bone contact** between cup and acetabulum

**Pitfall**

- **Dual radius** cup → can fracture the acetabulum

## Surface Coating Ask AI

- Covered in detail by **Dr. Mahmoud Fahim**
- Outer layer of cup → **highly porous** material → ↑ bone ingrowth
- Two options:
  - **Porous coated**
  - **Surface roughness** → ↑ bone ingrowth; not used now
- ± **Hydroxyapatite coating** → ↑ osteoconduction
- Porous coating method, most common and most used now → **plasma-sprayed titanium particles**
  - Explained in first lecture

## Liner and Modularity Ask AI

### Monoblock vs Modular Ask AI

Liner

- Cup comes as:
  - **Monoblock** → now obsolete (many problems)
  - **Metal shell** + liner → polyethylene or ceramic
- Monoblock = one single piece → **no backside wear**
  - Or ceramic cast inside cup → no micromotion between cup and liner

- Modular advantage → many **liner types:**
  - Captive
  - Offset
  - Lipped
- Later liner problem → **liner exchange** possible:
  - Infected
  - Loose
  - Polyethylene wear
- Screws through cup → relevant to **methods of fixation**

| | Monoblock | Modular |
|---|---|---|
| Structure | One single piece | Metal shell + polyethylene or ceramic liner |
| Backside wear | None | — |
| Liner types | — | Captive, offset, lipped |
| Liner exchange | — | Possible (infected, loose, polyethylene wear) |
| **Screws** through cup | Cannot be placed | Can be placed |
| Status | Obsolete | — |

### Liner Shapes Ask AI

- **Neutral**
- **Lipped** → lip of about 10 or 20 degrees → stability
- **Offset** → possibly combined with lip
- **Captive**
- Captive → **ring** constrains femoral head inside it

## Fixation Methods Ask AI

- Two methods → **press-fit** or line-to-line
- Press-fit → **under-ream** by about 1 mm → insert larger cup → cup held in place
- **Supplementary screw fixation** → if cup suspected not well press-fitted
- **Line-to-line** → screws can be used if:
  - No good initial bone contact
  - Cup not press-fitted after hammering
- Line-to-line or press-fit → **press-fit is better**
- Under-ream rim by 1–2 mm
- Under-reaming → stress on this, no more

**Clinical Pearl**

- **Press-fit is better** than line-to-line
- Press-fit → under-ream rim by 1–2 mm, then larger cup
- Not well press-fitted → supplementary screw fixation

## Acetabular Screws and Safe Zones Ask AI

- Screws placed after cup seated
- **Problems of screws** (three):
  - Source of wear particles
  - Fretting corrosion
  - Neurovascular injury
- Therefore → **four quadrants** around acetabulum:
  - Line from **anterior superior iliac spine** → ischium
  - Line **perpendicular** to it
  - → four zones; screws in **superior zone** *[Q1 · As spoken; safe zone is posterosuperior quadrant only; anterosuperior screws endanger external iliac vessels.]*
- Structures at risk:
  - **Sciatic** nerve → below
  - **Obturator** → in front
  - **External iliac bundle**

**Must-Know**

- Screw problems → wear particles, fretting corrosion, neurovascular injury
- Quadrants → ASIS–ischium line + perpendicular line → four zones
- Safe screw placement → **superior zone**
- At risk → sciatic below, obturator in front, external iliac bundle

## Survivorship: Cementless vs Cemented Ask AI

- Which survives longer → depends on surgeon's **school**
  - Some → cemented better; others → cementless
- Recent study, **cementless cup**:
  - 10-year survivorship → 96%
  - 20-year survivorship → about 90%
- **2013** study → about 27,000 hips
- Design → half cemented, half cementless; 10-year follow-up
- Result → **no difference in survivorship** between cemented and cementless
- Limitation → **older designs** of cemented and cementless cups
  - Further study needed → third-generation cups + HXLPE as used now
- Take-home → cementless cups possibly advantageous in many respects
  - **Technique used** → the key point
- Lecturer's conclusion → **no difference** between cemented and cementless

# How to choose your implant fixation

*(Choosing Between Cemented and Cementless Fixation in Cups and Stems)*

**Prof Dr Sherif Khaled**

**Watch Original Lecture**

*Lecture transcript processed with AI assistance.*

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# AI detected corrections (Mostly bad transcripts)

## Corrected - the text was changed; the transcript said

**C1** (p. 83) “as in the United States” → “as in the UK”

**C2** (p. 88) “literature from the United States” → “UK literature”

## Queried - kept as spoken, but looks wrong

**Q1** (p. 82) as spoken; fracture at 15 plus AVN 9 years later contradicts arthroplasty at age 15

## Opening Case: Fixation by Majority Vote Ask AI

- Per-case question: **cemented vs cementless** cup → audience vote
  - Same question for the stem → second vote
- Majority opinion followed → **cemented cup**; no personal decision
  - Stem probably cementless
  - Options listed: cemented cup, cementless cup, cementless stem
- Reasoning + three points raised by **Dr. Ahmed Hazem** → taken up later

## Young Adult with a Previously Nailed Femur Ask AI

### Intra-operative Problem Ask AI

Prior nail

- 38-year-old patient; **majority opinion** followed again
- Added **cerclage wire**
- No femoral fracture → wire placed **prophylactically** for fear of fracture
  - Reaming → **thin cortex** at this site
  - Canal previously instrumented: guide wire + **nail reamer** (prior nailing)
    - → previously fractured, **disrupted canal**
- Disrupted canal → no stem hold in **diaphysis** or metaphysis
  - Option: **cementless stem** → feasibility questioned
- Cementless feasible → **extend distally**
  - Component from the **revision** range
  - Distally extending **modular stems**
  - Audience asked for a solution
- Objection: distal (long) stem → ↑ **stress shielding**
  - Cementless advocates asked to solve this
- Answer left open

### Trade-offs of Each Option Ask AI

Prior nail

- **Cemented:** only drawback → likely loosening
  - This cup likely to loosen later; done 7 years ago; works in some cases
- Patient **lost to follow-up**
  - Cemented advantages:
    - **Easier**, simpler to perform
    - **Standard stem**, not the suggested long one
    - **Cheaper**
    - Antibiotic-loaded cement (probable)
- **Modular / cementless** advantage: age 38 → some prefer cementless
  - Argument not entirely correct: **England** → cemented fixation
    - Cemented in all cases from age 14 or younger
  - Cementless may still benefit this patient
- **Modular stem** option: Reef or Revitan
  - → acceptable **cementless implant** for the patient

### Audience Debate Ask AI

Prior nail

- Why not a **cementless cup**? → valid; lecturer cannot call it wrong
- Better choice → **cementless cup**
- Canal very small, narrow, thin → valid option, not considered at the time
- Case incomplete as shown → older, more **problematic** history
  - Pre-operative limb shortening (probable)
    - Cause: **sciatic nerve** possibly displaced downwards
- → **sciatic nerve** had to be explored (probable)
  - Harder case than shown; focus kept on **fixation technique**

### Registry Trend and Head Size Ask AI

Prior nail

- One opinion: **cement both components**
  - Current trend → more cemented, per **English National Joint Registry**
  - Question: why cement the cup?
- Probable **survivorship** at this level with hybrid, cementless, all-cementless
  - Question: this head size or a **larger head**?
- Surgeons judged this the **correct head size**
  - Unusual stem design; lecturer did not recognise it
  - Identified as **CPT**
- CPT confirmed — shoulder (probable) differs from CPT; probably not an **Exeter**
  - Cemented fixation point: large head probably had greater effect than cemented fixation itself

### Take-Home Message Ask AI

Prior nail

- Most opinions, incl. operating surgeon, if treated today → **cementless cup + cementless stem**
  - **Dr. Ahmed Galal:** cementless cup placed more medial + cementless stem
  - **Dr. Tarek:** short stem + cementless cup
- Others: **cementless** cup + cementless stem

## Post-Traumatic AVN in a 15-Year-Old Female Ask AI

- One opinion: would not operate → **refer** to a professor
- Age 15 → probable **acetabular fracture**; operated by lecturer, 2006 *[Q1 · as spoken; fracture at 15 plus AVN 9 years later contradicts arthroplasty at age 15]*
  - **Femoral neck fixation** also performed
- 9 years later → **AVN**
  - → **osteoarthritis** → needs total hip
- One vote for cemented; operated 9 years after AVN
- Interval ~8–9 years → i.e. 6–7 years after **fixation**; age cited as 15
  - **Dr. Ahmed Galal** → would do nothing for her

### Course Ask AI

AVN 15F

- Step 1 → **screw removal**; screw cultures → negative
  - Waited: screws suspected as pain source → pain persisted → **arthroplasty**
- Result not good: joint probably stiff; probable 3 cm **shortening**
  - Soft-tissue **releases** not needed → simple procedure
  - Majority followed → **cemented fixation**

### Cemented at 15? Ask AI

AVN 15F

- Cemented at 15 → acceptable per guidelines and **registries**
- Counter: cemented **easier to revise** → no excuse at age 15
  - Longer survival → **cementless**
  - Age 15 → aim for longer **survivorship**
  - Exception: surgeon trained in very good cemented technique, as in the **UK** *[C1 · transcript: "as in the United States"]*
- Only then → well-performed **cemented prosthesis** for her

## Post-Acetabular Fracture AVN with Superolateral Defect Ask AI

- 46-year-old male; **post-acetabular fracture AVN**
  - **Superolateral defect** on CT (CT not shown)
  - Vote: cemented vs cementless cup
- Reconstruction: **graft** vs ring
  - Defect too small for a ring → **femoral head graft**
- Graft fixation → **two screws**, not one

### Cup and Stem Choice Ask AI

46M defect

- **Cementless cup** in such a defect → yes
  - Previously and now → many such cases receive a cementless cup
  - Some cups (covered in this course) → work **without graft**
- These cups → much **easier**, reliable if trained; more expensive
  - Stem: **cementless**
- Reason → **Dorr type A**
  - Age 46 → either fixation acceptable; Dorr A bone → cementless

## Systemic Lupus with Osteoporosis Ask AI

- 36 years; **systemic lupus**
  - X-ray → **osteoporosis**
  - Vote on fixation type
- Audience votes → **cemented** cup
- Done: **cement for cup and stem**
  - Very osteoporotic → fear of losing a **cementless cup** (probable press-fit)
  - → **cemented cup** preferred in such cases

### Offset and Neck–Shaft Angle Ask AI

SLE

- Audience remark: **offset** very high
- Near-horizontal → not a **radiographic artefact**; placed deliberately
  - Always **imitates patient anatomy** (approx. 105, uncertain)
  - Egypt: anatomy often not reproducible → check available **implant sets** when planning
- Stem **neck–shaft angle** ≈ offset options
  - Each company → **standard** and offset versions
    - Offset version = **varised** or lateralized (name varies)
  - Each company → **spectrum** of options

  - → **pre-operative planning**: know what the case needs (next case)

# Inflammatory Arthritis in Young Females Ask AI

## Two Rheumatoid Cases Ask AI

RA

- **Juvenile rheumatoid arthritis**, 16-year-old female
  - Votes split: cemented / cementless
- Abstaining → not an acceptable **answer**
- **Wheelchair**-bound; bilateral involvement
  - JRA → accepted **indication** for cemented → cemented arthroplasty
- **Rheumatoid arthritis**, 30-year-old female → cementless
  - Reasons:
    1. **Age**
    2. **Good bone quality**
    3. **Dorr** ratio / type
- Same disease, opposite fixation:

| | JRA | RA |
|---|---|---|
| Patient | 16-year-old female | 30-year-old female |
| Status | Wheelchair, bilateral | — |
| Fixation | Cemented | Cementless |
| Reason | JRA = accepted cemented indication | Age, good bone quality, Dorr type |

## No Modular Stem or Dual Mobility in a Primary Ask AI

RA

- Not only very young or 46-year-olds → more cases follow
  - Primary case → no role for **modular** stem (Reef, Revitan)
- **Dual mobility** → only for problems needing dual mobility
  - Long stem better? → no
  - Most cases shown → **young females**
- Ages 15 and 16 → **females**, not males

## Why Not Arthrodesis Ask AI

RA

- **Rheumatoid** disease → arthrodesis not possible
  - **Arthrodesis** prerequisites:
    1. Normal spine
    2. Normal ipsilateral knee
    3. Normal contralateral hip
    4. **Non-progressive** disease
  - **Rheumatological** disease → no chance
- Counter: consider **arthrodesis** → if patient agrees and it fuses → good
  - Arthroplasty course → arthrodesis case to be shown → fixation choice asked
  - Aim → apply the **fixation lecture** to slide cases

- Arthrodesis case → **cemented total hip** planned
  - Arthrodesis avoided → would end up **converted** to total hip
- Everything must be considered
  - Contralateral **knee flexion deformity**
    - This side → **hip first,** then knee
    - Other side → **knee first,** then hip
  - Difficult case; outside the presentation's focus

**Exam Pearl**

- Arthrodesis needs normal spine, ipsilateral knee, contralateral hip
- Disease must be non-progressive → rheumatological disease excludes it
- JRA → accepted indication for cemented fixation
- Dual mobility → only for patients with problems needing it

## Coxa Valga: Planning and Implant Availability Ask AI

- Seven minutes left; many cases remaining
  - Vote: **cementless cup**? cementless stem?
- Many votes → **both possible**
- Routine **planning** → check available implants
  - Measure **distance** on the other (not fully normal) side
  - Assess: distance, cup, angle, **neck–shaft angle**
  - Other side → **coxa valga**
  - → cup and stem must match the high **shaft angle**
- **Varised stem** → not usable here
- Most stems in **Egypt** → 135° or 137.5°
  - Using one → wrong **offset,** altered greater–lesser trochanter distance
    - → patient **limping forever**
  - No planning → painful patient unable to walk → **unable to move**
- Implant unavailability → never lose the operation's **concept**
  - 145° stem needed but unavailable → tell patient → **wait** for suitable implant
- Final goal → **imitate real anatomy** with real implants
  - Here: cup + **cementless** stem
  - **Big head** → small patient
- Difficult case; earlier → would have **cemented**
  - Case from 2018 → experience increased since

**Pitfall**

- Most Egyptian stems: 135° or 137.5°
- Wrong neck–shaft angle → wrong offset → permanent limp
- Needed stem unavailable → wait, don't compromise
- Always plan with the implant sets actually available

### Cup in Protrusio Without Graft Ask AI

Coxa valga

- If worried → **graft** behind the cup
  - **Dr. Tarek Abdel-Shafy:** cup (probably resurfacing) in protrusio without graft
  - Large series, 2-year follow-up → cup–protrusio gap entirely **filled with bone**

### Offset and Limb Length Ask AI

Coxa valga

- **Offset** visible here
- **Offset** → must be imitated
  - **Limb length** easy here → other side diseased
    - Other side → needs surgery within 2–3 years
    - → limb length restoration not mandatory; offset is

## Indications: Cemented vs Cementless Ask AI

- Cemented indications drawn from the last few (**rheumatological**) cases
- Cemented **survivorship**: 97% at 10 years, 89% at 15 years

| Cemented | Cementless |
|---|---|
| Rheumatological diseases | Very young patients |
| Older patients | Good bone stock |
| Osteoporotic bone | Dorr type A |
| Dorr type C | All acetabular components (most evidence; Dr. Mahmoud Fahmy) |
| Low activity | — |

**High Yield**

- Cemented survivorship: 97% at 10 years, 89% at 15 years
- Dorr type A → cementless; Dorr type C → cemented
- Acetabular components → cementless per most evidence
- Osteoporosis, rheumatological disease, low activity → cemented

## Older Patients: Quick Cases Ask AI

- 62-year-old male, **osteoarthritis** → cemented; no disagreement
- **Dorr type C** (probably osteoporotic) → both components cemented
  - All his cemented stems → **loaded taper**
  - Prefers loaded taper over **composite beam**
- Has composite beam cases; still prefers loaded taper
  - 60-year-old female, **osteoarthritis** → very tempting for cementless
- Done → **cementless cup + cemented stem**
  - Next: 75-year-old male
- **Femoral neck fracture** at 75 → cemented bipolar

| Patient | Diagnosis | Fixation |
|---|---|---|
| 62-year-old male | Osteoarthritis | Cemented |
| Dorr type C | Probably osteoporotic | Cemented cup + stem, loaded taper |
| 60-year-old female | Osteoarthritis | Cementless cup + cemented stem |
| 75-year-old male | Femoral neck fracture | Cemented bipolar |

## Bipolar Hemiarthroplasty: Age Limit Ask AI

- **Bipolar** at 75 → safe
  - Younger patients → no bipolar
  - Age 60–70 → bipolar may not last beyond 10–15 years
    - → **acetabular erosion** after that
    - Some → eroded probably before 15 years (incl. their own studies)
- **Tripolar** cases → coming next
- Bipolar instead of total → financial?
  - At 75 or short life expectancy → depends on **physiological** status
  - Short life expectancy + **acetabular pain** → probably total
  - At 75 now → bipolar or total, probably to avoid dislocation
  - Implant must **hold** in this 75-year-old

**Clinical Pearl**

- Bipolar safe at 75
- Age 60–70 → bipolar may erode acetabulum within 10–15 years
- Avoid bipolar in younger patients

## Tripolar / Dual Mobility in the Elderly Ask AI

- Joint possibly done 16 years ago → must **hold**
  - Votes: all cemented / cementless / long stem / short stem
- Done: **long stem**, cemented, tripolar / dual mobility cup; age 75
  - **Revision** may be needed → correct limb length discrepancy
  - Corrects **limb length discrepancy** as much as possible → possibly over-restored
- Cause → raised **hip center**
  - Correct level → tip of **greater trochanter**
  - Slightly raised; proper **hip center** used
- 76 years old: cemented vs **cementless** → audience votes cementless
- **Tripolar** outcomes → very good results
  - Egypt: only **dual mobility** = cemented, single company (SERF); not routine
- Probable **tumour** case → cement

## Intra-operative Periprosthetic Fracture Ask AI

- **Intra-operative fracture** of the hip
  - Fixed → **cerclage wires**
  - Classified → **intraoperative type B**
  - Stem → cementless modular **Wagner**
  - Second option → **cemented**
- Third option → **distal fixation**
- Similar case (courtesy **Hatem Galal**) → same approach
  - **Periprosthetic fracture** → plate + cerclage wires

## Conclusions: Femoral and Acetabular Fixation Ask AI

### Femoral Component Ask AI

Conclusion

- Cemented **femoral component:**
  - **Elderly osteoporotic** patients
  - **Irradiated bone**
  - **Dorr type C** / stovepipe femur
  - **Deformed femur**

### Acetabular Component Ask AI

Conclusion

- **Acetabular components** → cementless
- Cementless cups favoured → components fail more under **shear force** (Fadi's lecture)
  - **Cement** → poor shear resistance
  - Indications: **younger patients**; older patients with good bone
  - Cementless acetabulum in all cases except:
- Poor bone → risk of damage on **impaction** → cemented component

**Must-Know**

- Components fail more under shear; cement resists shear poorly
- → cementless acetabulum in all cases except poor bone
- Cemented femur: elderly osteoporotic, irradiated, Dorr C / stovepipe, deformed

### Literature Bias Ask AI

Literature

- UK literature → always the **cemented camp** *[C2 · transcript: "literature from the United States"]*
  - UK surgeons → very well trained in **cemented** fixation; trend changing lately
- → cemented at any age → very long **survivorship**
  - **American literature:** nobody trained in cemented; younger generation never saw it
- → **biased literature** in both directions → read carefully

## Bearing Surfaces: Lecturer's Practice Ask AI

- Preferred: **ceramic on poly** or metal on poly
  - No metal on metal (probable)
  - No **ceramic on ceramic** → problem cases at ages 15, 12, 14 were all CoC

- Over ~20 years (uncertain) → **ceramic on poly**, comfortable
  - Older patients → **metal on poly** fine
  - **Registries:** metal on poly → ~30-year survival
  - Ceramic on poly → also 30-year, longer **survivorship**
- CoC → probably 16 years so far; none reached 30
  - Metal on poly → still **valid**
  - Ceramic on poly → no reason to fear
  - General preference → **hard-on-soft**
- **Wear** → theoretically lowest with ceramic-on-ceramic
  - Its **problems** → greater
  - Egypt → no good formal, specialised 8-year training
- CoC suits a **consultant** using the same setup for 40 years:
  - Same stem, cup, surgical setting, instruments, company
  - Same nurse, same assistance
  - Such a surgeon → CoC as he wishes; in **Egypt** → no

| Bearing | Lecturer's use | Registry survival |
|---|---|---|
| Metal on poly | Older patients | ~30 years |
| Ceramic on poly | Over ~20 years | ~30 years, longer |
| Ceramic on ceramic | Not used | Probably 16 years so far; none at 30 |
| Metal on metal | Not used (probable) | — |

# Perioperative Optimization

*(Pre-operative Risk Assessment, Infection Prevention and Patient Optimization)*

**Prof Dr Mohamed Saeed**

**Watch Original Lecture**

*Lecture transcript processed with AI assistance.*

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# AI detected corrections (Mostly bad transcripts)

## Queried - kept as spoken, but looks wrong

**Q1** (p. 98) as spoken; 30 mg vitamin D is implausible for mice, and the dose has no stated referent

# Rationale and Framework of Optimization Ask AI

- Aim → better performance → better arthroplasty outcomes
- Three guiding questions:
  - **Why** perform patient optimization
  - **How** to perform it
  - **What** to do to optimize the patient

## Why Optimize Ask AI

- Improves patient outcomes
- Reduces complications and readmissions
- Better outcomes → direct patient benefit

## How — Lasting Practice Change Ask AI

- **Protocols** and methods → improved process → good evidence
- Evidence → shows the process is beneficial
- Target: lasting changes in practice
- Improve three domains:
  - Ourselves
  - The process
  - The technology used
- Applied across the **perioperative periods:** pre- → intra- → post-operative

## Multidisciplinary Team Ask AI

- **Multidisciplinary team** delivers care:
  - Arthroplasty surgeons
  - Anesthesiologists
  - Infectious disease management
  - Nurses
  - Therapists
  - Nutritionists
  - Administrative co-workers
- Whole hospital cooperates → good operative outcome
- Goal: optimize patient outcome after arthroplasty

## Pre-operative Framework Ask AI

- Pre-operative steps:
  - **Checklist** of patient risk factors
  - Assign **level of infection risk**
  - Formulate **medical optimization plan**
- **Bundles** = optimization tool; pre-operative example named in lecture as "workforce" bundle

## Pre-operative Risk Factors Ask AI

- Pre-operative period → discussed in detail

### Non-modifiable (Patient-related) Ask AI

- Non-modifiable risk factors:
  - Inflammatory arthritis
  - Immunosuppression
  - **Diabetes mellitus**
  - Prior surgery on the joint
  - Previous joint infection
- Other comorbidities listed alongside:
  - Metabolic
  - Allergic
  - Cardiac
  - Renal disease
  - Neurological disease
- Effect → increased risk → worse joint replacement outcome
- Patients with these factors → do worse than others

### Modifiable Ask AI

- Definition: changeable → improved outcomes
- Modifiable risk factors:
  - **Poor dentition**
  - **Body mass index** > 40
  - **Malnutrition**
  - **Diabetes mellitus**
  - **Tobacco use**
  - **MRSA screening**
  - **History of open wounds or infections**
  - **Anemia**
  - **Obstructive sleep apnea**
- All → worse operative outcomes

**High Yield**

- Modifiable factors = targets of pre-operative optimization
- BMI threshold: > 40
- Diabetes mellitus listed among both factor groups

## Management of Modifiable Risk Factors Ask AI

| Factor | Consequence | Management |
|---|---|---|
| Poor dentition | **Septic focus** → ↑ infection risk | Dentist referral → dental clearance |
| BMI > 40 | DVT, readmissions, infections | Do not operate; weight loss or bariatric surgery |
| Uncontrolled DM (high **HbA1c**) | Wound complications, infections; longer stay; ↑ mortality | — |
| Smoking > 10 cigarettes/day (half a pack) | More significant post-op complications; more wound complications | — |
| Obstructive sleep apnea | Longer hospital stay; post-op complications | — |

- Diabetes control marker → **Hemoglobin A1C**
- Smoking → direct relationship with post-operative complications

**Clinical Pearl**

- BMI > 40 → do not operate
- Poor dentition → dentist before arthroplasty
- Obesity → weight loss or **bariatric surgery** first

## Malnutrition Ask AI

- Major risk factor for infection

### Mechanism Ask AI

Malnutr

- Impaired wound healing
- Impaired **collagen formation**
- Impaired inflammatory response
- ↓ **Lymphocyte** production
- ↓ Macrophage function
- ↓ Ability to fight infection

### Assessment Ask AI

Malnutr

- Assess before the procedure
- **Albumin** < 3.5 g/dL → malnourished
- Other markers → more expensive, harder to obtain
- Follow-up with **anthropometric measurements**
- Any method → **nutritional assessment** before surgery

### Evidence Ask AI

Malnutr

- Bohl et al., 2018
  - ~49,000 arthroplasty patients

  - **Hypoalbuminemia** → complications, readmissions, longer hospital stay
- Gu et al., 2017
  - 11,000 arthroplasty patients
  - 83% of infected patients → irrigation and **drainage** for infection
  - Only 5% improved — attributed to malnutrition

**Exam Pearl**
- Malnutrition marker: **albumin** < 3.5 g/dL
- Hypoalbuminemia → complications, readmissions, longer stay (Bohl 2018)

## Intra-operative Optimization Ask AI

- Sequence before theatre: workup → risk assessment → medical optimization → physician clearance → admission → operating room

### Skin Preparation Ask AI

- At home: **no shaving** — not the night before
- **Chlorhexidine** wash
- **Alcohol-based** skin preparation
- Antiseptic draping

### Theatre Attire Ask AI

- **Body exhaust suit** — new arthroplasty attire, used worldwide
  - **Ventilation system** outside the room
  - Complete **mask** → isolated from patient and environment → no contamination

### Instruments and Closure Ask AI

- **Glove exchange** before implantation
- **No flash-sterilized instruments**
  - Forgotten items not brought in by flash sterilization
  - All instruments regularly sterilized
- After wound closure → **dilute povidone-iodine** solution wash
- **Silver-impregnated dressing**

**Pitfall**
- Shaving the night before at home → avoid
- Flash-sterilized instruments → not used
- Gloves not changed before implantation → contamination risk

## Post-operative Optimization Ask AI

- **Dedicated joint unit** → ambulation and nutrition
- Early rehabilitation + ambulation → shorter stay

### Early Mobilization Evidence Ask AI

- Peterson et al., 2019
  - Mobilization from post-op day 2 → three hours
  - Then three more hours every day
  - → better outcomes, eating and activity; reduced length of stay
  - → greater post-operative value for the patient

### Post-operative Bundle Elements Ask AI

- **24 hours** of antibiotics
- **Chlorhexidine wash**
- **Aspirin** → lower-risk patients
- **Physiotherapy** available in the unit
- Wound follow-up via **phone photographs**

## Evidence for Optimization Bundles Ask AI

- Van Horn et al., 2019 → significant improvement in patient outcomes
- Bundle effective in practice → reduced infection risk

| Study | Design | Result |
|---|---|---|
| 2018 study | ~3,114 patients; 3 years before vs 3 years after bundle | **SSI** ↓ 62%; **PJI** ↓ 92% |
| Gottschalk et al., 2017 | Single center | Infection 12.9% → 1.9% (↓ 87%); length of stay ↓ 20% |
| Marker-only bundle | Before: 520 patients; after: 4,075 patients | Complications: 35.14 → 14%; 33 → 15%; 42 → 14%; 4.4 → 1.3 |

- Centers built own bundles from parts → shared with every site
- Marker-only bundle:
  - Markers: **hemoglobin A1C**, BMI, **hemoglobin**, **albumin**
  - Assessment only → no intervention, surgery not stopped
- All → significant reductions in complications and rates

**High Yield**

- Bundle → SSI ↓ 62%, PJI ↓ 92% (2018)
- Gottschalk 2017: infection 12.9% → 1.9%
- Assessment alone → significant complication reduction

## Vitamin D Deficiency and Infection Ask AI

### Emerging Modifiable Targets Ask AI

- Under study for pre-/post-operative protocols:
  - **Vitamin D** — not yet in protocols, but used
  - **Patient education**

### Design Ask AI

Vit D

- **Hegde et al., 2018 — mouse model**
- Vitamin D–sufficient vs vitamin D–deficient mice
  - 40 mice → deficient diet
  - 20 mice → sufficient diet
- Dose: 30 mg *[Q1 · as spoken; 30 mg vitamin D is implausible for mice, and the dose has no stated referent]*
- Deficient group → infection inoculated at stainless-steel implant → followed three weeks
- Sacrificed → studied **neutrophil infiltration** and **bacterial burden** in cultures

### Findings — Deficient Group Ask AI

Vit D

- ↑ **Bacterial bioluminescence** and neutrophil fluorescence
- ↑ **Colony-forming units**
- ↓ **Activated macrophage recruitment**
- Vitamin D replacement → significant effect on bacterial burden and neutrophil infiltration

## Patient Education Ask AI

- 2018 study: pre-operative education + phone follow-up for arthroplasty

### Pre-operative Phone Education Ask AI

Education

- Phone education starts **one week** pre-op
  - Calls: one week → four days → four days → two days before operation
- Content:
  - Description of the operation
  - Preparation for the operation
  - **NPO** and relevant instructions

### Post-operative Calls Ask AI

Education

- Start post-op day 1 → continue two weeks
- Topics:
  - Wound care
  - Medications
  - Pain control
  - Symptoms of concern
  - Mobilization goals
- End point: mobility level 4 in orthopedic clinic
- After two weeks → surgeon visit → outcome assessment

### Results Ask AI

Education

- ↑ **Patient satisfaction** in intervention group
- 87% → calls helped preparation for surgery
- 100% → kept more informed
- 97% → would participate again
- ↓ Unplanned calls and visits to orthopedic surgeon and clinic
- Patient satisfaction = end goal

## End Points of Optimization Ask AI

- Reassurance of the patient
- Improved patient satisfaction
- Improved patient outcomes
- **Reimbursement** from insurance companies
- After the program → ↓ infection rates, ↓ overall complications, ↓ costs

# Preoperative evaluation & Patient Counselling

*(History, Examination and Counselling Before Hip Arthroplasty)*

**Prof Dr Ihab Desouky**

**Watch Original Lecture**

*Lecture transcript processed with AI assistance.*

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## The Road from Clinic to Operating Room Ask AI

- Pathway: **outpatient clinic** → operating room
- Three questions answered along the whole road
  - Is surgery **indicated** now?
  - Is this patient ready for surgery now?
  - Expected outcomes → do we have the capacity to achieve them?
- Sequence of evaluation
  - **Patient interview** (history taking)
  - **Data analysis** → technical considerations
  - **Counselling** → consenting, immediately before surgery

**High Yield**
- Is surgery indicated now?
- Is the patient ready now?
- Expected outcomes, and capacity to achieve them?

## Patient Interview Ask AI

### Basic Data Ask AI

History

- Age, sex, **activity level**, occupational habits
- **Chief complaint**
  - Usually pain
  - Others → difficulty walking / limping

### Past History Ask AI

History

- **Past medical history**
  - Previous hospital admissions
  - Stability of chronic conditions
  - Comorbidities → hypertension, **diabetes mellitus**, anticoagulant use
  - All → affect subsequent management decisions
- **Past surgical history**
- Cardiac and pulmonary diseases
- **Anticoagulant therapy** → affects:
  - Type of anaesthesia
  - Placement of neuraxial block
  - Prophylaxis given beforehand → patient at risk of **DVT**
- Psychiatric (mental) conditions

### Social and Functional History Ask AI

History

- **Social history**
  - Lives alone vs with family
  - Expected support
- **Daily activities**

  - Working or not
  - Sports participation
- Patient **expectations**
- All taken before examination

## Patient Factors Affecting the Decision Ask AI

- **Age**
  - Younger than 50–60 years → need longer-lasting implant
  - Risk multiplied: young + **inflammatory** or psychiatric conditions
- **Sex**
  - Women → small **acetabulum** → smaller cup
  - No chance of larger cup / head
  - **Dislocation** rate about 10–12% higher in women than men
    - Cause → smaller acetabular component
- **Activity level** → must be known
  - May be limited → reduced bone stock, **osteoporosis** (previous lecture)
  - Affects mobility problems

## Pain Assessment and Contraindications Ask AI

### Characterising the Pain Ask AI

Pain

- **Pain** limiting patient's activity → must be understood
- Day pain vs **night pain** → important tools
- First, most important question: pain source
  - **Hip** itself
  - Referred → **lumbar spine**
  - Referred → sacroiliac joint
- Response to previous treatment
- Further questions
  - Pain severity → changes daily activities
  - Rest pain during the day
  - **Night pain** → very significant
  - **Analgesic** use and its duration
  - Hip analgesics, anti-inflammatory drugs, assistive devices
- Previous surgery around the hip; previous procedures

### Contraindications Ask AI

Pain

- Skin infections / lesions relevant to surgery
  - More than three foci
  - Large **open wounds / ulcers**
- **Neuropathic** condition
- Permanent **paralysis**
- All → **contraindications** to hip surgery

# Physical Examination Ask AI

## Inspection and Gait Ask AI

Exam

- Order: **inspection** → gait examination
- **Limping,** if present
- **Attitude** of limb → standing and sitting
- **Previous incisions**
- **Sinuses**
- Skin infection in same limb
- **Leg ulcer**
  - All → dealt with before surgery

## Palpation, Motion and Power Ask AI

Exam

- **Palpation** → bony prominences
- **Range of motion**
  - Normal range of motion (figure)
- **Abductor** muscles → must have good power
  - Muscle power assessment
- **Neurovascular status**
- Finally → clear limb length analysis

## Limb Length Assessment Ask AI

LLD

- **Apparent shortening**
  - From **pelvic obliquity** and scoliosis
  - No true shortening in the limb
  - Cause may be the **spine**
- **True limb lengths** → detect true **limb length discrepancy**
- **Fixed flexion deformity**
  - Measuring true length → sound limb in same degree of flexion
  - Allows correct true-length measurement
- **Block test** → apparent length
  - Wooden blocks; patient stands on blocks
  - Pelvis levelled on blocks
  - Large difference → may be true discrepancy from hip pathology

**Clinical Pearl**

- Apparent shortening → pelvic obliquity, scoliosis; cause may be spine
- True lengths → confirm true discrepancy
- Fixed flexion deformity → sound limb placed in same flexion

## Special Tests Ask AI

Exam

- Hip special tests → **Thomas test,** Trendelenburg (abductor) test
- **Trendelenburg test** → tests the abductors
  - Watch the side of the **anterior superior iliac spine (ASIS)**
  - Patient stands slowly on one leg → pelvis observed

  - Pelvis does not rise → **abductor insufficiency**
  - Insufficiency → addressed during surgery

## Scoring Systems Ask AI

- **Scoring system** → patient's general function
- Two types

| | Patient-reported outcome | Observer / physician-reported outcome |
|---|---|---|
| Filled by | Patient, on paper | Doctor |
| Use | Follow-up of improvement | Physician-assessed items |
| Example | Arabic, English, German, other translations | **Harris Hip Score** |
| Key point | Must be in patient's own language | Deformity, range of motion → filled by physician |

- **Patient-reported outcome**
  - Patient fills paper per his answers → followed afterward
    - Shows his improvement
    - Study (2019) → score must be translated into patient's language
    - Translations: Arabic, English, German, several others
    - Must be in patient's **own language**
- **Observer / physician-reported outcome**
  - Filled by the doctor
  - **Harris Hip Score** → includes deformity, range of motion
  - Deformity, range of motion → filled by physician

## Radiographic Evaluation Ask AI

- **Radiographic evaluation** → nature of hip pathology
- **Bone quality** → comparison with contralateral side
- **Radiographic templating**
- **Bone defects** assessment → two parts
  - **Loss of substance**
  - Defect in **bone alignment**
- Beyond bone loss → **spinopelvic motion**

## Spinopelvic Mobility Ask AI

### Normal Mechanics Ask AI

Spinopelvic

- Ask about recent **spine surgery** and recent spine problems
- Standing → **lumbar lordosis** + pelvis tilts anteriorly
- Sitting → **posterior pelvic tilt** → allows flexion

| | Standing | Sitting |
|---|---|---|
| Lumbar spine | Lordosis maintained | **Flattening** of lordosis |
| Pelvis | **Anterior pelvic tilt** | Posterior pelvic tilt |
| Acetabulum | — | Opens anteriorly |

- Anterior acetabular opening on sitting
  - Prevents **anterior impingement**
  - → prevents **posterior dislocation**
- After **total hip arthroplasty** → same sequence, standing and sitting (figure)
- Lateral **X-ray** → standing and sitting positions

### Disturbed Mobility Ask AI

Spinopelvic

- Lumbar spine problem, e.g. **fusion**
  - Pelvic tilt locked anteriorly during sitting
  - → **anterior impingement** → **posterior dislocation**
  - Occurs even with correct cup orientation
  - Cause → disturbed **spinopelvic mobility**
- Opposite direction
  - **Lost lumbar lordosis** → no anterior tilt on standing
  - Anterior part of head uncovered → **anterior dislocation**
- More stability needed → **dual-mobility** arthroplasty
  - Better than conventional

| | Lumbar fusion | Lost lumbar lordosis |
|---|---|---|
| Position at fault | Sitting | Standing |
| Pelvic tilt | Locked anteriorly | No anterior tilt |
| Mechanism | Anterior impingement | Anterior head uncovered |
| Dislocation | **Posterior** | **Anterior** |

**Pitfall**

- Correct cup orientation → still dislocates if spinopelvic mobility disturbed
- Fusion → posterior dislocation; lost lordosis → anterior dislocation
- Solution → dual-mobility arthroplasty

## Medical Fitness and Implant Planning Ask AI

### Medical Fitness Ask AI

Fitness

- Preoperative medical reports, consultations → covered by **Dr. Mohamed Saeed**
- **ASA classification**

| ASA class | Decision |
|---|---|
| I, II, III | Operable after **optimisation** |
| 4 or 5 | Not fit → unsuitable for **elective surgery** |

### Data Analysis and Technical Considerations Ask AI

Planning

- All data → **radiographic** + **clinical** → assessed
- Determines **approach** and **implant**, according to:
  - Age
  - **Bone quality**
  - Previous surgery
  - **Availability of implants**

## Patient Counselling Ask AI

### Definition and Goals Ask AI

Counsel

- Second part of the lecture
- **Definition:** providing information about performing surgery
  - **Orally** or in **written form**
  - To patient or his **representatives**
- Clear and understandable
- Patient knows what the procedure involves → clarity essential
- **Complications** that may occur
  - Measures to prevent them, as far as possible
- Simple, plain words
- Goal → patient realises role of surgery in improving his life

### Preoperative Patient Education Ask AI

Counsel

- **Patient education**, preoperative stage
  - **Disease** → his condition, problems of the natural course
  - **Benefits** of surgery → improving his condition
  - **Risks** → related to operation itself and to prosthesis
    1. Dislocation
    2. Fracture
    3. Limb length discrepancy
    4. Nerve injury
    5. Infection
- Patient signs **consent** before surgery
- **Anaesthesia**
  - Regional (spinal)
  - **General anaesthesia**
- **Postoperative pain control**
- **Postoperative rehabilitation**
  - Ongoing **physiotherapy** programme → restores function

  - Condition → patient committed to it
- **Patient expectations** → discussed before talking about the operation

## Home Preparation and Preoperative Medications Ask AI

### Family Support and Home Modification Ask AI

Home

- Functional challenges can be overcome → patient needs support
- **Family support** group around the operation → family caring for patient
- **Home modifications** → easy walking without falls
  - Remove obstacles (loose rugs)
  - Install **handrails** → no falls
  - **Walking aid**
- **Assistive devices** → activities without falls; many other tools

### Preoperative Medications Ask AI

Drugs

| Drug | Action before surgery |
|---|---|
| **Antiplatelet drugs** | Stop 7–10 days before |
| **NSAIDs** | Stop 7–10 days before |
| **Methotrexate** | Continue as is |
| **Hydroxychloroquine** | Continue as is |
| Other drugs | Discuss with **rheumatology** specialist |

## Patient Instructions Before Surgery Ask AI

- Continue regimen before surgery
- Food rich in **iron** and **calcium**
- Stop **smoking** → 6 months before and 6 months after surgery
- Day before surgery
  - Remove **nail polish**
  - Shower → antibacterial soap over the body
  - Sleep on clean sheets
  - Good night's sleep
  - Nothing to eat or drink after designated time
  - No lotions
  - No **shaving** before surgery

**Must-Know**

- Antiplatelets, NSAIDs → stop 7–10 days before
- Methotrexate, hydroxychloroquine → continue
- Smoking → stop 6 months before and 6 months after
- No shaving, no lotions before surgery

## Take-Home Message Ask AI

- **Preoperative evaluation** for arthroplasty affects:
  - Decision → operate or not
  - **Timing**, implants, incision / approach
  - Risk of future complications
- Answer the **three questions** in preoperative assessment
- Honest, clear information to patients
- Accept these responsibilities → best outcomes and preparation

# Rehabilitation in THA

*(Hip Precautions, Early Mobilization and Functional Recovery)*

**Prof Dr Molham Mahmoud**

**Watch Original Lecture**

*Lecture transcript processed with AI assistance.*

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# AI detected corrections (Mostly bad transcripts)

## Corrected - the text was changed; the transcript said

**C1** (p. 114) “cementless: immediate weight bearing” → “cemented: immediate weight bearing”

**C2** (p. 117) “wait 4 to 6 months before driving” → “wait 4–6 weeks before driving”

# Rehabilitation in THA: Overview Ask AI

- **Rehabilitation** → key, often forgotten part of **total hip arthroplasty (THA)**
  - Common attitude: operate → send to physiotherapy → leave the rest
  - Understandable, but must change
- Definition → **restoring function** after surgery
- Performed by a **team**
  - Orthopaedic surgeon = **head of the team**
  - Surgeon responsible for patient until the very end
- Starts from first patient contact → pre-operative + post-operative parts
- **Hip arthroplasty** → among the best procedures for pain relief and function
- Aim → pain relief + function within expected time course
  - Ideally via a **rehabilitation protocol**
- Components
  - **Preoperative education** and exercises
  - Post-operative **functional activity**
  - In-patient **post-operative exercises**
  - In-patient rehab → **home discharge** → **outpatient rehab**
  - **Avocational activities**

**High Yield**

- Rehab = team; surgeon = **team head**
- Starts at first contact, not after surgery

# Preoperative Education and Exercises Ask AI

## Education Ask AI

- Benefits
  - **Prevents complications** → patient understands early movement and early rehab
  - **Reduces anxiety**
    - Major operation → prepare patient → less anxiety before and after surgery

## Exercises Ask AI

- Hip muscles usually impaired → **preoperative conditioning** required
- Significant **pre-op vs post-op function correlation**
  - Poorer pre-op function → lower post-op score than good pre-op function
  - Poor pre-op function → poorer post-op function

**Exam Pearl**

- Pre-op function predicts post-op function

## Functional Activity: Weight Bearing Ask AI

- Two main goals
  - **Weight bearing**
  - Adherence to **hip precautions** (dislocation precautions)
- Key determinant → **cemented vs cementless** implant
  - **Cemented** → immediate weight bearing possible *[C1 · transcript: "cementless: immediate weight bearing"]*
  - **Cementless** → in the past, no immediate full weight bearing → gradual progression

## Hip Precautions Ask AI

- Many factors → covered in a dedicated lecture
- Practical message → leg positions to avoid → prevents dislocation
- Positions → **depend on surgical approach**

### Posterior / Posterolateral Approach Ask AI

Posterior

- Includes posterior variations
- Avoid
  - **Internal rotation**
  - **Adduction**
  - **Flexion** beyond 70–90°
- Combination = post-op dislocating position
- Easy triggers → low seat, bending to pick something up
  - No excessive hip flexion in any of these
- Maintain **hip angle**
- Low seat / picking up → tendency to over-flex hip
  - Solution → move the whole body, or use a **reacher** (same idea as Dr. Ehab)
- Avoid **crossing the legs** → adduction + internal rotation

### Anterior / Anterolateral Approach Ask AI

Anterior

- Opposite of posterior → dislocating movements
  - **External rotation**
  - **Hyperextension**
  - **Adduction**

| Approach | Movements to avoid |
|---|---|
| Posterior / posterolateral | Internal rotation, adduction, flexion > 70–90° |
| Anterior / anterolateral | External rotation, hyperextension, adduction |

### Commonest Dislocating Positions Ask AI

- **Crossing the legs** (one leg over the other)
- Lying in **external rotation**

**Must-Know**

- Precautions depend on **surgical approach**
- Posterior → avoid internal rotation, adduction, flexion > 70–90°
- Anterior → avoid external rotation, hyperextension, adduction
- Commonest → **crossing legs**, lying in external rotation

## Functional Restriction and Assistive Devices Ask AI

### Functional Restriction Devices Ask AI

- **Abduction pillow**
- **Knee immobilizer**
- **Abduction brace**
  - Simple items (**pillow**) → sufficient
  - Cumbersome **abduction brace** → poor compliance
  - **Randomised controlled trial** → abduction brace: no proven benefit, only complicates matters

**Pitfall**

- **Abduction brace** → no benefit on RCT + poor compliance; pillow suffices

### Assistive Devices Ask AI

- **Walker, cane, crutches**
- Benefits
  - **Stability**
  - Less load on the operated hip
  - Prevent **falls**
  - Wider base of support
  - **Psychological and physical confidence** → moves without falling
- Choice depends on
  - **Stage of rehabilitation**
  - Patient **ability** to use device
  - Patient **age**

## Post-operative Exercises Ask AI

- Start → **immediately**
  - Operated in the morning → **mobilisation** + exercises in the **afternoon**
- Exercise **chart** → lists all exercises
  - Given to **physiotherapist** or patient
  - Simple; covers lower-limb muscles **from ankle to hip**
  - Each exercise easy to perform
- Dose → 3–4 times/day, 10 repetitions each
- Progression → **distal to proximal**
  - **Ankle pumps** → dorsiflexion / plantar flexion + ankle rotation (internal / external)

  - **Knee bending** + **quadriceps isometrics** → **hip abduction** — all supine
  - Patient gains control →
  - **Weight bearing** → standing abduction, extension, flexion
    1. Flexion → leg not above the **waist**
    2. Initial maximum ≈ 70°, even if patient can flex further

**High Yield**

- Exercises start **same day** of surgery
- 3–4 times/day × 10 repetitions
- Progression **distal** → **proximal**, supine → standing

## Home Discharge Ask AI

- **Discharge criteria**
  - Walks 15–30 m with an appropriate **assistive device** → acceptable; without one → excellent
  - **Transfers** independently
  - Manages some **activities of daily living** at home
  - **Understands precautions**
  - **Medically stable** → above all
- After discharge → **home therapy** continues
  - **Post-operative rehabilitation** not to be forgotten after leaving
  - **Home rehabilitation** → not widespread in Egypt; available abroad
- Exercise stage complete → next stage: avocational activities

**Clinical Pearl**

- Discharge → walks 15–30 m, transfers, ADLs, knows precautions, **medically stable**

## Avocational Activities Ask AI

- Definition → hobbies, sports, activities beyond ordinary daily tasks
- **No randomised controlled trial**
- Evidence → **survey** of **American Association of Hip and Knee Surgeons (AAHKS)**
  - Each surgeon gave his preference → compiled recommendations
  - ≈ 550 surgeons responded
  - Wait at least 3–6 months before **sports activities**
- Sports chart → three categories

| Category | Examples | Rule |
|---|---|---|
| **High-impact** | Jogging, squash | Not allowed; no reason to do them |
| **Low-impact** | Golf, swimming, walking | Allowed; no problem with load |
| **Intermediate** | — | Surgeon-dependent; prior experience; time since surgery |

**High Yield**

- AAHKS survey (≈ 550 surgeons), not RCT
- Sports → wait ≥ 3–6 months
- **High-impact** (jogging, squash) → not allowed

## Driving Ask AI

- Key avocational activity
- Return to driving → **independence** + **psychological relief**
- Questions to answer
  - Can he operate the **pedals**?
  - Car → **automatic or manual**?
  - **Reaction time** → most important
    - Prolonged **braking time** / slow leg use
- Recommendation → wait 4–6 weeks before driving *[C2 · transcript: "wait 4 to 6 months before driving"]*
- **Left THA** → faster return
  - Left leg not needed for driving
  - Right leg → operates the pedals

**Exam Pearl**

- **Reaction time** → most important driving question
- Left THA → faster return to driving

## Sexual Activity Ask AI

### Importance Ask AI

- Very important for the patient who needs it
- 75% of arthroplasty candidates → difficulties with sexual activity
- Same 75% → significant improvement in **sexual function**
  - After the arthroplasty
- Very **sensitive** topic
  - Only 10% ask their doctor (Western countries)
  - 90% want advice without asking
  - Don't wait for patient to ask
  - Don't assume irrelevant because of age or other factors

### Recommendations Ask AI

- Resume sexual activity 1–3 months after arthroplasty
- Patient → **passive** partner ("P-passive"), not the active one
- Avoid positions with excessive hip flexion → beyond
  - 70°
- Positions by sex

| Patient | Position |
|---|---|
| Female | Supine; passive |
| Male | Leg in extension, or standing; passive |

## Clinical Lesson: Indirect Questions Ask AI

- Lecturer's early THA case
- Follow-up for **suture removal** at week 2–3
  - Man from Upper Egypt, ≈ 75 years, building contractor
  - Asked: "Can I take a bath?" → wound perfect → told yes
  - Next morning → hip **dislocated**
  - "Bathing" = euphemism for intercourse
- Lesson → patient asks indirectly, in simple terms
  - Patient won't ask directly → **you must explain it**

## Sexual Counselling Ask AI

- Pictures not necessarily needed
- **Sexual counselling** → very important
- Start with **open discussion** → patient + partner

**Clinical Pearl**

- Only 10% ask; 90% want advice without asking
- Raise the topic proactively → **sexual counselling**
- Patient = **passive** partner; avoid flexion > 70°
- Resume 1–3 months post-op

## Summary Ask AI

- Rehab starts pre-operatively
- Continues after **discharge home** with follow-up
- **Avocational and sexual activities** → very important

# Guidelines for VTE prophylaxis

*(Risk Assessment, Agents and Duration of Thromboprophylaxis)*

**Prof Dr Mahmoud Abdel Karim**

**Watch Original Lecture**

*Lecture transcript processed with AI assistance.*

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# AI detected corrections (Mostly bad transcripts)

## Queried - kept as spoken, but looks wrong

**Q1** (p. 125) as spoken; half dying overstates hip fracture mortality (about 10% at 30 days, 20-30% at 1 year)

## Opening Case: Clopidogrel Before Surgery Ask AI

- Patient on **Plavix (clopidogrel)** alone → dose taken this morning
- Injury → needs **replacement,** not fixation
- Questions posed:
  - VTE prophylaxis plan?
  - Stop the Plavix?
  - What is known about Plavix?
  - Core question → **when** to operate
- Answer revisited at lecture end

## Scope and Sources Ask AI

- ~Half of orthopaedic surgeons → same answer to opening case
- Scope:
  - **Elective THA and TKA**
  - **Trauma** — hip fracture
  - **Controversies** — e.g. opening case
- Basis → **NICE guidelines**
  - **NICE** = National Institute for Health and Care Excellence
- Training place → shapes tools carried into practice
  - Know what you know, and from where
- Other sources:
  - **ACCP** — American College of Chest Physicians
  - **Cochrane group** — very important source
- NICE versions:
  - Earlier → 2010
  - Current → March 2018

## Why VTE Prophylaxis Matters Ask AI

- Theoretical topic, but very important → not to be neglected
- **VTE events** in orthopaedic surgery — DVT, PE:
  - No prophylaxis → `40–60%`
  - With prophylaxis → `~1–2%`
  - Europe / developed countries → reported `~0.1–0.2%`
- Some cases still lost
- VTE → **preventable cause of death**
- First step → **risk assessment:**
  - **VTE risk**
  - **Bleeding risk**
- Both often needed in same patient
- High **bleeding risk** → may limit prophylaxis

**High Yield**

- No prophylaxis → VTE 40–60%
- With prophylaxis → ~1–2%
- VTE → preventable cause of death
- Risk assessment → VTE risk + bleeding risk

## VTE Risk Factors Ask AI

- Full table in guideline resources → no need to memorise
- Five to remember:
  - **Previous clot** — patient or relative, especially first-degree
    1. Yes → "unfortunately, sir…"
    2. → high risk of major clot, above average
    3. → **most important factor**
  - **Active cancer** or cancer treatment
  - **Age** > 60 (sometimes quoted > 40)
  - **Obesity** — subject of panel debate
  - Known **thrombophilias**

**Must-Know**

- Previous clot (self or first-degree relative) → most important
- Active cancer or cancer treatment
- Age > 60
- Obesity
- Known thrombophilias

## Bleeding Risk Factors Ask AI

- Second assessment → **risk of bleeding**
- Long list → prefer a **thinking algorithm**
  - Asked in American College training programme since 2006
- Factors:
  - **Active bleeding**
  - **Hepatic disease** → be careful
  - Acquired **bleeding disorders**
  - Inherited bleeding disorders:
    1. **Haemophilia**
    2. **Von Willebrand disease**
    3. Outside orthopaedic expertise

## General Measures: Hydration and Mobilisation Ask AI

- Elderly relatives get clots → partly inadequate **fluid intake**
- Intake → 2–3 L/day

  - Unless restricted by **cardiological** status
  - Never allow **dehydration**
- Postoperative **mobilisation** — even in bed:
  - Move upper and lower limbs → ankle pump exercises
  - Warn patient in advance

## Elective Total Hip Arthroplasty Ask AI

### Risk Balance Ask AI

THA

- Two risks weighed → **thrombosis risk** vs bleeding risk
  - THA, TKA, all procedures here → thrombosis risk higher
  - → VTE risk usually exceeds **bleeding risk**

### Pharmacological Prophylaxis Ask AI

THA

- Most cases → **LMWH**
  - Generic names used
- Guidelines:
  - 2010 → duration 28 days
  - Recent years → **aspirin** option, incl. ACCP
- **Aspirin** dose:
  - Initially → 150 mg
  - Now → 75–150 mg
  - Duration → ~1 month (28 days)
- Routine → **LMWH** from start × 28 days
- Speaker's preference → **LMWH**
  - **Aspirin** → reserved for specific cases
  - Selection → after **risk stratification**
- Worldwide → aspirin **approved** for THA and TKA
- Alternatives:
  - **Rivaroxaban**
  - **Apixaban**
  - **Dabigatran etexilate**

### Mechanical Prophylaxis Ask AI

THA

- Combine pharmacological + **mechanical** prophylaxis
  - e.g. **anti-embolism stockings**
- Contraindication to drugs → **mechanical alone** from start
  - Typical in trauma → active bleed elsewhere → decision needed
- Same approach if VTE risk high

## High VTE Risk with High Bleeding Risk Ask AI

- Both risks high → **multidisciplinary group**
  - Hepatologist
  - Neurologist

  - Haematologist
  - Vascular surgeon
- **Combined meeting** → best decision
- **Involve family:**
  - e.g. LMWH given → heavy bleed → must stop
  - → clots may occur
- No avoidance → disclose openly

## Elective Total Knee Arthroplasty Ask AI

- Same approach as THA; shorter duration:
  - **Aspirin** 75 or 150 mg × 14 days — NICE wording
  - **LMWH** × 14 days
  - Plus **anti-embolism stockings** / other mechanical methods
- Same alternatives → rivaroxaban, apixaban, dabigatran etexilate
  - Generic names only
- Contraindication → start **intermittent pneumatic compression (IPC)**
  - Unavailable in most local hospitals
  - Continue → **until patient mobile**

| | THA | TKA |
|---|---|---|
| LMWH | 28 days | 14 days |
| Aspirin | 75–150 mg × 28 days (~1 month) | 75 or 150 mg × 14 days |
| Alternatives | Rivaroxaban, apixaban, dabigatran etexilate | Same |
| Mechanical | Anti-embolism stockings | Stockings / other mechanical |
| Drugs contraindicated | Mechanical alone | IPC until mobile |

**Exam Pearl**

- THA → 28 days prophylaxis
- TKA → 14 days prophylaxis
- Aspirin → 75–150 mg, now an approved option
- Always combine with mechanical prophylaxis

## Hip Fractures Ask AI

### VTE Risk and Mortality Ask AI

Fracture

- **Fractures** → very high stakes; ~half of patients die (as quoted) *[Q1 · as spoken; half dying overstates hip fracture mortality (about 10% at 30 days, 20-30% at 1 year)]*
- Saying → "come from the hip, go from the hip"
  - Anecdote → speaker's patient died before surgery
  - Advice → prophylaxis × one month → persistent **VTE risk**

- Bleeding risk also present → calculate, discuss with patient

### Postoperative Agents Ask AI

Fracture

- **LMWH** → 6–12 h post-op; ~12 h
  - Patient left 24 h → clot → common; share responsibility
- **Fondaparinux** → 6 h post-op
  - Condition → low **bleeding risk**

### Timing and Preoperative Prophylaxis Ask AI

Fracture

- Trauma patient → may take time to prepare
- Elective → no **preoperative** prophylaxis needed
- Best practice → surgery within 36–48 h; now **36 h**
- Surgery delayed → start prophylaxis
  - Stop **Clexane** 12 h pre-op
  - Stop **fondaparinux** 24 h pre-op
- Surgery delayed → **mechanical** prophylaxis until surgery (ACCP)

| Agent | Post-op start | Stop pre-op |
|---|---|---|
| LMWH (Clexane) | 6–12 h (~12 h) | 12 h |
| Fondaparinux | 6 h, if low bleeding risk | 24 h |

### IVC Filter Ask AI

Fracture

- Often written → "don't forget **IVC filter**"
- Since 2012 → apparently limited to specific indications, not primary prophylaxis

### Evidence and Caution Ask AI

Fracture

- DVT evidence → no certainty for or against
  - Study; invest in training
  - Follow guidelines **dogmatically**
  - Track new evidence
- Some agents → efficacy possibly offset by **bleeding complications**

**High Yield**

- Hip fracture → surgery within 36 h
- LMWH → 6–12 h post-op; stop 12 h pre-op
- Fondaparinux → 6 h post-op if low bleeding risk; stop 24 h pre-op
- Delay → mechanical prophylaxis until surgery
- Duration → one month

## Perioperative Antiplatelets Ask AI

### Aspirin Ask AI

Aspirin

- Patient on aspirin → stop before surgery? **Never**
  - Primary prevention (no specific disease) → continue
  - Cardiovascular indication → continue

- Involve **cardiologist** from start → "no need to stop aspirin"
- Don't stop aspirin or Plavix on **own judgement**
  - Advice against guidelines → **their responsibility,** not yours
- **Spinal / epidural** → allowed on aspirin
- Start **routine preoperative regimen**
- **Platelets** prepared → in case of bleeding
- Final decision → surgeon's own

## Clopidogrel (Plavix) Ask AI

Plavix

- Question → **continue or discontinue** Plavix on admission
  - Revision surgery
  - Stent + trauma
- Stopping unilaterally → serious complications
  - **Bare-metal stent**
  - **Drug-eluting stent**
- **Neuraxial block** on clopidogrel
  - → **contraindicated**
- Plavix stopped or not → operate **early** (next day), not after a week
  - Safe technique + confident **haemostasis** → same or next day
- **General anaesthesia** → proceed
  - Anaesthetist view → GA fine, no problem
  - Requires quick, confident technique
- Stent > 6 months old (corrected from 6 weeks):
  - Stop **Plavix**
  - **Continue aspirin**
- Inform patient of risk; **platelets** prepared

| | Aspirin | Clopidogrel (Plavix) |
|---|---|---|
| Stop before surgery | Never — primary prevention or CV | Not unilaterally; stent > 6 months → stop, continue aspirin |
| Spinal / epidural | Allowed | Contraindicated |
| General anaesthesia | — | Fine |
| Timing of surgery | — | Early — same / next day |
| Prepare | Platelets | Platelets |

**Pitfall**

- Stopping aspirin or Plavix on own judgement
- Stopping Plavix with a stent → serious complications
- Neuraxial block on clopidogrel → contraindicated
- Waiting a week on Plavix → operate early instead

## Orthogeriatric App Ask AI

- **Orthogeriatric App** — by AO Trauma
  - Android and iPhone → recommended download
- Covers:
  - **Osteoporosis**
  - **Delirium**
  - **Perioperative pain management**
  - **Anticoagulation**
- Anticoagulation section:
  - **Perioperative thromboprophylaxis**
  - **Management of pre-existing** anticoagulation
  - Next page → yes / no questions
  - Agent list → select → when to stop, when to give, what to add
  - Plus risk assessment

## Opening Case Resolved Ask AI

- Professor of anaesthesia's mother:
  - Fracture at 9 a.m.; **Plavix** at 9 a.m.
  - Operated at 3 p.m. → did well
  - **General anaesthesia**
- Incorporate into practice
- Prophylaxis → one **month**

# Lateral Approach

*(Direct Lateral (Hardinge) Approach: Technique and Surgical Anatomy)*

**Prof Dr Hazem Abdel Azeem**

**Watch Original Lecture**

*Lecture transcript processed with AI assistance.*

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# Introduction and Historical Perspective Ask AI

## Hip Surgery Then and Now Ask AI

- Past hip practice → very heavy patients, many difficulties
  - Classic scenario → **fracture neck of femur**, patient left bedridden
- Speaker's early career → 1974 or 1975 start
  - Probable standard then → **hemiarthroplasty** for fracture neck of femur
- Arthroplasty since → developed; teaching expanded
  - Solutions now for hip problems once thought unsolvable
- Childhood hip conditions threatening later life
  - Previously → diagnosis possible, **treatment options** limited
- Speaker's pride in progress of hip surgery at this university

## Egyptian Hip Society Ask AI

- Announcement → April, 8th or 9th
  - Annual meeting of the **Egyptian Hip Society**
  - Society founded ~8 years ago → forum for communication and development
  - Next conference → Cairo
    - Attendees invited to participate, contribute ideas and presentations

# Lateral Approaches to the Hip Ask AI

- **Lateral approaches** → a group of quite different approaches
- Common feature → **centered on the greater trochanter**
  - Usually directed anteriorly
- Members of the group
  - **McFarland approach** (McFarland and Osborne)
  - **Modified McFarland approach**
  - **Hardinge approach**

# Hardinge Approach: Position and Incision Ask AI

## Position Ask AI

Hardinge

- **Hardinge approach** → most commonly used lateral approach → covered first
- Session focus → surgical technique, not theory
  - Indications and contraindications → not discussed
- Position options
  - **Lateral position**
  - Supine position
    - Supine → sandbag under the buttock; limb slightly tilted
- Speaker's preference → **lateral position**
  - Easier for trainees to get used to

### Incision Ask AI

Hardinge

- Length → usually >10 cm
- Runs over the **greater trochanter**
  - ~Half of incision over trochanter (centered on it)
  - Other half likely extending proximally
- Mark **anterior** and **posterior borders** of the greater trochanter
  - Trochanter → middle of the field
- Incise skin → subcutaneous tissue
- Visible in wound → **tensor fasciae latae**

## Hardinge Approach: Superficial Dissection Ask AI

### Fascial Layer Ask AI

Hardinge

- **Gluteus maximus** fibres → from posterior side, inserting posteriorly
- Interval between TFL and gluteus maximus
  - Contains **fibro-fatty tissue** or appears as a yellowish line
  - Follow this line → deeper
- Self-retaining retractor → then incise the **fascia**
- Fascial incision → extended distally ~10 cm below tip of greater trochanter
- Orientation → imagine the destination
  - Area exactly opposite → **base of the neck**
  - **Head of femur** → at the pointer's position
  - Gives impression of where the approach leads

### Trochanteric Bursa Ask AI

Hardinge

- **Trochanteric bursa** → usually easy to deal with
  - Exception → thick **trochanteric bursitis** containing fluid
    - Undesirable → possible bleeding or contamination
    - Management → quick excision
- Then clear the tendinous part in front

### Gluteus Maximus Insertion Ask AI

Hardinge

- Part of **gluteus maximus** → inserts onto posterior part of greater trochanter
- Options
  - Retractor beneath it → preserve it
  - Or extend the fascial incision distally
- Either way → facilitates retraction
- Anatomy here → **varies between individuals**
- Retract TFL and gluteus maximus → **trochanter** seen beneath

### Abductor Exposure Ask AI

Hardinge

- Identify **gluteus medius** tendon and **vastus lateralis**
  - Essential → next step is the **gluteus medius split**

# Gluteus Medius Split and Superior Gluteal Nerve Ask AI

## The Split Ask AI

Hardinge

- **Superior gluteal nerve** lies in this region → direct the split accordingly
- No single fixed split pattern
  - Gluteus medius fibre groups → **anterior**, middle, part of posterior
  - Anterior group distinct from middle group

## Trendelenburg Concern Ask AI

- Goal → preserve **abductor function**
  - Assessed by → **Trendelenburg test**
  - Weakness → **Trendelenburg gait**
- Claim: lateral approach causes postoperative Trendelenburg gait → **not correct**
  - Not difficult to avoid
- Therefore → split carried downward (distally)
- Pass between fibres → some gluteus medius fibres encountered → carry split distally
- Image description
  - Part that was **reflected**
  - **Abductor part**
  - Tenotomize → push anteriorly
- **Gluteus minimus** → in depth of the wound
- Major fear → subsequent **Trendelenburg gait**
  - Even if transient, for some hours
  - Usual cause → **nerve concussion** (neuropraxia)

## Safe Zone Ask AI

SGN

- Anatomical **safe zone** → from greater trochanter upward
  - Extent → roughly one-third of the distance
  - Within it → superior gluteal nerve avoided
- Course of **superior gluteal nerve**
  - Probably above the piriformis → **between gluteus medius and gluteus minimus**
  - Main trunk → runs a distance on gluteus minimus muscle
  - Enters **posterior part** of muscle → branches proceed to anterior part
- Preserve **medius–minimus plane** → nerve preserved
  - Avoid excessive retraction, traction, stretching
- Always stay in the plane between gluteus medius and minimus

**High Yield**

- Trendelenburg gait after lateral approach → avoidable, not inevitable
- Usual cause → neuropraxia (nerve concussion) of superior gluteal nerve
- Nerve → between gluteus medius and minimus; enters posterior part
- Protection → stay in medius–minimus plane, no excessive traction
- Gluteus medius split → carried distally

# Hardinge Approach: Tenotomy and Capsule Ask AI

## Inverted-T Tenotomy Ask AI

Hardinge

- After **gluteus medius** split
  - **Gluteus minimus** found → probably not split; runs from above downward
- **Tenotomy of gluteus medius tendon** → anterior part
- Distal tenotomy also needed → **inverted T** shape
  - Opens space between gluteus medius tendon and underlying **capsule**
- **Capsule** now seen beneath
  - Gluteus medius previously covering it from above → now retracted out of the way

## Capsule Management Ask AI

Hardinge

- Capsular opening → surgeon's preference; any type acceptable
  - **T capsulotomy**
  - **Capsulectomy**
  - **Square capsulotomy**
  - Inverted pattern → transverse at the neck, vertical above

| | Past | Current |
|---|---|---|
| Total hip arthroplasty | **Excision of capsule** + synovectomy first | **Without capsulectomy,** except where needed |
| Total knee arthroplasty | Included synovectomy | — |
| Debridement | Much wider field | Narrower |
| Options | — | Capsulotomy, synovectomy, **labral excision** as needed |

# Dislocation, Acetabular Exposure and Closure Ask AI

## Dislocation Ask AI

Hardinge

- Patient → lying on the side
- **External rotation** of the hip + simultaneous control of the leg
  - Leg held in a sterile leg bag or similar
  - **Assistant** → moves the leg from the other side
- Alternative → leg brought over the table edge → **not preferred**

## Knee Caution Ask AI

- Caution in hip surgery patients
  - Elderly without reserve, or **knee osteoarthritis**
  - Leg twisted across the other + force by hand → **knee pain**
    - Persists long postoperatively → troubles walking
- Possibly only the assistant applies the manoeuvre, from above

**Pitfall**

- Forceful twisting of leg during dislocation → persistent knee pain
- At-risk → elderly without reserve, knee osteoarthritis
- Leg over the table edge → not preferred

### Acetabular Exposure Ask AI

Hardinge

- After dislocation
  - Neck osteotomy probably completed → work on **acetabulum**
- Two or three pointed retractors
  - Placed between labrum/acetabulum and capsule
  - Many surgeons do not know where to place them
- Layers → **labrum**, socket beneath, **capsule** attached above it
  - Retractor site → **between labrum and capsule**
- Incision → allows **extension** in every needed area

**Clinical Pearl**

- Acetabular retractors → between labrum and capsule
- Two or three pointed retractors

### Femoral Preparation and Closure Ask AI

Hardinge

- Then **femoral side** preparation → facilitates later steps
- After preparation and **reduction** → everything back to normal
  - Unless a **capsulectomy** was performed
- Repair sequence
  - **Capsule** (if capsulotomy performed)
  - **Gluteus minimus**
  - **Gluteus medius group** → reattached
    - Onto the **trochanter**
  - Close in layers
- Closure → layer by layer

## McFarland Approach Ask AI

### Concept Ask AI

McFarland

- Speaker learned it and continued using it
- **Simple approach** → does not damage muscles
  - **No tenotomies**
- Position → **lateral position**
- Incision
  - Not necessarily centered on trochanter
  - For a **limited hip procedure** → centered on trochanter, as in Hardinge
- Incision carried in same line → **tensor fasciae latae**
  - → **iliotibial band** distally

### Original McFarland Ask AI

McFarland

- One unit taken → **gluteus medius** + tensor fasciae femoris + band
- **Gluteus medius–vastus lateralis** conjoint tendon → kept in continuity

### Reflecting the Common Tendon Ask AI

McFarland

- Through iliotibial band incision → reflect **common tendon** from greater trochanter from above
- Two options
  - **Subperiosteal** reflection (raising the periosteum)
  - **Osteotomy** → part of greater trochanter taken
- Then clear gluteus and TFL
  - From above
- Exposure → upper part of the **neck**
- Upper part of **head** → covered by layers
  - **Gluteus minimus** → capsule beneath → head beneath capsule
- Image review
  - Line of incision of **gluteus medius**
  - **Trochanter**
  - **Tensor fasciae latae**
- Reflection of common tendon → either **subperiosteal**
  - Or **osteotomy of greater trochanter** with the common tendon
- View → **gluteus medius, common tendon, vastus lateralis**
  - Reflected anteriorly

## Modified McFarland Approach Ask AI

### Rationale Ask AI

Mod McF

- Original McFarland → **whole gluteus medius** taken anteriorly
  - Endangers **nerve supply** → arrives from posterior part
- Solution → **modified McFarland;** incision made anteriorly
  - Between **anterior third** and **posterior two-thirds** of gluteus medius
  - Posterior two-thirds → left in place (as in Hardinge)
  - Anterior part → reflected forward

### Technique Ask AI

Mod McF

- Split → made at the corresponding site
- Flap reflected
  - **Anterior part of gluteus medius**
  - Part of vastus lateralis, likely its posterior part
- Successive steps beneath → **gluteus medius**

| | Hardinge | Original McFarland | Modified McFarland |
|---|---|---|---|
| Position | Lateral (speaker) or supine | Lateral | Lateral |
| Incision | Centered on greater trochanter | Centered on trochanter for limited procedure | Same line, extensile |

| | Hardinge | Original McFarland | Modified McFarland |
|---|---|---|---|
| Gluteus medius | Split; anterior part tenotomized | **Whole** muscle reflected anteriorly | **Anterior third** reflected; posterior two-thirds left |
| Tenotomy | Inverted-T | None | None |
| Continuity | — | Gluteus medius + conjoint tendon + vastus lateralis | Gluteus medius + vastus lateralis sleeve |
| Nerve risk | Superior gluteal nerve in split | Nerve supply endangered | Nerve supply spared |

## Extensile Exposure and Extended Trochanteric Osteotomy Ask AI

### Extensile Value Ask AI

Mod McF

- **Extensile** from above downward → proximally and distally
  - Usable in **revision** surgery
- Extendable → probably to the femoral shaft
  - Relevant for shaft view → e.g. **long stems**
  - Extension follows the same structures
- All → continuous lateral structures kept linked

### Extended Trochanteric Osteotomy Ask AI

ETO

- Detailed in next lecture → **Dr. Mahmoud Abdel Karim**
- Requirements for **extended trochanteric osteotomy**
  - Osteotomy and release of **trochanteric fragment**
  - **Muscle group** remains **attached** to fragment
    - Not taken out in the hand and put back loose → in continuity
- This approach → fragment kept as a **flap** during osteotomy
  - As seen in the diagram
- Osteotomy fragment → **vastus lateralis** part attached
- Enter the **canal** → work → return **fragment flap** to its place
- Now performed widely → **revisions** increased greatly
  - Not very difficult
  - Used in difficult situations, likely with cemented stems
- Prerequisite → **extensile exposure**
  - **Modified McFarland** approach
  - Or **posterior approach** → has some limitations

### Speaker's Experience and the Sleeve Principle Ask AI

- This approach → more advantages than posterior approach
- Long experience with **McFarland / modified McFarland**
  - Since assisting late professor (El-Hommos), with modifications
  - Work with Dr. Maher in England
  - Many cases in Egypt, including health insurance cases
- **Extensile exposure** → any level, limited or extended, needs nothing else

  - Extended as needed → **muscle sleeve** remains continuous
  - Distally or proximally → sleeve in continuity
- **Distal extension** → after **osteotomy of the neck**
  - Probably retract trochanter to another position
  - Reflect posterior flap instead of **anterior flap**

**Must-Know**
- Modified McFarland → anterior third of gluteus medius reflected with vastus lateralis
- Extensile proximally and distally → usable in revision
- ETO → trochanteric fragment kept attached to muscle group as a flap
- Muscle sleeve → kept in continuity

## Disadvantages and Complications Ask AI

### Exposure Limitations Ask AI

Lateral

- Disadvantage → other side probably not included in the exposure

### Gluteus Medius Complications Ask AI

Lateral

- Reopening needed on 2–3 occasions
  - **Gluteus medius tendon tear**
  - **Gluteus medius denervation**
- Not **100% safe**
  - Gluteus medius complications → whether tendon detached or kept in continuity
  - Dependent on **technique**

### Trochanteric Osteotomy Nonunion Ask AI

- **Osteotomy of greater trochanter** → may fail to unite (**nonunion**)
  - Do not rely on **suture only** → more secure fixation

### Leg Length Ask AI

Lateral

- Another disadvantage
  - Cannot observe leg lengths → no **leg length equalization**
  - Possible in **posterior approach**

**Exam Pearl**
- Lateral approach complications → gluteus medius tear, denervation
- Trochanteric osteotomy → nonunion risk; more than suture fixation
- Leg length equalization → posterior approach, not lateral

# Posterior Approach

*(Positioning, Incision and Soft-Tissue Repair in the Posterior Approach)*

**Prof Dr Mohamed Abo Elsoud**

**Watch Original Lecture**

*Lecture transcript processed with AI assistance.*

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## Introduction Ask AI

- Topic: **posterior approach** to the hip
- Context: follows Dr. Hazem's lateral approach talk
- First seen performed by **Dr. Tarek Abdel Shafy**
  - Speaker then a resident, just back from the army
- Format: surgical video + running commentary
- Style → *Science and Faith* programme, late **Dr. Mostafa Mahmoud**
- Upcoming lecture: **Choice of the Approach**
  - Literature → which approach is better
  - Lateral approach advantages vs posterior approach disadvantages

## Patient Positioning Ask AI

- **Lateral position** — no alternative
- **Pelvic support**
  - Anterior → **symphysis pubis** or anterior superior iliac spine (ASIS)
  - Posterior → **sacrum**
    - Not on the lumbosacral junction
- Leg moves freely anteriorly + posteriorly → full **hip flexion**

**Pitfall**

- Leg striking the anterior post → **false impression** of easy dislocation
- Posterior support on lumbosacral junction → wrong; sacrum only

## Skin Incision Ask AI

- **Classical incision:** curved posteriorly
  - Centred on the **greater trochanter**
- Upper (proximal) + lower (distal) limbs — as in most hip approaches
- **Extensile** → proximal or distal extension as required

## Superficial Dissection Ask AI

- Skin → subcutaneous tissues → fascia over **gluteus maximus** + fascia lata
- **Fascia lata** opened → diathermy or knife, per preference
- **Gluteus maximus** fibres → split bluntly by hand, in fibre line
- Video = speaker's own case
- Next layer: **fat over the sciatic nerve** + bursal surface (term unclear in recording)
- **Constant bleeder** at this angle
  - Muscular branch → gluteus maximus supply
- Sciatic nerve usually lies within this fat

### Bursal Layer Handling Ask AI

Bursa

- Options (bursa — term unclear in recording)
  - Simply opened
  - Scissors passed beneath → opened over them with **diathermy**
- Diathermy applied onto scissors → **scissor tip points toward sciatic nerve**

## Short External Rotators and Sciatic Nerve Ask AI

- **Gauze** → sweeps fat + sciatic nerve posteriorly
  - Left in place → protects sciatic nerve
- Routine posterior approach → sciatic nerve **not exposed**
- **Short external rotators** now clear
  - Obturator internus tendon + two gemelli
  - Below → fleshy **quadratus femoris**
  - **Piriformis** hidden → beneath posterior border of gluteus medius
- Piriformis exposure → retract **posterior edge of gluteus medius**
- Identification
  - Fleshy muscle → **quadratus femoris**
  - Tendon with muscle above + below → **obturator internus**
    - Superior + inferior gemellus
- Same patient, gauze removed → **sciatic nerve** + obturator internus clearly seen
- Retractor → holds posterior border of gluteus medius
- → **Piriformis tendon** revealed (next video)
- Muscle deep to piriformis tendon → **gluteus minimus**

## Release of Rotators and Capsule Ask AI

- **Gluteus minimus** → protect; never cut
- Piriformis tendon identified
- Rotators released → **diathermy** or knife
  - **Tagging sutures** on rotators → surgeon preference
- **Rotators + capsule** opened together
  - Gluteus medius → not opened
  - Gluteus minimus → not injured
- **Capsule** = white tissue beneath → opened too
- **Upper quadratus femoris** released → lesser trochanter + femur well seen
- Capsule opened → gluteus minimus visible
- **Hohmann retractor** beneath gluteus minimus → retracts + protects it
- Upper capsule → **capsulotomy** completed
- **Synovitis** → gush of fluid/synovium on capsular opening
- Hohmann passed **superiorly** → carries gluteus minimus with it
  - → **superior capsule** seen → capsulotomy completed
- Upper quadratus femoris completed + opened → **lesser trochanter** seen
- **Quadratus femoris** incision completed → superiorly + inferiorly
- Opened capsule visible

- Glistening **iliopsoas tendon** → on lesser trochanter

## Dislocation and Neck Exposure Ask AI

- **Dislocation** → in total hip
- Fracture neck of femur → only **stump of the neck** delivered

| | Lateral approach | Posterior approach |
|---|---|---|
| Dislocation manoeuvre | External rotation + figure-of-four | Maximum internal rotation |

- This case: fracture neck of femur → neck stump delivered
- Landmarks: **lesser trochanter** + iliopsoas tendon
- **Blunt Hohmann** → beneath iliopsoas tendon
- Second Hohmann above neck stump → mostly **sharp** → neck well exposed
- **Rotator + capsule stump** → always remains in piriformis fossa
  - However close to greater trochanter the release
- **Piriformis fossa** = stem entry point
  - → stump removed → **box osteotome** → femoral canal
- Arthritis case: femoral head, lesser trochanter, Hohmann
  - Hohmann on **upper neck** → rotator stump removed same way

**Exam Pearl**

- Lateral approach → external rotation + figure-of-four
- Posterior approach → **maximum internal rotation**
- Rotator/capsule stump in piriformis fossa → blocks stem entry → remove

## Neck Cut and Head Extraction Ask AI

- **Neck cut** → two cuts
  - **Horizontal** → along rotator stump line
  - **Vertical** → second cut
- Horizontal cut → **greater trochanter** must not fracture — key point
- Osteotome in horizontal line alone → exits → **greater trochanter fracture**
- **Vertical cut** → protects greater trochanter
- Residual bone piece → **safety**, not timidity
  - → removed with box osteotome
- Fracture neck → head still inside → **corkscrew** → head extracted
- **Ligamentum teres** → usually attenuated in elderly
  - This case: very thick → scissors → easy head delivery
- Head delivery → leg manipulation, head taken forward (wording unclear in recording)
- Head still tethered → **inferior capsule** + ligamentum teres
  - Divided → diathermy or scissors
- Elderly, very tough tissues → **scissors**
- **Anterior capsule** + ligamentum teres still attached → head delivered

**High Yield**

- Neck cut = horizontal + **vertical** cut
- Vertical cut → prevents greater trochanter fracture
- Leftover bone → removed with box osteotome
- Fracture neck → head retrieved with **corkscrew**

## Anterior Capsular Resection Ask AI

- **Anterior capsular resection** — very important step
- Older arthroplasty patients → **anterior capsule** often very hypertrophied
  - → femur held stuck against acetabulum
  - → femur cannot go forward → acetabulum not seen
- **Acetabular reaming** → hand must be free anteriorly
  - Otherwise → reaming in **retroversion**
- Anterior capsule = structure holding femur back

| Removal method | Technique |
|---|---|
| From below | Cut with **scalpel** |
| Hohmann-assisted (easier) | Hohmann between **gluteus minimus** and anterior capsule |

- Capsule in this case → very hypertrophied
- Hohmann → takes **minimus** forward
- **Kocher** → holds anterior capsule for resection
- White capsule → Kocher slips → re-applied
- Whole anterior capsule resected → **anterior mobilisation of the femur**
- Anterior **Hohmann** → takes femur forward
  - Femur then "forgotten" → leg straight, no further attention
- Hohmann placement → **labrum–capsule plane** (per Dr. Hazem)
- Anterior + posterior Hohmanns → both between labrum and capsule
- Posterior retractor seen entering between white **labrum** and capsule

## Labrum and Inferior Capsule Release Ask AI

- Capsule + retractor within it identified
- Capsular release completed
  - **Inferior capsule** → incised
  - **Labrum** → cut + removed (per Dr. Tarek Abdel Shafy)
    - Mostly removable at cup preparation

| Labrum | Tool | Reason |
|---|---|---|
| Anterior | Diathermy or scalpel | — |
| Posterior | **Scalpel** preferred | Diathermy heat → sciatic nerve |

- Posterior labrum removed → residual tissue also removable
- Once removed → view opens; tissue brought downward
- **Inferior capsule** → incised
- Anterior tissue removed → comes out very easily
- **Transverse acetabular ligament** → appears at inferior acetabulum
- Inferior **Hohmann** → also possible

> **Clinical Pearl**
> - Posterior labrum → **scalpel**, not diathermy
> - Diathermy near sciatic nerve → onto scissors, tip toward nerve
> - Gauze left posteriorly → protects **sciatic nerve**

## Acetabular Exposure and Preparation Ask AI

| Retractor | Placement |
|---|---|
| Posterior Hohmann | Between labrum and excised capsule |
| Second Hohmann | Anterior → femur taken anteriorly |
| Inferior Hohmann | Inferior acetabulum |

- **Transverse acetabular ligament** identified

### Surgeon Position Ask AI

Surgeon

| Hip | Surgeon position | Benefit |
|---|---|---|
| Right | Behind the patient | — |
| Left | Other side | Straight hand; clearer reaming view |

- This case: right hip → same side

### Acetabular Preparation Ask AI

Cup

- **Acetabular reamers** → cup bed prepared
- Reaming → **acetabular cup** inserted
- ± 1–3 **acetabular screws** as needed (unclear in recording)
- Cup alignment → **transverse acetabular ligament** at lower cup margin

## Posterior Repair Ask AI

- Femoral side completed → **posterior repair** mandatory
- Repaired: **piriformis**, rotators, capsule

| Fixation | Rationale |
|---|---|
| **Gluteus medius tendon** | More durable → withstands external + internal rotation |
| Drill holes in bone | Possible; rotation, esp. internal → pull-out |

- → Some surgeons prefer gluteus medius tendon
- Repair complete → joint reduced + covered, no longer visible
- All **posterior structures** repaired → non-absorbable sutures preferred

## Take-Home Message Ask AI

- **Posterior approach** → easy + reproducible
- **Pelvic support** → patient does not roll forward or backward
  - Video: support slightly high, on abdomen
  - Correct → pubis or anterior superior iliac spine
- **Preserve gluteus minimus**
  - → avoids posterior + lateral approach disadvantages combined
  - i.e. no added **abductor** injury
- **Retractor placement** → protects sciatic nerve + anterior structures (vessels)
- **Anterior capsular release** → anterior mobilisation of the femur
  - Esp. elderly with hypertrophied capsule

**Must-Know**

- Pelvic support → pubis/ASIS anteriorly, sacrum posteriorly
- **Gluteus minimus** preserved throughout
- Anterior capsular resection → avoids reaming in retroversion
- Posterior repair → gluteus medius tendon, non-absorbable sutures

# Direct Anterior Approach DAA

*(Patient Selection, Technique and Learning Curve of the Anterior Approach)*

**Prof Dr Sherif Khaled**

**Watch Original Lecture**

*Lecture transcript processed with AI assistance.*

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# AI detected corrections (Mostly bad transcripts)

## Corrected - the text was changed; the transcript said

**C1** (p. 156) “abduction first” → “adduction”

**C2** (p. 159) “2 crutches for 2 months” → “2 crutches for 2 weeks”

## Queried - kept as spoken, but looks wrong

**Q1** (p. 152) As spoken; learning curve of 5-10 years conflicts with 'first 100 cases' and 'first two or three years'.

**Q2** (p. 158) As spoken; standard posterior stability test is flexion + internal rotation + adduction, not abduction.

## Intervals of the Direct Anterior Approach Ask AI

- Second lecture of the DAA series
- Two intervals:

| Interval | Between | And |
|---|---|---|
| Superficial | Sartorius | Tensor fasciae latae (TFL) |
| Deep | Rectus femoris | Tensor fasciae latae (TFL) |

- Remaining steps → as in other approaches

## Patient Selection Ask AI

- DAA → applicable to all patients
  - Some patients better suited → surgeons who use it say so
  - Some surgeons build a whole patient-selection programme around it
- **Ideal patient:**
  - **Primary hip** → not used in revisions
  - BMI < 30
  - Small-built → i.e. female
  - Non-osteoporotic bone
  - Preferably **coxa valga** / valgus neck
- **Learning curve** → operate only on such cases:
  - First five to ten years *[Q1 · As spoken; learning curve of 5-10 years conflicts with 'first 100 cases' and 'first two or three years'.]*
  - Or first 100 cases → in females
- Reason → muscles weaker in females
- Trained group: the lecturer, Ahmed Hazem, Dr. Fouad, Mahmoud Abdel-Karim
- First two or three years → stay with favourable cases

**High Yield**
- Ideal DAA patient: primary hip, BMI < 30, small-built female
- Non-osteoporotic bone; coxa valga / valgus neck preferred
- Not used in revisions
- Learning curve: favourable cases only, first 100 cases

## Concept and Advantages Ask AI

- **Minimal incision** (7 cm, even 2 cm) → not the idea at all
- Core idea → **intermuscular / internervous planes**
  - Nothing cut → surgeon goes between muscles
  - Only **anterior capsule** opened
  - Posterior structures respected:
    1. Posterior capsule

        - Short external rotators
        - Sciatic nerve
        - Glutei
- Therefore:
    - **Dislocation** → much lower, almost zero
    - **Trendelenburg gait** → zero
    - **Heterotopic ossification** → almost zero
    - **Postoperative restrictions** → none

## Disadvantages and Requirements Ask AI

- Disadvantages → special instruments + slow learning curve
- Ordinary instruments → DAA impossible; dedicated set mandatory
    - **Offset reamers** / **offset inserters** → not essential; simple one acceptable
    - Without the instrument set → nothing can be done
    - **Six retractors** → important; work still difficult with them
- Ordinary instruments → acetabulum perhaps manageable, femur not
    - **Technically demanding** approach
    - Demanding **learning curve**
- **Special table**?
    - Previously avoided → believed to need special table
    - Team learned DAA in Belgium 4 years ago from operating surgeons

## Positioning, Draping and Skin Incision Ask AI

### Positioning Ask AI

- **Ordinary table** only
    - Support placed → supports the patient
    - Leg: one third above table break, two thirds below
        - Reason → leg goes into **figure-of-4**
    - Drawing: obese patient; leg on a table; patient near table edge

### Draping Ask AI

- Some centres → special drapes for ease
- Here → ordinary lateral draping, drapes side to side
- Assistant on one side → leg on the other

### Skin Incision Ask AI

- Landmark → ASIS (as shown)
    - Start → three fingerbreadths distal + three fingerbreadths lateral
- Some Belgian surgeons → **bikini incision**
- Team's **classic incision** → **extensile**, extendable distally
- Second option → **transverse** incision
    - Experienced surgeons only
    - Team: not yet after 4 years; not before 10 years

- Incision oblique, about 8 cm

## Superficial and Deep Dissection Ask AI

### Superficial Dissection Ask AI

- Skin + subcutaneous tissue → cautery dissection → **TFL**
- TFL appearance → as on slide
- Muscle visible through fascia
  - **Fascia opened in the middle** (over muscle) → away from the lateral femoral cutaneous nerve
  - Fascial flap held
- Medial (front) flap → finger along TFL → falls into **TFL–sartorius interval**
- Interval used → **two retractors,** above and below femoral neck

### Neck Anatomy and Retractor Placement Ask AI

- **Femoral neck fracture** → no neck → retractor unusable
  - Therefore → femoral neck fracture **not an indication**
  - Hence valgus neck / long neck preferred → retractors usable
- Short neck / **protrusio:**
  - Protrusio → practically impossible
    - Team's protrusio cases → very, very difficult
    - First or second retractor cannot be placed → do not do it
  - Look for neck valgus

**Pitfall**

- Femoral neck fracture → no neck → retractors cannot be placed
- Femoral neck fracture is not an indication for DAA
- Protrusio / short neck → practically impossible; avoid

### Deep Dissection Ask AI

- Deep fascia opened with scissors → **yellow fat** beneath
- Cautery dissection → **anterior aspect of acetabulum**
- **Iliocapsularis** → attached over anterior acetabulum
  - Its fascia opened → **curved retractor** beneath it
  - Pushed in one direction only (as shown)
  - Effect → lifts iliocapsularis → exposes anterior acetabulum
  - Also → protects **femoral nerve**
    - Femoral nerve lies superficial to iliocapsularis → no risk

### Capsule Ask AI

- **Capsulectomy** → as in posterior approach (Dr. Hazem)
- As in lateral approach → capsulotomy or capsulectomy
  - Some → **H-shaped** capsulotomy
  - Some repair capsule; others do not
    - Non-repair rationale → **anterior capsule** has no importance

## Neck Osteotomy and Head Removal Ask AI

- **Neck osteotomy** → single osteotomy possible
- Difficult cases → **double osteotomy**
  - Middle fragment removed (corkscrew) → head comes out easily
  - Sometimes too small to extract head as in lateral / posterior approaches
- Head removed → **corkscrew**
- Next retractor → toward the obturator

## Acetabular Preparation Ask AI

### Exposure Ask AI

Acetabulum

- **Three Hohmann retractors** → needed for acetabular view:
  - Inferior
  - Posterior wall
  - Anterior wall
- Viewing in this direction → very accurate

### Cup Orientation Ask AI

Acetabulum

- Cup orientation:

| Parameter | Reference | Angle |
|---|---|---|
| **Anteversion** | From the floor | 15° |
| **Inclination** | Handle opened from midline | 40° |

### Reaming Ask AI

Acetabulum

- **Straight vs curved reamer** → no difference
  - Straight reamer photographed → shows it is not very important
  - Importance of **canal finder** + straight reamer → seen later
- Acetabular component inserted

## Fixation Philosophy Ask AI

- All surgeons observed abroad → **cementless,** not cemented
- Team's cemented cases → more difficult
  - Cementing the cup → no problem
  - **Cementing the femur** → difficult
- **Cementless cup and stem** → reduces operative time
  - Reason → **fast-track surgery**
  - Discharge → same day or next day
  - Cementless → easier
  - Cementless → faster

# Femoral Exposure and Preparation Ask AI

## Exposure Ask AI

Femur

- **Femoral side** → the difficult part
- Trick: adduction (stated as abduction) → Hohmann **lateral to greater trochanter** *[C1 · transcript: "abduction first"]*
- **Bare area:**
  - Gluteus medius attachment → not at tip; 1.5 cm lateral to tip
  - Area between tip and gluteus medius attachment → **bare area**
  - Retractors placed here
  - Very small area → bone cannot be **osteoporotic**
  - Team has fractured this area before
  - Some say area has no value (tendons independent) → still try hard to preserve it
- All retractors placed → small capsular structures released
- **Hook** on femur → pulled to confirm femur comes out
- Reason → **greater trochanter** lies inside, behind acetabular rim
  - Supine patient → femur must be brought out from under acetabulum → very difficult
  - Hook required → another reason osteoporosis excluded from indications
- Femur pulled outward → structures **released**
  - Until hook lifts femur a sufficient distance
  - Before this → do not try; impossible or very difficult
- Femur up → **figure-of-4:**
  - Healthy leg lifted
  - Operated leg in figure-of-4 beneath it → rested on table side

**Exam Pearl**

- Bare area: between greater trochanter tip and gluteus medius attachment
- Gluteus medius inserts 1.5 cm lateral to the tip
- Osteoporotic bone → bare area fracture risk

## Preparation Ask AI

Femur

- Femur rasped → **offset rasp**, a little easier
- Retractors:
  - **Hohmann** → bare area, between trochanter and gluteus medius
  - **Medial retractor** → on the **calcar**
  - Leg → figure-of-4
- **Box osteotome** → **canal finder**
  - Without it → canal location never predictable
  - Canal → very strange place, very strange direction
  - Canal finder inserted → shows the way → canal followed

## Calcar and Releases Ask AI

Femur

- Cut high → **calcar planed (reamed);** flat calcar → cut precise
- Release small capsular structures:
  - Superior → release

  - Inferior → release
  - **Posterior** → do not release
    - Releasing them → compromises stabilizers + **posterior capsule**
- Posterior capsule released → approach loses its benefit

**Must-Know**
- Never release the posterior structures
- Posterior capsule released → DAA loses its benefit

# Trial Reduction and Intraoperative Assessment Ask AI

## Trial Reduction Ask AI

- **Trial reduction** → traction + internal rotation → reduces very easily
  - Force needed → problem: do not force; go back and find it
  - Reduction should be very easy

## Impingement and Telescoping Ask AI

- Impingement checked first; then telescoping:

| Test | Manoeuvre | Checks |
|---|---|---|
| Impingement | Internal rotation | **Anterior impingement** |
| Impingement | External rotation | **Posterior impingement** |
| Telescoping | Distal (as usual) | **Vertical offset** |
| Telescoping | Lateral | **Lateral offset** |

- Femur comes laterally → **offset stem** needed
  - Extended-offset / lateralized (high-offset) stem

## Limb Length Discrepancy Ask AI

- **LLD assessment** → very easy
- Reason → patient supine, both legs in front of surgeon
  - **Medial malleoli** → equal length?
  - **Galeazzi test** → hips $45°$ + knees $45°$ flexed → compare in flexion

## Combined Version Ask AI

- Check → **cup parallel to head (neck) cut**
- Parallel → **combined version** excellent
  - Cup inclination → very accurate
  - **Combined anteversion** → ideal

**Clinical Pearl**
- Trial reduction: traction + internal rotation → very easy
- Force needed → stop and look for the problem
- Cup parallel to neck cut → ideal combined anteversion

### Stability Testing Ask AI

- Stability tests:
  - External rotation + adduction
  - External rotation + abduction
  - External rotation + abduction + extension
  - Flexion + internal rotation + abduction *[Q2 · As spoken; standard posterior stability test is flexion + internal rotation + adduction, not abduction.]*
- Sequence and reverse:
  - Extension + external rotation → test adduction, abduction, extension
  - Leg flexed → internal rotation + abduction
  - Then the reverse

### Final Limb Length and Offset Ask AI

- Finally → limb length checked; in front of surgeon (supine)
  - Advantage of DAA → checking limb length
  - And offset

| Approach | Limb length method | Ease |
|---|---|---|
| Posterior / lateral | **Schanz pin** in pelvis + **sterile ruler** | Harder |
| Posterior / lateral | Mark on greater trochanter (stitch) → measured repeatedly | Harder |
| DAA | Limb length seen directly + **Galeazzi** test | Much easier |

- Posterior / lateral example → lengthen 2 cm: distance 9 cm → must end 11 cm

## Closure and Postoperative Care Ask AI

### Closure Ask AI

- **Tranexamic acid** 1 g → before starting closure
- TFL fascia → closed **continuously**
  - Then skin + subcutaneous tissue
- **No drain**
  - Exceptions → muscular patients, bleeding

### No Restrictions Ask AI

- **Postoperatively** → no restrictions
  - Patient can pray, flex, put on shoes, turn in bed
- Clinical example → 58-year-old female
  - **Alzheimer's disease / minimal cognitive impairment**
    - Minimal cognitive impairment → term used in younger patients

### Illustrative Case Ask AI

- **Femoral neck fracture**; patient is a doctor
  - Suggestion → give her precautions; concern → poor compliance
  - Team → arthroplasty + precautions against dislocation

- Sleeps **curled up (baby position)** since operation day
- **Never dislocates** → the idea of DAA
  - Suits patients needing it → **no restrictions;** hip will not dislocate
  - Reason → intact stabilizers:
    1. Posterior capsule
    2. Rotators
    3. Glutei
- Only **anterior capsule** opened → almost no value in stability

### Rehabilitation Ask AI

- **No physiotherapy** needed
- **2 crutches** for 2 weeks (stated as months) → **1 crutch** for a period → then off *[C2 · transcript: "2 crutches for 2 months"]*

## Complication: Femoral Nerve Neurapraxia Ask AI

- One case of **femoral nerve neurapraxia** → one year to recover
  - Very serious complication; happened to the team
  - Patient profile:
    1. Short neck + **protrusio**
    2. About 38 years old
    3. Very obese
  - Nerve conduction study → all wrong; all directions reversed
  - Recovered; no other complications
- More severe (permanent) injury → would not have been easy

**Pitfall**

- Femoral nerve neurapraxia → one year to recover
- Occurred in short neck + protrusio + very obese patient
- These features → outside DAA indications

## Take-Home Message Ask AI

- **Intermuscular and internervous** → natural interval around the hip
- **Ordinary table** → no special table required
- Lower dislocation rates + faster recovery
- **No restrictions** on movement → unlike other approaches
- Patient → very satisfied

# Extended Trochanteric Osteotomy ETO

*(Indications, Types and Technique of Trochanteric Osteotomies)*

**Prof Dr Mahmoud Abdel Karim**

**Watch Original Lecture**

*Lecture transcript processed with AI assistance.*

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## Introduction and Philosophy Ask AI

- Core question → femur decided to be opened → how to open it
- Aim → know how to open such a case
- **MIS** (minimally invasive surgery) → intention to do a good job
  - Not only the technique or incision used
- Revision goal → adequate exposure → facilitates the operation
- Speaker learning the **anterior approach** with Dr. Sherif
  - Not yet used in a single private patient → reflects current opinion
  - Use for own patients → only after confidence + sufficient cases

## Indications of Trochanteric Osteotomy Ask AI

- Improved exposure → **extensile approach**
- Removal of **well-fixed cement**
- Removal of a well-fixed **femoral component**
- Difficult hip, even in primary settings
  - Old ankylosis / fusion
  - Cluster of adhesions
  - **Heterotopic ossification (HO)**
- **Trochanteric malposition**

## Types of Trochanteric Osteotomy Ask AI

- Covered previously; beyond the scope of this lecture

| Type | Cut | Muscle carried |
|---|---|---|
| **Standard** (A, B in figure) | Trochanteric | Gluteus medius only |
| **Sliding** | Slightly lower | Gluteus medius + vastus lateralis |

- Topic of this lecture → **extended trochanteric osteotomy (ETO)**
- Dr. Hazem → introduced its characteristics and situations needing it
- Approach (figure) → **posterior** or **lateral**

## Extended Trochanteric Osteotomy: Definition and Indications Ask AI

### Definition Ask AI

ETO

- Osteotomy of the **lateral one third of the femur** → removes that cortical layer
  - Nickname → **"coffin lid"**
- Opens the femur → straight access to the canal
- >90% of slides → writings of **Wayne Paprosky**
- His material at **ICJR** meetings → makes the procedure much easier

### Indications Ask AI

ETO

- Removal of **well-fixed cement**
  - Long cement mantle not otherwise removable, e.g. infection / sepsis
- Reasonably **well-fixed cementless stem** → ETO sometimes required
  - Sometimes combined with cement removal
- Well-fixed cement + **femoral varus deformity** → ETO → **neutral reaming**
  - Reaming in varus / valgus (usually varus) → femoral **perforation**
- **Correction of proximal femoral deformity**
- Removal of **intramedullary hardware**, e.g. a nail
  - Paprosky case → shown later
- Improves **acetabular exposure**
- Allows **abductor tensioning**

### Current Indications by Frequency Ask AI

ETO

- Indications above → **complex primary** setting
- Paprosky (2018) ranking:

| Rank | Indication |
|---|---|
| 1 — most common | **Infection** (speaker has done such a case) |
| 2 | **Periprosthetic fracture**, Vancouver type B2 |
| 3 | **Femoral loosening** |
| 4 — least common | **Femoral varus deformity** |

**Exam Pearl**

- Most common current ETO indication → **infection**
- Second → periprosthetic fracture, Vancouver B2
- Least common → femoral varus deformity

## Contraindications and Advantages Ask AI

### Contraindications Ask AI

ETO

- **Cemented femoral revision** → cement may extrude into osteotomy site → fixation difficult
  - Practice declining → used only if no other solution
  - Almost **a lost art apart from Exeter**; uncertain if still done on femoral side
- **Non-union rate** → concern for some authors; arguable either way

### Advantages Ask AI

ETO

- **Quite versatile**
- Easy once a few done → learn by attending someone who performs it
- Decreased operative time; decreased complications
- **Cement removal** → dangerous stage, even in the best hands
  - Use **image intensifier**; be careful
  - ETO → makes it much easier

- Improves **acetabular exposure** (view shown)
- Starts as a window → not merely a window → **extensile exposure**
- Decreased operative time, **blood loss**, risk of complications
- Decreased risk of a second problem → perforation / fracture
- Addresses **abductor tensioning**

## Preoperative Planning Ask AI

- **Accurate templating** → very, very important
- Common errors (all stages of training):
  - Osteotomy too short or too long
  - No room left for the stem distally
  - Proximal fracture
  - Failure to remove the implant
- **Preoperative templating** → determines **osteotomy location** → improves exposure
- Length must allow:
  - Component extraction
  - Adequate fixation of planned components
- **Femoral deformity** → osteotomy at the **apex** if correcting it

### Osteotomy Length Ask AI

ETO

- No fixed number
- Minimum 10 cm from the **greater trochanter tip**
- Minimum 4–6 cm **diaphyseal fixation** of planned prosthesis
- Check available sizes with the company
- In practice ~12 to 15 or 16 cm → varies → derive from **templating** and planning

**Must-Know**

- ETO length → at least 10 cm from greater trochanter tip
- Distal diaphyseal fixation → at least 4–6 cm
- Typical length → 12 to 15 or 16 cm, from templating
- Deformity correction → osteotomy at the apex

### Intraoperative Sequence and Backup Ask AI

- Stem outside planned limits → do not proceed without **backup**
  - Different implants available
- Perform ETO **before dislocation** (Dr. Tarek)
  - In this case mandatory → dislocation → femoral fracture
- Small / osteoporotic bone → shortening or fracture may be more likely
  - May still be the only option; in some cases the only solution
- Stem removed proximally → loose cement or long mantle → remove **distal cement**
- Aim → exposure

## Approach Ask AI

- Speaker prefers the **posterior approach** → not difficult
- Must also master posterior approach extensions (possibly **Kocher-Langenbeck**) and related exposures
- Skill acquired with time → all must learn it
- Anterior approach users → must be well trained to **convert** if needed
- One very experienced surgeon's practice ≠ applicable to all
- Posterior approach → versatile, good exposure
- ETO also via **lateral** or anterolateral approach → Daniel, Barry, other groups
- Speaker's preferred technique → posterior approach

## Surgical Technique Ask AI

### Exposure Ask AI ETO

- Preserve the **musculotendinous sleeve** (Dr. Mohamed Abou El-Saoud)
  - Above → gluteus medius
  - Below → vastus lateralis
- Use a **simple retractor**
- No complete stripping to bare bone (Dr. Hazem) → "it's not an art"
- **Minimal stripping** ~2 cm → **Hohmann retractor** → start osteotomy

### Measuring and Marking Ask AI ETO

- Measure **according to the template**
- ~12 to 15 or 16 cm → always measure
- Mark → **methylene blue**, diathermy, or anything else

### Posterior Longitudinal Cut Ask AI ETO

- Posterior longitudinal limb → **saw**
  - Saw from the start → if experienced
  - Alternative → **drill holes** first → prevents saw **deflection** → fracture
  - Multiple drill holes; add one at the corner → avoids **fragmentation** / comminution
- Connect the drill holes
- Corners → must be **rounded**
- Sharp / square corner → **fracture**; quite common
- Abroad → **pencil burr** rounds the corner
- Alternative → osteotome or drill holes → round as much as possible

### Anterior Limb Ask AI ETO

- Two options (Dr. Tarek with Dr. Hazem):

| Option | Method | Caution |
|---|---|---|
| Complete anteriorly | Saw, **pencil burr**, etc. | **Anterior vessels** |
| Controlled fracture | Simple **osteoclasis** | — |

- Sequence → complete **longitudinal posterior portion** → anterior part with saw from the front

- Alternative (Dr. Tarek) → move the limb → same step
  - Especially **osteoporotic bone** → cannot take much → may fracture
- Speaker's preference → osteotome or drill hole → **curve** the cut proximally → drill holes
- Connect them, even from the front
- **Vastus lateralis** in front → do not **devascularize** the fragment
  - Minimal handling; no tissue stripping

### Levering the Fragment Ask AI

ETO

- Insert multiple **osteotomes** → 2 or 3
- Very gentle; do not rush; not aggressive
- Before levering → release **anterior pseudocapsule** → 100%
  - Not released → fragment fractures → pieces retained
- Final product delivered → existing stem exposed

**Pitfall**

- Sharp / square corner → fracture → round it
- Saw without drill holes → deflection → fracture
- Anterior saw cut → anterior vessels at risk
- Stripping vastus lateralis → devascularized fragment
- Unreleased anterior pseudocapsule → fragment fracture

### Stem Extraction Ask AI

ETO

- Remove the **stem** → different instruments and steps, incl. **Gigli saw**
- Instruments often unavailable locally → learn tricks
- Removal not guaranteed
- Different instruments → flexible / back-cutting **osteotomes** (instrument wish-list slide)
- Train within own institution → develop beyond available instruments

## Prophylactic Cerclage and Fixation Ask AI

### Prophylactic Distal Cerclage Ask AI

- **Cerclage distally** → before anything else → avoids regret
- Commonly forgotten → reaming proceeds
  - Even in a primary using a **long stem** for any reason
- Missing cerclage → split → **fracture**
- Most important point of the lecture

**High Yield**

- **Prophylactic cerclage** / cable distally → before reaming
- Applies also to primaries with a long stem
- Omitted → femur splits → fracture

- Then **broaching**, etc.
- **Advancement** → remove a piece distally → described long ago
  - Indications decreased a lot; trick worth mentioning

### Fixation of the Osteotomy Ask AI ETO

- **Cables** and wires → at least 2 to 3
- No cables / limited budget (lesson from Dr. Maher):
  - **Double-wire cerclage**
  - Prophylactic cable distally → very important
- Sequence → prophylactic cerclage first → 2 to 3 more proximally, per case

### Figure-of-Eight Trochanteric Cerclage Ask AI GT

- Trick published in **JBJS American 2003**
- Often needed → **greater trochanter cerclage** → figure-of-eight technique
  - **Transverse cerclage** → passed **under the muscle**
  - Cerclage over the muscle → muscle necrosis
- Wire under muscle → up over the **trochanter** → vertical limb
- New configuration → **excellent compression** + stability
- Very important; appearance shown

## Complications Ask AI

- Often not discussed
- Each complication → how to avoid it
- Tight **anterior pseudocapsule** → must release it
- **Proximal migration** may occur
  - Standard osteotomy → only gluteus medius attached
  - **Sliding osteotomy** → vastus lateralis also attached → **tethering effect** → prevents migration
  - Still might occur in some cases

## Case Examples Ask AI

### Case 1 — Infected THA (Speaker) Ask AI

- Presented **infected** → stem removal uncertain → needed ETO
- Preoperative templating advised; intraoperative **image** always obtained (instrument / marker) to confirm
  - Especially with a long stem
- **Heavily infected on acetabular side**
- **Femoral component** → quite well fixed (seen after opening)

> **Clinical Pearl**
> - Templating → plus intraoperative image to confirm length
> - Especially with a long stem

- Finished product; **intraoperative** view → no problem anywhere
- **Triple cerclage**
- Double wires if cables unavailable → **cables** very expensive

### Case 2 — Intramedullary Implant (Paprosky) Ask AI

- **Wayne Paprosky's** case, not the speaker's
- ETO → very useful, quite valid → **intramedullary implant** extraction

### Case 3 — Childhood Deformity Ask AI

- **Deformity** from early childhood
- Procedure → partial ETO + **completed medial osteotomy** → correction
- Paper → nearly six years of **follow-up**

## Approaches: Comparative Evidence Ask AI

- Approaches only → evidence piece left to Mohamed
- 2015 paper → compared the approaches discussed
- Scoring → plus / minus or number of pluses → **efficacy** + superiority of results

| Domain | Components scored |
|---|---|
| **Acetabular exposure** | Anterior, superior, posterior |
| **Femoral exposure** | Metaphysis, canal, lateral cortex |

- **Trans-trochanteric** approaches → 3 pluses in everything
- Next → **posterior approach**

## Further Reading and Conclusion Ask AI

### Further Reading Ask AI

ETO

- Details + further reading → shared for whoever wants

| Resource | Type | Authors | Source |
|---|---|---|---|
| "Extended Trochanteric Osteotomy in Complex Primary Total Hip Arthroplasty" | Paper — very important | Craig Della Valle, Wayne Paprosky | **JBJS American 2003** |
| "Surgical Treatment of Hip Arthritis", chapter 37 | Book — very good | Paprosky | — |
| "Extended Trochanteric Osteotomy" | **Campbell's video** | Crockarell, Gregory Dabov | YouTube |
| "Extended Trochanteric Osteotomy in Complex Total Hip Arthroplasty" | Video interview | **Craig Della Valle** (works with Paprosky) | ICJR website |

- Campbell's video → watched by speaker in 2003 as assistant lecturer
- Both videos → everyone advised to watch

### Conclusion Ask AI

- ETO → very **versatile** concept
- Very good results → if **adequately planned** + small details followed

- **Preoperative templating** → very, very, very important
- **Instruments** and backup → manage a lost item or major problem

# Choice of the approach

*(Comparing Posterior, Lateral and Anterior Approaches and Choosing Between Them)*

**Prof Dr Mohamed Abo Elsoud**

**Watch Original Lecture**

*Lecture transcript processed with AI assistance.*

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# AI detected corrections (Mostly bad transcripts)

## Queried - kept as spoken, but looks wrong

**Q1** (p. 177) As spoken; lateral approach lowers but does not abolish dislocation or femoral fracture; absolute claim overstated.

## Introduction: Determinants of Approach Choice Ask AI

- Original title **Choice of the Approach** → reworked at course organization
  - By Dr. Sherif Khaled and Dr. Mohamed Gabba
  - New title: **Literature and Registry Data Review**
- Difficult topic → careful assimilation of evidence required
- Generally held view, summarizable in ~five slides
  - Approach choice in **total hip arthroplasty (THA)** → highly controversial
  - THA highly successful → **regardless of surgical approach**
  - → Choice largely surgeon opinion and preference
- Determinants of choice
  - **Surgeon preference** → main determinant
  - Dislocation risk
  - Obesity
  - Previous incisions
  - Previously implanted hardware
- Each surgeon → favored approach → defends it

## Classical Advantages and Disadvantages of Each Approach Ask AI

### Posterior (Posterolateral) Approach Ask AI

Posterior

- Advantages
  - Easy
  - Not bloody
  - Very quick
  - **Abductor mechanism** not violated
  - **Extensile** → good femoral and acetabular exposure
    - → Extendable for revision surgery
- Disadvantage → **high dislocation rate**

### Direct Lateral Approach Ask AI

Lateral

- Advantages
  - Easy
  - Excellent **acetabulum** exposure
  - Dislocation rate → lower than posterior
- Main disadvantage → **postoperative limp**, Trendelenburg gait

### Direct Anterior Approach (DAA) Ask AI

DAA

- Newer approach → "younger sibling" of lateral approach
  - Described as modification of lateral approach
  - Enters hip from front
- Main advantage → **less muscle damage** → faster recovery
- Main disadvantage → **technically difficult** → more problems
- Associated **neuropraxia** → lateral femoral cutaneous nerve

### Classical View at a Glance Ask AI

| Approach | Advantages | Disadvantage |
|---|---|---|
| Posterior | Easy, not bloody, quick, abductors intact, extensile | High dislocation rate |
| Direct lateral | Easy, acetabular exposure, dislocation < posterior | Limp, Trendelenburg gait |
| Direct anterior | Less muscle damage → faster recovery | Technically difficult, LFCN neuropraxia |

- Open question → evidence behind each claim
  - Correct, incorrect, or weakly supported?

## Patient-Reported Outcomes Ask AI

- **Patient-reported outcomes (PROs)** → patient satisfaction with operation

### JBJS 2014 — Total Hip Replacement Ask AI

THA

- Journal of Bone and Joint Surgery (JBJS), 2014
- Slightly better PROs → **posterior approach**
- Attributed to **abductor weakness** with lateral approach

### JBJS 2019 — Hemiarthroplasty Ask AI

Hemi

- Fracture patients, elderly > 75 years
- Approach → **no apparent effect** on outcome
  - Elderly satisfied with hemiarthroplasty itself, regardless of approach
- Contrast: THA → posterior slightly better for PROs

## Functional Recovery: Direct Anterior vs Posterior Ask AI

### CORR 2018 RCT — 101 Patients Ask AI

DAA

- **Randomized controlled trial (RCT)**, Clinical Orthopaedics and Related Research (CORR)
- Smaller than larger series discussed before
  - ~2,000 and 42,000 patients → likely Swedish registry
- Comparison → **mini-posterior approach** vs anterior approach
- Findings
  - Slightly faster early recovery with DAA → up to first 2 months
  - PROs, radiographic parameters, complications → equal

### Journal of Arthroplasty 2017 RCT — 70 Patients Ask AI

DAA

- DAA vs posterior approach
- Outcome scores → no difference
- **Functional recovery** better with DAA
  - Putting on socks
  - Tying shoes
- **Hip flexion activities** better with DAA → at 6 weeks

- Flexion strength, by contrast
  - **Hip flexors weaker** with DAA → dissection close to them
  - Flexion activities still better → up to 6 weeks

### Meta-analysis 2018 — ~1,000 Patients Ask AI

DAA

- **Systematic review and meta-analysis,** Journal of Arthroplasty
- DAA patients
  - Fewer **narcotics**
  - Less pain
  - Better functional outcome
- Explanation → accompanying RCTs
  - Lower **CRP levels**
  - MRI and (probably cadaveric) studies → less **muscle edema** with DAA
    1. Less muscle affection
    2. Less tissue damage

### Meta-analysis 2015 Ask AI

DAA

- Second systematic review and meta-analysis
- Better **functional rehabilitation** → DAA
- Cost → much **longer surgical time** with DAA

## Gait and Trendelenburg Gait Ask AI

- Question: does direct lateral approach truly produce **Trendelenburg gait**?

### Cochrane Systematic Review 2010 Ask AI

Gait

- **Cochrane Database**; large number of patients (exact number not recalled)
- Trendelenburg gait, lateral vs posterior → **not significantly different**
- Dislocation rate, posterior vs lateral → not different
- Authors → **data insufficient** → no firm conclusion
  - Trendelenburg gait unsettled either way

### Gait Analysis RCT 2018 — 30 Patients Ask AI

Gait

- 10 anterior, 10 posterior, 10 lateral
- Lateral approach → higher **pelvic tilt**
  - Pelvis drops on opposite side + **ipsilateral trunk lean** → Trendelenburg gait
- Reported as affecting patients' outcomes (hedged)
- Measured → first 6–12 weeks

### Smartphone Accelerometer RCT 2020 Ask AI

Gait

- **Smartphone accelerometer**
  - Carried while walking → counts steps, records gait
- RCT, ~70 patients, DAA vs posterior
  - **Very minimal differences** → in favor of DAA
  - Limited → first month

**Exam Pearl**

- DAA functional advantage → early only: 2 months, 6 weeks, 1 month
- Hip flexors weaker after DAA, yet flexion activities better to 6 weeks
- Cochrane 2010 → Trendelenburg gait lateral vs posterior not significantly different
- Gait analysis 2018 → lateral approach: pelvic drop, ipsilateral trunk lean

## Dislocation Ask AI

- Key question → dislocation higher or lower → which approach better?

### Acta Orthopaedica 2014 — Hip Fracture Ask AI

Fracture

- Hip fracture patients > 75 years
- Posterior approach → markedly higher **risk of re-operation due to dislocation**
- Fracture patients → PROs good and similar across approaches
  - Yet posterior → much higher re-operation risk → dislocation
- Supporting papers → 2016; 2008 (Acta Orthopaedica)

### Lateral vs Posterior and Augmented Repair Ask AI

Posterior

- JBJS paper → **direct lateral approach**
  - → Dislocation risk lower than posterior
- Second paper, Acta Orthopaedica → **augmented posterior repair**
  - → Decreases dislocation rate/risk
  - Still does not reach anterolateral or lateral approach rates

### Swedish Hip Register 2019 — 156,000 Patients Ask AI

Posterior

- Two periods → 1999–2006 and 2007–2014
- Endpoints
  - Risk of re-operation due to dislocation
  - Risk of **all-cause re-operation**
- 1999–2006 → dislocations higher with posterior approach
- 2007–2014 → dislocation risk similar, lateral vs posterior
  - Attributed → **augmented posterior repair**
- Posterior approach → lower risk of all-cause re-operation
  - Dislocation set aside, all causes included

### 2016 Study — Posterior vs DAA Ask AI

DAA

- Dislocation risk, posterior vs DAA → probably **no difference**
  - Probable trade-off → more femoral complications with DAA

### 16,000-Patient Comparison Ask AI

Dislocation

- Posterior vs lateral vs DAA
- Posterior dislocation rate → 10 times **lateral approach**
- Posterior dislocation rate → 2 times **DAA**
- Anterior approaches → increased **femoral failure** risk (femoral fractures)
- Conclusion → **lateral approach safest**
  - No femoral fractures, no dislocations *[Q1 · As spoken; lateral approach lowers but does not abolish dislocation or femoral fracture; absolute claim overstated.]*

- → Lowest rate of mechanical complications

### Dislocation Evidence at a Glance Ask AI

| Study | Population / comparison | Dislocation finding |
|---|---|---|
| Acta Orthop 2014 | Hip fracture > 75 years | Posterior → higher re-operation for dislocation |
| JBJS | Lateral vs posterior | Lateral lower |
| Acta Orthop | Augmented posterior repair | Lower, still > anterolateral/lateral |
| Swedish Hip Register 2019 | 156,000; 1999–2006 | Posterior higher |
| Swedish Hip Register 2019 | 156,000; 2007–2014 | Lateral = posterior |
| 2016 study | Posterior vs DAA | Probably no difference |
| 16,000 patients | Posterior vs lateral vs DAA | Posterior 10× lateral, 2× DAA |

**Must-Know**

- Elderly hip fracture → posterior approach → higher re-operation for dislocation
- Augmented posterior repair → lowers posterior dislocation risk
- Swedish 2007–2014 → dislocation risk lateral = posterior
- Posterior approach → lower all-cause re-operation (156,000 patients)
- Anterior approaches → increased femoral failure risk

## Other Complications Ask AI

- Beyond dislocation → rates of other complications?

### Journal of Arthroplasty 2019 — Three Groups Ask AI

Lateral

- ~2,000 patients
- Lateral group
  - Longer **surgical time**
  - More general anesthesia
  - Longer hospital stay
  - Surprising → expected less
- Major and minor complications → **no difference** among 3 groups
  - Lateral, posterior, DAA

### RCT 2017 — 70 Patients Ask AI

DAA

- RCT, 70 patients
- Same findings
- **Minor adverse events** slightly higher → DAA
  - Cause → neuropraxia of **lateral femoral cutaneous nerve** (lateral cutaneous nerve of thigh)
  - Very **transient** → ~12 weeks

### Traction Table and Femoral Fracture 2017 Ask AI

DAA

- ~500 patients; **traction table** use vs non-use
- Endpoint → incidence of **femoral fractures** with DAA
- Traction table vs none → no difference
- Fractures in both groups → low rate, ~2.5%
- Fracture vs no-fracture patients → no difference, including **gender**
- Fracture patients → very small proportion

## Readmission Ask AI

- **Journal of Orthopaedics**, 2020 → readmission after DAA at 30 and 90 days
- Overall readmission rates → low
- **Obese patients** → higher risk of early wound complications → readmission
  - Superficial wound problem
  - **Hematoma formation**
  - Similar issues

## Long-Term Implant Survival and Revision Rates Ask AI

- Question → long-term **implant survival** better with any approach?

### Norwegian Registry Thesis 2008 Ask AI

Survival

- **Doctoral (PhD) thesis**, Norway, Norwegian Registry
- Lateral approach + trochanteric osteotomy → **best long-term survival**
  - i.e., **digastric trochanteric osteotomy** (explained by Dr. Hazem)
  - Probably → implant positioned very well
- Posterior approach → **highest revision rate** → dislocation
- Trochanteric osteotomy vs none → no difference in **abductor strength**
- Abductor strength improves up to 12 months
  - Marked improvement → first 6 months
  - Continued improvement → 6–12 months
  - → **Trendelenburg gait** improvement expected up to one year

### Swedish Registry 2019 — 38,000 Patients Ask AI

Survival

- Posterior vs lateral → **no difference**
  - Reoperation rate
  - Implant survival

### Swedish Registry 2017 — Mini Approaches Ask AI

Mini

- **Mini approaches** → increased risk
  - More mini approaches → higher reoperation rate
  - → Lower implant survival
- Probable cause → implant not in optimal position
  - Less-exposed (mini / direct anterior) approaches → lower survival

### Swedish vs American vs Australian Registries Ask AI

Registries

| | Sweden | United States / Australia |
|---|---|---|
| Commonest approach | Posterior (probable) | Posterior (probable) |
| DAA use | Essentially none | A proportion of surgeons |
| Revision rates | Better | Worse |
| 5- and 10-year implant survival | Better | Worse |

- **Observational** finding only
- Authors → possibly related to non-use of DAA in Sweden

## Learning Curve of the Direct Anterior Approach Ask AI

- Practical question → switch from lateral/posterior to DAA?
  - Motive → early recovery, current trends
- Instructive paper → highly experienced **posterior approach** surgeon switched to DAA
  - First 1,000 cases reviewed
- Published 2018
- Formal **training** + fellowship → proficiency → then 1,000 DAA cases
  - Comparable to Dr. Ahmed Samir
- **Total OR time** (setup + surgery) → 30% initial increase
- **Procedure time** (surgery only) → equal to posterior
  - → After first 400 cases
- Operative time shorter than posterior → after ~150 further cases (hedged)
- Total OR time equal to posterior → 900 cases
- **Early femoral fractures**
  - Femoral fractures and femoral complications → higher in early learning curve
- → **Very steep learning curve** (1,000 cases)

## Learning-Curve Milestones Ask AI

DAA

| Milestone | Cases |
|---|---|
| Total OR time, initial phase | 30% higher |
| Procedure time = posterior | First 400 |
| Operative time < posterior | ~150 further (hedged) |
| Total OR time = posterior | 900 |
| Early femoral fractures / complications | Higher in early curve |

**Pitfall**

- DAA → very steep learning curve, even after fellowship
- Procedure time matches posterior only after 400 cases
- Total OR time matches posterior only after 900 cases
- Femoral fractures higher early in learning curve

## Revision Surgery Ask AI

- **Extended trochanteric osteotomies** → revision surgery (also discussed by Dr. Hazem)
- **Review article** (not original research), 2015
- Revision → may need entirely different approach
  - Careful **preoperative planning** (Dr. Mahmoud Abdel Karim)
- Options beyond posterior or lateral
  - Possibly direct exposures
  - **Digastric trochanteric osteotomy** approach
  - **Extended trochanteric osteotomy** approach
- Author → trochanteric osteotomy (and possibly direct) approaches best
  - → Allow all problems to be solved

## Summary: Take-Home Message Ask AI

- **THA** → successful, largely unrelated to approach
  - Most cases → over 95% satisfied
  - Provided everything else done correctly
  - Surgeon uses approach he performs well
- Settled points
  - Elderly **hip fracture** → direct lateral approach → best results
    - Genuinely reduces dislocation rate
  - Posterior → careful repair of **capsule and short external rotators**
    - → Dislocation rate greatly reduced
    - Some authors → rate similar to direct lateral
- **Trendelenburg gait** → still a problem
  - Not definitively settled

  - Certainly more frequent with lateral approach
- DAA → **very long learning curve** → results close to posterior and lateral
- Better DAA functional recovery → **time-limited,** early period only
- **Obese** or revision patient → posterior approach preferred
- Best approach for each surgeon → the one he performs well

**High Yield**

- THA successful regardless of approach → over 95% satisfied
- Elderly hip fracture → direct lateral approach → lowest dislocation
- Posterior + capsule/short external rotator repair → dislocation near lateral
- Lateral approach → more Trendelenburg gait; DAA → early-only recovery gain
- Obese or revision patient → posterior approach

# Complex primary THA; Acetabular side

*(Acetabular Anatomy, Defect Classification and Reconstruction Options)*

**Prof Dr Ahmed Hazem**

**Watch Original Lecture**

*Lecture transcript processed with AI assistance.*

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## Lecture Objectives Ask AI

- Complex primary THA material → scattered across sources
- Objectives
  - What is a **complex primary** case?
  - How to **recognise** it → complex vs routine primary
  - How to evaluate it → management **algorithm**
- Aim → algorithm, or at least handling principles

## Acetabular Anatomy Ask AI

### Cave Analogy Ask AI

Anatomy

- Acetabulum ≈ **cave**
  - Two pillars → **anterior column** + posterior column
  - Columns join → **dome** (superior region)
  - Columns → anterior wall + **posterior wall**
  - Build sequence: anterior + posterior columns → walls close the front

### Terminology Ask AI

Anatomy

- Same structures → basis of classification throughout
- **Rim** → peripheral margin
- **Floor** = medial wall → middle of acetabulum
- Anterior wall + posterior wall + dome → surround acetabular entrance
- Terminology used throughout lecture

## Definition of Complex Primary Acetabulum Ask AI

- Native acetabulum → **not a true hemisphere;** irregularities
- Reaming → hemispherical → accepts hemispherical cup
  - Cemented or cementless
- Cannot be made hemispherical

→ **additional step** → reconstruct normal hemispherical shape

- Definition → **not clearly stated** in literature
- Working definition → any additional step to reconstruct a **hemispherical acetabulum**
  - Site: wall, column, dome or floor

→ **complex case**

**High Yield**

- Complex primary → any **additional step** to restore a hemispherical acetabulum
- Site of that step → wall, column, dome or floor
- No clear literature definition

## Scope Ask AI

Scope

- This lecture → **bone side** only
- Non-bone side (likely implant) → Dr. Mohamed Younis
- Complex case → not only a bone problem
- Many conditions → deficiency of walls, columns, rim, dome, any part

# Classification of Acetabular Deficiency Ask AI

## AAOS Classification Ask AI

AAOS

- **AAOS classification** → used in this lecture
  - Developed 1989; American Academy of Orthopaedic Surgeons
  - Defects → several categories
- **Type I — segmental deficiency** → loss of part of rim
  - Peripheral or central
  - Peripheral → anterior, posterior or superior segmental loss
  - Same peripheral/central subdivision → applied to cavitary type
- **Type II — cavitary deficiency** → loss of volume within bone
  - Rim preserved; inside enlarged with missing area
  - Central → part of **floor** missing; rim present → contained
  - Peripheral → anterior, superior or posterior
- Segmental and cavitary → two separate entities

| | Segmental (Type I) | Cavitary (Type II) |
|---|---|---|
| Containment | **Uncontained** — missing piece, not surrounded | **Contained** — enlarged, containment retained |
| Rim | Piece lost | Preserved |
| Defect | Rim segment | Loss of volume |
| Subtypes | Peripheral (anterior, posterior, superior) / central | Peripheral (anterior, superior, posterior) / central |

- **Type III — combined deficiency** → cavitary + segmental in one acetabulum
  - Example → superior rim loss + substance loss
  - Complex because

→ part of **rim** missing

- **Type IV and V** → atypical in complex primary
  - **Type IV — pelvic discontinuity** → acetabulum/pelvis in two unconnected pieces
    - Example → post-traumatic cases
  - **Type V — arthrodesis** → not deficiency but **excess bone**
    - Included → cannot be treated as routine

## Alternative Radiological Classification Ask AI

Radiologic

- Mentioned → so as not to upset its proponents
- Primarily a **revision** classification → osteolysis + deficiency of revision
  - Some surgeons → use in primary

  - Likely the Paprosky classification
- Simple; relies mainly on **radiology**
  - **Type 1** → minimal deficiency; rim intact, no migration; small cavitary defect
  - **Type 2** → defect with some superior involvement, not large
  - **Type 3** → **migrated** defect → cup migrated superiorly
- This lecture → keeps to AAOS

# Recognising a Complex Case Ask AI

## History Ask AI

History

- First clue → **patient's history**
- **Childhood onset** → "many pelvic operations when very young"

→ most likely **post-dysplastic** → deficiency in 100% of cases

  - Deficiency → superior or anterior
- **Post-traumatic** → usually deficient (in our country)
  - Exception → fracture fixed, and fixed well
  - Planned hip replacement → proper evaluation → recognised as complex, not simple
- **Rheumatological** → rheumatoid arthritis; possibly psoriatic arthritis
  - Usually **no deficiency**
  - Exceptions → steroid use / osteoporosis; long-standing disease

→ head migrates into acetabulum → **protrusio** (central cavitary defect)

- **Osteoarthritis** → usually routine primary
  - Long-standing + **osteoporosis** → usually not routine → check for deficiency

**Pitfall**

- **Post-dysplastic** hip → deficiency in 100% of cases
- Post-traumatic → deficient unless fracture well fixed
- Long-standing RA or **osteoporosis** → protrusio
- Long-standing OA + osteoporosis → not routine

## Examination Ask AI

Exam

- **Examination** → little information on deficiency

## Required Imaging Ask AI

Imaging

- **Radiology** → most important; tells deficiency or not
  - Detailed radiology → Dr. Mahmoud Abdel-Karim (afternoon)
- Radiographs → AP + lateral + **Judet views**
- Judet views added
  - Good for **columns** assessment (e.g. anterior column)
  - True AP / true lateral → very difficult
    - Deformed pelvis, **pelvic inclination**, other factors

→ all available views → complete picture

- **CT scan** → very important → defines defects precisely
  - Axial cuts + **coronal cuts**

**Clinical Pearl**

- Judet views → **columns** assessment
- Deformed pelvis → true AP / true lateral hard → add all views
- CT → axial + **coronal** cuts in every complex case

# Plain Radiograph Assessment Ask AI

## Reference Lines Ask AI

X-ray

- Familiar lines → related to **centre of the hip**

| Line | Structure | Normal | Abnormal |
|---|---|---|---|
| Anterior wall line | Anterior wall; continuous with inferior border of superior pubic ramus, crosses acetabulum | Just in front of (medial to) hip centre | Further → **anterior wall deficient** |
| Posterior wall line | Posterior wall | ≈ centre of head, ≈ lateral to it | Less → **posterior wall deficient** |
| Dome / sourcil | Roof | Starts + ends in same **sclerotic zone** | Open (e.g. positive acetabular index) → **dysplastic** |
| Teardrop | Floor | Follows floor | — |
| Ilio-pubic line | Anterior column | — | — |
| Ilio-ischial line | Posterior column | — | — |

- All lines → must be checked; assessment not always easy

## Templating the Hip Centre Ask AI

Template

- Confirm X-ray correctly taken → **horizontal reference line**
- → three basic reference lines
- Measure roof-to-reference distance → lines → where **roof** should be
- Midpoint between the two → mark area → measure distance
- Lines simulated on **contralateral side** → normal acetabular relationships
  - Detail → Dr. Mahmoud (afternoon)
- Point at intersection → ilio-ischial line + ilio-pubic line + **teardrop** (floor)
- Centre point → vertical + horizontal distances, simulated from contralateral side

→ **centre of hip**

= centre of cup

- Cup → placed at desired **inclination**
- Bone missing over templated cup → **deficient bone**

## Case Examples Ask AI

Cases

- Case 1 → reconstructed
  - Reason → **superior + posterior** deficiency

- Case 2 → deficient acetabulum or not?
  - Undecidable until lines drawn
  - Lines drawn → **not deficient** → routine primary
  - Components excellent; only slight **medialization** → good cup seating

## CT Assessment Ask AI

### Axial Cuts Ask AI CT

- **Axial cuts** → show column deficiency
  - Anterior / posterior wall → present or absent
  - Anterior and posterior columns
  - **Floor thickness** → how far to ream → adequate cup medialization

### Coronal Cuts Ask AI CT

- **Coronal cuts** → superior coverage
  - Cup uncovered > 50% → **reconstruction** needed
- Basic principles of radiological assessment

### 3D CT Ask AI CT

- **3D CT** → better idea of defect
- Example
  - Anterior view → good anterior column; protruding **anterior osteophytes**
  - Lateral view → **superior dome** deficiency
  - Inferior / posterior view → good posterior column + good floor

## Goals of Acetabular Reconstruction Ask AI

- **Short term** → reduce **joint reaction forces**
  - High hip centre of rotation → normal position
  - Normal **biomechanics** restored

→ very good long-term results; less wear and loosening

- **Long term** requirements
  - Good **fixation** (cemented or cementless) → good host-bone contact
  - Adequate **cup positioning**
  - Rigid fixation of cup → **host bone**

## Cavitary Defects — Management Ask AI

### Small Cavitary Defect Ask AI Cavitary

- **Small cavitary defect** → ignore
- Case: **posterior wall + posterior column** fracture, very well fixed → small residual defect
  - Cup ~1 mm larger
  - That alone → very good outcome
- Anterior approach case → small cavitary defect
  - First → confirm **no anterior deficiency** on X-ray or CT

  - Then → proceed anteriorly → place cup

### Large Cavitary Defect Ask AI

Cavitary

- **Larger cavitary defect** → e.g. protrusio
  - **Morselized graft** (bone chips) → autograft or allograft
  - Pieces ≈ 3–5 mm
  - Impact → **impaction bone grafting** → good support
  - Condition → cup contact with ≥ 50% **host bone** → good fixation

### Extensive Cavitary Defect Ask AI

Cavitary

- Additional devices
  - Cup with **external flanges** → prevents cup migrating onto graft
  - More **screws**
  - Devices
    - **Müller ring** → graft-protecting device
      - Prevents excessive graft extrusion
      - Protects graft until **incorporation**
- Most important → 50% of **porous surface** against host bone

**Must-Know**

- Small cavitary → **ignore**
- Large cavitary → **impaction bone grafting** (3–5 mm chips)
- Extensive → flanges, screws, **Müller ring**
- Always → 50% of porous surface on **host bone**

## Segmental Defects — Management Ask AI

- **Segmental defect** = uncontained → assess location + size

### Location Ask AI

Segmental

| Segment | Role | If deficient |
|---|---|---|
| Anterior / inferior | Good rim still surrounds acetabulum | Tolerated → may be ignored |
| Superior / posterior | Main cup support; intact → **press-fit** / stable fixation | Cannot be relied on → **acetabular augmentation**; must be reconstructed |

- Superior / posterior defect → never neglected; region supports cup
  - No fixation without augmentation
  - Different reconstruction methods for this region
- Anterior / inferior ignored → only with very good quality of remaining **supporting bone**

### Reconstruction Methods Ask AI

Segmental

- Methods
  - **Structural allograft**
  - Structural graft fixed with **screws**
  - **Augments**

- Key principle → **uncontained** → **contained**
- Approach used in practice

**Exam Pearl**

- **Superior + posterior** segments → main cup support → must reconstruct
- Anterior + inferior → may ignore, if remaining bone good
- Segmental principle → **uncontained** → **contained**

## AAOS Types III–V — Management Ask AI

### Combined Defects Ask AI

Combined

- **Combined defects** → uncommon in primary; more in revision
- Each defect separately

→ segmental (uncontained) → **contained**

→ then cavitary defect

### Pelvic Discontinuity Ask AI

Discont.

- **Pelvic discontinuity** → un-united / non-united acetabular fractures
  - Very common post-traumatic
- Principle → **unstable** → **stable** construct
- Fix discontinuity first → stable → then cup or **ring/cage**
  - No cup straight away

### Hip Fusion Ask AI

Fusion

- **Hip fusion** → not bone deficiency → problem is **excess bone**

## Summary — Management Algorithm Ask AI

- Complex primary → not precisely defined → **additional step** to reconstruct acetabulum
- Classification → **AAOS**
- Evaluation → history, examination, **radiology**
- Radiographic evaluation → good X-rays + axial and coronal **CT cuts**

| Defect | Management |
|---|---|
| Cavitary, small | Ignore |
| Cavitary, large | **Impaction grafting** |
| Segmental | Structural graft / augment; anterior + inferior → may ignore; superior + posterior → must reconstruct |
| Combined | Segmental first → convert to cavitary → then cavitary defect |
| Pelvic discontinuity | Stabilize columns first → then treat as segmental or cavitary |

# Complex primary THA; Femoral side

*(Femoral Deformity, Retained Hardware and Reconstruction Strategies)*

**Prof Dr Mohamed Gobba**

**Watch Original Lecture**

*Lecture transcript processed with AI assistance.*

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## Complex Primary THA: The Mindset Ask AI

- Opening case: condition for eight months; radiograph at decision to operate
- Clearly **not a normal femur** → specific mindset required
  - Question: how to get this patient through surgery?
- Particular features → think ahead → prepare additional equipment
- Additional steps not used in routine **primary total hip**
- **"Complex"** = not an ordinary total hip
- Three commonest femoral-side problems
  - **Failed intertrochanteric fixation**
  - Hip **fusion (arthrodesis)** → patient wants conversion to total hip
  - **DDH**

## Failed Intertrochanteric Fixation: Preoperative Planning Ask AI

### Broken Hardware Ask AI

ITF

- Many challenges; first → **broken hardware**
- On reaching implant → remove hardware first
  - Some operations: removing construct alone → two hours
  - Prolonged removal → surgeon did not prepare well
- Plan exactly in advance how hardware will be removed
- Anticipated problems → plan before theatre; be well prepared
- Proper **preoperative planning** mandatory

### Imaging Ask AI

ITF

- **AP X-ray** alone not enough
  - Add a traction view
  - Add **CT**
- CT → location of **united** vs ununited fracture parts
  - → changes type of implant used
- Prepare all before surgery; no intraoperative discovery

### Stability Ask AI

ITF

- Greater **stability** than usual required in this situation
  - Option giving more stability → **dual mobility**
  - Not simply the standard **28 mm head** used routinely

### Bone Quality and Defects Ask AI

ITF

- **Bone quality** on films → imagine bone left after **plate** removal
- **Defects**: present? management? **graft** needed?
  - All answered before going in
- Fracture may be **united** → healed in malunited position

### Canal Block Ask AI

ITF

- **Proximal femur**: pieces of **cortical bone** may block the canal
  - Equipment set → team prepares **high-speed burr** to remove obstructions
  - Otherwise → stem **rasps** cannot be introduced
- Rasps designed to remove hard **cancellous bone,** not cortical bone
- Rasp impacted hard against cortical bone → femoral **fracture**

**Pitfall**

- Cortical pieces blocking proximal canal → rasp impaction → femoral fracture
- Prepare high-speed burr in equipment set

### Infection Ask AI

ITF

- **Exclude infection** before taking patient to theatre
- Details and steps: Dr. Mahmoud Abdel Karim, **PGI** lecture

### Hardware Removal Set Ask AI

ITF

- Complete **instrument tray** solely for removing screws, including broken ones
  - Many surgeons unaware it exists
- No tray → no entry into **complex primary** case
  - Unavailable → **postpone the patient**
- Otherwise → two hours removing plate alone
  - → full concentration lost for rest of operation
- Hardware removal = **preliminary step** leading into main procedure

**Must-Know**

- Hardware removal set unavailable → postpone the patient
- Hardware removal = preliminary step, not the operation

## Failed Intertrochanteric Fixation: Intraoperative Technique Ask AI

### Long Stem and Screw Holes Ask AI

ITF

- **Long-stem prosthesis** → must **bypass the screw holes**
  - Bypass distance → **two cortical diameters**
- Method
  - Most distal screw hole → draw line → measure **canal diameter** at that level
  - Stem passes beyond by `2` cortical diameters, e.g. `4 cm`
- Purpose → avoid **stress riser**
  - Stem tip at level of hole → femur fractures

**High Yield**

- Long stem bypasses most distal screw hole by two cortical diameters
- Stem tip at screw-hole level → stress riser → fracture

### Dislocation Ask AI ITF

- Take time → adequate release before dislocating
- Direct dislocation → **osteoporotic bone** fractures
- Fracture = last thing wanted in this operation
  - Surprises already expected on entry and **canal preparation**
  - Goal → finish case safely

### DHS in Situ at Dislocation Ask AI ITF

- Dilemma: remove **lag screw** and plate first, or leave them?
- Lag screw **cut out** and impinging on **acetabular dome** → leave it too?
- Planned decision → written in advance at **planning meeting**

| Situation | Hardware before dislocation |
|---|---|
| Lag screw cut out | Remove plate and screws first, then begin |
| Femoral head badly destroyed or ununited, hardware in place | Remove plate and screws first, then begin |
| Lag screw in good position | Keep in → facilitates rotating femoral head |

### Canal Block and Soft-Tissue Repair Ask AI ITF

- **Canal block** (as above) → be prepared
- Have **Ethibond** or FiberWire available
  - Greater trochanter often involved more than expected
  - → repair of capsule and **gluteus medius**
  - = main effort at this stage

## Conversion After Failed Fixation: Evidence Ask AI

- Studies compare **total hips after DHS** vs after PFN
  - Conversion after PFN → **more complications** than after DHS
  - Converting PFN → expect somewhat higher complication rate
- Other studies → **worse outcomes** after intramedullary (IM) devices
- Above = minimum preparation for **primary total hip after failed intertrochanteric fixation**

## Arthrodesis Conversion: Indications Ask AI

- **Arthrodesis** conversion: fused hip → total hip
- Very uncommon in our practice; not routine
- **Indications**
  - **Mechanical back pain** → e.g. fused at age 11, now cannot walk from back pain
  - **Generalized loss of function** → wants to work and be active
  - **Failed fusion**
  - **Shortening**
  - **Pain in adjacent joints**
  - **Social problems** → e.g. girls of marriageable age cannot live normally

- All factors weighed; decision neither easy nor simple

## Arthrodesis Conversion: Technical Challenges Ask AI

### Landmarks and Hardware Ask AI

Fusion

- Identify **anatomical landmarks**
  - Normal hip → evident
  - Here → **distorted anatomy,** especially after previous surgery
- **Remove the hardware**

### Fusion Takedown and Neck Cut Ask AI

Fusion

- Takedown **osteotomy** → **intraoperative fluoroscopy** for exact position
- Anatomy very distorted, especially after hardware removal
  - → fluoroscopy when making the **neck cut**

### Acetabular Reaming Ask AI

Fusion

- Choose **center of rotation** → begin **reaming** there
  - **Smallest reamer** first → best estimate
  - Usually a marker at presumed center of rotation
  - Identify anterior and posterior columns → go between them
  - Ream until seated in bone → leave reamer → **image intensifier** shot
  - Accept, or too high / too medial → correct → continue enlarging
- Never ream, insert **cup,** then discover wrong position on X-ray

### Soft-Tissue Release Ask AI

Fusion

- Consider iliopsoas **tenotomy**
  - Muscles not moving → all **fibrosed** → motion problems

### Femoral Component Ask AI

Fusion

- Usually **porous-coated** stem → metaphyseal fixation
- Must **bypass the stress risers**
  - Previous plate
  - Previous osteotomy

### Abductors Ask AI

Fusion

- **Abductors** = key problem in arthrodesis → assess from beginning
- Completely replaced by white (fibrous/fatty) tissue → **transfer** options
  - Vastus lateralis transfer
  - **Gluteus maximus** transfer
  - Details outside this lecture
- Deficient even after reconstruction → future **instability**
  - Plan from start → **dual mobility** or constrained implant

**High Yield**

- Abductors = key problem in arthrodesis conversion
- Deficient after reconstruction → dual mobility or constrained implant

## Arthrodesis Conversion: Outcomes and Counselling Ask AI

- Evidence: two or three papers → same conclusion
- Total hip groups compared by **functional scores**

| Group | Functional outcome |
|---|---|
| Primary THA | High scores; patients very satisfied |
| THA after revision | Somewhat lower |
| THA after fusion | Not satisfied, even after total hip |

- Conclusion → conversion patients
  - **Inferior functional outcomes**
  - Higher complication rates
- Most important factor → **patient expectations**
  - E.g. mild back pain, simple analgesics, years unchanged
  - Symptoms neither progressing nor improving
  - Imagines a magical state → walking normally like ordinary people
- Counsel patient → does he really need the total hip?
- **Systematic review** → complication rate more than 54%
- Lower patient expectations → very important

**Clinical Pearl**

- Fusion conversion → complication rate above 54% (systematic review)
- Lower patient expectations before offering conversion

## Total Hip Arthroplasty in DDH Ask AI

- Detailed coverage → lecture by Dr. Mohamed Younes; brief note here
- **Indications for surgery**
  - **Pain**
  - Interference with **ADL**
- Dislocated since birth → abnormal **hip anatomy**
  - **Excessive femoral anteversion**
  - Femur: very high anteversion + very **small canal**
- → **modular stem** → femoral version set → stem implantable

## Subtrochanteric (Rotational/Shortening) Osteotomy Ask AI

### Trial Before Osteotomy Ask AI

DDH

- After exposure → **broaching** femur to maximal size
- **Trial prosthesis** inserted → fits well
- Fit confirmed, but 3–4 cm still to shorten

  - → **subtrochanteric shortening osteotomy**
- After this step → remove trial

### Osteotomy Ask AI

DDH

- Level: below **lesser trochanter** (subtrochanteric)
- Configuration: **chevron** or transverse; many methods
- Essentials
  - Osteotomy
  - Iliopsoas tenotomy
  - Separate the fragments
- No femoral resection yet → amount determined after this step

### Determining Resection Length Ask AI

DDH

- Trial prosthesis in → **assistant** applies traction on femur
  - → tension → fragments in final position
- Mark **overlap** with diathermy = excess segment
- Excess removed → femur brought together
- Result → best tension
  - Avoid **over-tensioning** or under-tensioning
  - → desired **soft-tissue tension**
- Images: before vs after operation

## Opening Case Revisited Ask AI

- End of lecture → revisit opening X-ray

### Fixation Level Ask AI

Case

- Options
  - **Metaphyseal** fixation
  - Metaphyseal-diaphyseal fixation
  - Diaphyseal-diaphyseal fixation
- Region very badly damaged
- Company contacted → **long stem** prepared = first planned point

### Neck Cut and Second Osteotomy Ask AI

Case

- Second point → site of **neck osteotomy**
- Usual neck cut? → answered by **templating**
- Rasp in canal → would exit through the **trochanter**
  - Trochanter became medial → overhanging the canal
  - Not acceptable → cannot start with rasp in this area
- Decided at planning → **second osteotomy** before starting
- Second osteotomy → window opened → canal directly in front
- After prosthesis insertion → **fragment fixation**
  - Fragment shaped at an angle → fits onto prosthesis

## Conclusion Ask AI

- **Complex primary THA** = big topic → never enter unprepared
- Self-check before operating
  - Required **skills**
  - Necessary **knowledge**
  - Available equipment and anaesthesia support
- Know your **limits**
- Closing quote
  - "A good surgeon knows how to operate."
  - "A better surgeon knows when to operate."
  - "And the best surgeon knows **when not to operate**."

# THA after dysplasia; Soft tissue releases

*(Classification of Dysplasia and Soft-Tissue Release Strategies)*

**Prof Dr Mohamed Youness**

**Watch Original Lecture**

*Lecture transcript processed with AI assistance.*

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# AI detected corrections (Mostly bad transcripts)

## Queried - kept as spoken, but looks wrong

**Q1** (p. 208) Transcript garbled here; a high hip centre usually shortens the abductor lever arm, which raises joint forces

**Q2** (p. 208) Same point: a high hip centre usually shortens the abductor lever arm, it does not lengthen it

## THA in the Dysplastic Hip — The Challenge Ask AI

- Key questions → difficulty → can it be overcome → does patient benefit?
  - Comprehensive problem → **femur** and acetabulum
  - Ultimate aim → patient benefit
  - Patient → young, **high-demand**
- **Soft tissue** problems → mostly present; must be addressed
  - Wants active life, long-term use of joint
  - Wants no **early revision**

## Pathological Anatomy on the Radiograph Ask AI

- Single X-ray → summarizes all problems of the condition
  - Acetabulum → not **normal anatomy**; not dealt with easily
  - Femur–acetabulum relation → not that of a normal joint
  - Problems expected on both sides
    - **Dysplastic acetabulum** → abnormal appearance
    - Femur → hypoplastic, small
- **Neck** → shortened, not in true orientation
  - Intra-op → don't expect to find the neck in front of you
- **Medullary canal** → small, narrow
  - Cortex → probably very thin
  - Marked **anteversion**
  - Trochanter → not in normal position

## Classification Ask AI

- First step → **stratify** type of dysplasia
  - First classification → Hartofilakidis

### Hartofilakidis Classification Ask AI

- **Hartofilakidis** → A, B, C by degree of dislocation

| Type | Description |
|---|---|
| A | Subluxation (dysplasia) |
| B | Low dislocation |
| C | High dislocation |

### Crowe Classification Ask AI

- **Crowe classification** → most widely used
- Four groups → by amount of head dislocated outside acetabular socket

| Group | Head outside acetabulum |
|---|---|
| 1 | < 50% |
| 2 | More (approx. 50–75%) |
| 3 | Up to approx. 100% |
| 4 | Femur high up; no relation to acetabular socket |

- **Group 4** → worst

**High Yield**
- Crowe → most widely used classification
- Basis → % of head outside acetabular socket
- Group 4 → femur high, no relation to socket

### Radiographic Examples Ask AI

- X-ray series → simpler → more complex degrees
- **Crowe IV** → femur rides high in pelvis, away from acetabular socket
- Further X-ray examples

## Indications for Surgery Ask AI

- **When to operate** → very important point
  - Typical → patient in forties or fifties with dysplasia
  - Questions → recent history; operate on every **acetabular dysplasia**?
- Tell patient today → operation + joint replacement?
  - Ordinary complaint vs incidental **DDH** / acetabular dysplasia
  - Surgery → not the only option
  - Management → very **conservative**
- Reason → considerable technical problems and complications
  - Required → **pain** + disability
    - Ambulation difficulty
    - Major functional problems
  - Many problems → still may not **justify** surgery
  - **Indication** → must be very clear at assessment
- Goal → patient benefits 100%
  - Hip replacement → good results
- **Total hip** for acetabular dysplasia → only when benefit is 100%
  - Must confirm significant level of **disability**
  - Gait problem

**Must-Know**
- Dysplasia alone → not an indication
- Indication → pain + significant disability
- Technical problems, complications → considerable

## Femoral and Acetabular Deformity Ask AI

- Problems in brief → both sides

| Femoral side | Acetabular side |
|---|---|
| Head → deformed, displaced | Deficiency |
| Neck → shortened | Oblong, vertically oriented |
| Trochanter → posterior | False acetabulum |
| Medullary canal → narrow | Deficient bone stock for cup |

- Cup in **false acetabulum** vs true → makes a difference
  - Restore **biomechanics** → patient outcome depends on it

**Pitfall**
- Open, start preparing → cup placed in false acetabulum
- Correct target → true acetabulum

## Soft Tissue Problems Ask AI

- **Soft tissue** → most important issue
  - Forces **extra procedures** during THA in dysplastic hip
- All muscle groups around hip → involved
  - **Abductors** → weak, contracted
  - Rectus femoris
  - Iliopsoas
  - Hamstrings
- Cause → long-standing shortened state in dislocated position
- **Capsule** → elongated, thickened
  - Part interposed where head should sit
- Knowing these points → be prepared
- Unknown → unprepared → problems (Dr. Ahmed, Dr. Gouda)
- Otherwise → unnecessary steps or poor results
  - You cannot see what you do not know

## Goals of Reconstruction Ask AI

- **Assessment** → very important
  - Expected → shortening
  - Restore normal anatomy → regain length → plan compensation
- Regain length → without stretching muscles
  - **Neurovascular structures** → at risk
- Principles known before surgery → goals for good results

### Acetabular Side Ask AI

Acetabulum

- First goal → restore **center of rotation**
- Large difference → early results + **long-lasting** prostheses
- Normal center reached → **cup** stable initially and long term
- Reconstruct acetabular bed
- → **stability** in correct position

### Femoral Side Ask AI

Femur

- After acetabular side → femoral side
- Femoral side restores
  - **Offset**
  - Abductor mechanism
  - Limb length

**Must-Know**

- Restore center of rotation
- Stable cup → initially and long term
- Femoral side → offset + abductor mechanism
- Femoral side → limb length

## Preoperative Planning and Special Sets Ask AI

- **Preparation and planning** → avoid unnecessary steps
  - Standard **primary** set → cannot be relied on
  - Special set → fits anatomy and deformity
  - Anticipate problems → **special implants** + special sets
- Goal → don't make the difficult case harder
  - **Surgical technique** → crucial for good results
- **Special sets** → used
- Sound **principles** → know why each technique, why each step
  - Practical steps → release small muscles → bring acetabulum to its place
- Depends on → acetabulum location + plan for **acetabular reconstruction**
  - **Bicanic** (2014) → current concepts in acetabular and femoral reconstruction
- Different concepts → reach **four pillars** → good results

## Acetabular Reconstruction Ask AI

### High Hip Center Ask AI

Acetabulum

- Early studies → cup placed high, before tackling true acetabulum
- Reason → **contracture** + soft tissue problems around hip
  - Cup → slightly high center
  - Femur reconstructed onto it → good results reported
- Later study → **stresses** and forces on acetabular component
- Long-term results → not as that school claimed

  - Higher center → ↑ **lever arm** of surrounding muscles (abductors) *[Q1 · Transcript garbled here; a high hip centre usually shortens the abductor lever arm, which raises joint forces]*
  - → ↑ stresses on construct → failure

**Exam Pearl**

- High hip center → ↑ abductor lever arm *[Q2 · Same point: a high hip centre usually shortens the abductor lever arm, it does not lengthen it]*
- → ↑ stresses on construct → failure

### Placing the Cup in the True Acetabulum Ask AI

- Options → cup in **true position**

| Technique | Key point |
|---|---|
| Bulk graft | Cup in true position |
| Partly uncovered cup | Up to 30–40% uncovered, within true acetabulum → osseointegration |
| Augments | — |
| Different shells | Reconstruction |
| Cotyloplasty | Cup placed medially |

**High Yield**

- Up to 30–40% of cup → may be left uncovered
- Condition → cup within true acetabulum

## Soft Tissue Release and Balancing Ask AI

- Study (2012) → **soft tissue release** + balancing with acetabulum in true position
- **Stepwise technique**
  - Not everything around hip at once
  - Assess release needed → reduced, stable hip
- Release sequence → stop once result good
  - Step 1 → adductors (probable) + part of **superior attachment of gluteus maximus** at greater trochanter
  - Step 2 → **iliopsoas**, part of rectus, sartorius from ASIS
  - Step 3 → **piriformis** + hamstrings

**Clinical Pearl**

- Release stepwise, not all at once
- Stop once hip reduced, stable

## Femoral Reconstruction Ask AI

- Then → **femoral reconstruction techniques**
  - **Distractor**
  - **Trochanteric osteotomy** → different modalities
  - **Subtrochanteric osteotomy**
- Shortening → e.g. Dr. Gouda's **subtrochanteric osteotomy**
  - **Trial** in position → determine segment to resect → compensate length
  - Then modular components placed
- Segment → **modular fixation**
- **Distal femoral osteotomies** + shortening

## Key Message and Conclusion Ask AI

- Second key point → know **pathological anatomy** + expected problems
  - → prepare **special instruments** + special sets
- → avoid many technical problems
  - Correct planning → good results → patient benefits

# Dysplastic hip

*(A Case-Based Approach to the Dysplastic Hip)*

**Prof Dr Fouad Zamel**

**Watch Original Lecture**

*Lecture transcript processed with AI assistance.*

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# AI detected corrections (Mostly bad transcripts)

## Corrected - the text was changed; the transcript said

**C1** (p. 220) “white line” → “radiolucent (white) line”

# Mildly Dysplastic Hip: Case 1 Ask AI

## Context Ask AI

Case 1

- Pathology, **pathological anatomy**, acetabular reconstruction → covered earlier by Dr. Mohamed Younis
  - This session → case-based discussion
  - First case → simple, from private practice

## Presentation Ask AI

Case 1

- Woman, presenting at 51–52 years
  - Complaint → **limp**
- Early presentation; very active, horse rider; hip pain
  - Dysplastic or not? → little **dysplasia**, visible on full radiographic views
- Diagnosis → hip arthritis secondary to **mild dysplasia**
  - Mild dysplasia → mechanics not much disturbed, despite long-standing course

## Planning Ask AI

Case 1

- Plan → **total hip replacement**
  - Little **subluxation** → some difficulty possible
- Question raised: **CT** needed?
  - First step → list risks expected in a dysplastic case
  - Mildly dysplastic acetabulum → little difficulty expected
- Opposite hip → replaced 1 year earlier
- Second hip replaced → no problems
  - Acetabulum → essentially **normal**
  - Uncovering / lateralization → arguable, minimal

# Principles of Reconstruction Ask AI

## Center of Rotation and Bone Stock Ask AI

- Main principles → repeated throughout (also covered by Dr. Younis)
- Main mechanical principle → cup at **anatomical (true) center of rotation**
  - Cup at center of rotation → work with bone present there
  - Best bone stock → at true **center of rotation**, contrary to expectation
  - Holds despite region being unloaded
- Bone and acetabulum never fully loaded → still best **bone stock**
- Higher position → slightly more bone, not much more
  - Best bone around center of rotation → use it for any cup type

## Sequence of Goals Ask AI

- Reconstruction order:
  - Restore **center of rotation** → place cup there first
  - Secure cup **fixation**
  - Finally → restore **global offset**
- Global offset → affects **leg length** and offset → affects **stability**

  - These → goals of hip replacement in dysplasia
  - First goal → **contained cup**
  - Good, contained cup → good hip, good patient outcome
- Achieved in Case 1 → not a problem
- Not always this straightforward across the spectrum

**High Yield**
- Cup at anatomical (true) center of rotation
- Best bone stock → at true center of rotation, though unloaded
- Order: center of rotation → cup fixation → global offset
- Global offset → leg length + offset → stability
- First goal → contained cup

## Spectrum of Dysplasia Ask AI

- Wide range of radiographic appearances
- Range of dysplastic hips → different grades
  - Classification → not discussed here
  - Dysplasia starts → changes in **acetabulum** and **proximal femur**, hip still at center of rotation
- Increasing severity → head rides higher → additional problems
  - End stage → head completely dislocated, never reduced
- What not to do → cup in high **false acetabulum** (example case, discussed at end)
  - Correct site for cup → **true acetabulum**

**Pitfall**
- Cup placed in high false acetabulum → wrong
- Cup belongs in true acetabulum

## Subluxated Dysplastic Hip: Case 2 Ask AI

### Radiographic Findings Ask AI

Case 2

- Change in **proximal femur**
  - **Subluxation** → **Shenton's line** broken
- Bone deficient inferiorly
  - Subluxation → present, not major

### Proximal Femur in Dysplasia Ask AI

- Proximal femur findings:
  - Increased **anteversion**
  - **Narrow canal**
- Proximal femoral anatomy → always abnormal in dysplasia
  - Common misconception → problem acetabular only
  - Problem → proximal femur as well
- Clinically → marked **internal rotation** (anteversion)

- Intraoperatively → **excessive anteversion**, consistently found
  - Stem positioned by native femoral version → stem in excessive anteversion
  - **Small canal** → confirmed

**Pitfall**

- Dysplastic femur → excessive anteversion, always
- Stem positioned by native femoral version → stem in excessive anteversion
- Problem not only acetabular → proximal femur abnormal too

### Iliopsoas and Capsule Ask AI

- Capsule in dysplastic hip → not short
  - Two parts; one part adherent above acetabulum → needs **release**
- Plan → **total hip**; type to be chosen

### Cup Sizing Ask AI

Case 2

- Need for special stems, special cups, small cup?
  - Cup size determined by → **anteroposterior dimension**
  - Relevant part → lower / anterior wall, not upper part

### Acetabular Reconstruction Ask AI

Case 2

- Young patient → **cemented reconstruction**, acetabular side
  - Reason → very small cementless components unavailable
  - Not because cemented cups work better
  - Preferred bearings → **highly cross-linked polyethylene** or **ceramic**
- Other techniques acceptable; shown later
- **Bone graft** from small femoral head → ample bone
- Good-quality graft piece
  - Cemented reconstruction → speaker's former technique; different technique now
  - Performed 12–14 years ago
  - No graft **resorption**; graft lies posteriorly
- Femoral canal → very narrow

## High Dislocated Hip: Case 3 Ask AI

### Acetabular Side Ask AI

Case 3

- High dislocated hip with **false acetabulum**
  - Plan → **total hip**
- Planning → list expected problems, divide them
  - Main problems → acetabular side
- Identifying **true acetabulum**:
  - False acetabulum vs lower part
  - Lower part → **anterior wall**, posterior wall behind
  - Anterior dimension → not narrow
  - Small cup → not needed
- **Mushroom-shaped** femoral head → large anterior dimension

  - Large anterior dimension → larger cup required
  - Questions: reduction and component problems?
- This bone stock → accepts larger cup
  - Cup here → good size, any design

### Femoral Side Ask AI

Case 3

- **Shortening osteotomy** → acceptable option
  - Technique must be defined
- Different procedures; amount of shortening → per surgeon
  - Special stems?
- Options:
  - Direct shortening
  - Extensive shortening
  - **Cemented stem** or **cementless stem**
- Choice → technique best known, best results in surgeon's hands
- This case → **femoral head** harvested; large amount of bone

### Reconstruction Ask AI

Case 3

- Graft fixed with **screws** superiorly; **cementless cup**
  - Operated 14–15 years ago; doing well

## Completely Dislocated Hip: Case 4 Ask AI

### Problems Ask AI

Case 4

- Completely dislocated hip
  - **Leg-length discrepancy**
  - Very large displacement → **sciatic nerve** problems
- Capsule origin → guide to true acetabulum
- Capsule position → site of **false articulation**
  - False socket → very small, small diameter

### Locating the True Acetabulum Ask AI

- True acetabulum → difficult to find
  - Follow **ligamentum teres**
  - Intraoperative **fluoroscopy** → easy alternative
- Walls → flat, **saddle-shaped**
  - Anterior and posterior columns → suitable position
  - Columns obscured → covered by bone / soft tissue
- Area closed off by bone overlying columns
  - True acetabulum → very, very small
  - Do not ream as in a normal hip
- Previous case → large acetabulum, because loaded
  - This case → very small acetabulum
- Walls → not clear
  - Landmarks → follow **sciatic notch** and **iliopsoas**
  - True acetabulum → between them

  - Overlying soft tissue → resembles exostosis / protrusion
- Not a true exostosis, not large → marks route to true acetabulum

| | Loaded acetabulum (Case [illegible]) | Unloaded true acetabulum (Case [illegible]) |
|---|---|---|
| Size | Large | Very, very small |
| Walls | Anterior and posterior walls identifiable | Not clear; flat, saddle-shaped |
| Reaming | Standard | Not as in a normal hip |
| Landmarks | Lower part, anterior wall | Ligamentum teres; sciatic notch and iliopsoas; fluoroscopy |

### Floor Thickness Ask AI

Case 4

- Floor **thickness** → speaker's technical trick:
  - Small **drill** through the area only
  - Drilling → estimate of thickness
- Thickness cannot be palpated → drilling easy, estimates thickness in centimeters
- Reconstruction / **grafting** → required
- Patient from **Yemen;** operated many years ago

**Clinical Pearl**

- True acetabulum → follow ligamentum teres
- Alternative → intraoperative fluoroscopy
- Between sciatic notch and iliopsoas → true acetabulum
- Floor thickness → small drill hole

## Sciatic Nerve and Lengthening Ask AI

- Cup at true acetabulum + reduction of **femoral head** → risk
  - Risk → **sciatic nerve** injury, at reduction
  - Lengthening > 3.5–4 cm → risk of sciatic nerve injury
- Intraoperative check:
  - **Palpate sciatic nerve** → feel tension with leg pulled down
  - Nerve much tighter at reduction → problem

**Must-Know**

- Lengthening > 3.5–4 cm → sciatic nerve injury risk
- Intraoperative test → palpate sciatic nerve, leg pulled down
- Nerve much tighter on reduction → problem

## Cemented Reconstruction with Bulk Autograft Ask AI

- Technique → **meticulous cemented technique**
- Combined with **shortening osteotomy**

  - Osteotomy + reconstruction → **bulk femoral-head autograft** fixation
  - Learned from **Dr. Maher** many years ago; still favoured, useful
  - No autograft reconstruction revised by speaker yet
  - Some cases possibly revised by others
- No graft case revised since return home 20 years ago
  - First revision → now
  - Graft reconstruction → works well; this patient > 8–9 years out
- Follow-up → **nine-year follow-up**
  - **Cemented reconstruction** → works well
  - **Standard poly, 28 mm heads** → work
  - Prerequisite → **meticulous technique**

## Subtrochanteric Shortening: Case 5 Ask AI

- Woman, 36 years; current hip problems
  - Hip reconstructed; fixation with 6 screws (6-hole DCP)
  - Technique → **rasp (broach)** in proximal femur first
- Rasp inserted before femoral **osteotomy** → then **subtrochanteric shortening**
- Femur back in place → reduce → measure fragment **overlap** → resect that amount
  - Some techniques → trial stage before cutting
  - Speaker → no trial stage

**Exam Pearl**

- Rasp in proximal femur before osteotomy
- Subtrochanteric shortening → reduce, measure overlap, resect overlap

## Leg-Length Discrepancy and Pelvic Tilt Ask AI

- Additional problem beside **leg-length discrepancy** → **pelvic tilt**
  - **Long-standing pelvic tilt** → usually not easily corrected
  - Patients struggle to correct tilt
  - Even after shortening → often end up with more length

## Bilateral Dysplasia: Case 6 Ask AI

### History Ask AI

Case 6

- Woman from **Port Said**; both hips affected
- Left hip replaced 4 years earlier
  - Postoperative dislocation → reduced → walking
  - Told bed rest 3 months → walked 3 months → dislocated again → reduced again
- Then hip **spica** for a while → mobilized → no further dislocation
  - Painful, but walking on it for almost 4 years
  - Question: cup stable or unstable?
  - **Osseointegration** → superior part

- At least on x-ray
  - Other hip → **Crowe** group 3–4, at most 3

## Side Selection Ask AI

Case 6

- Operate? Which side first?
- Question anticipated by **Dr. Abu Saud**
  - Most choose left first
- Reason → otherwise second side has a problem
  - Right more painful than left → right done first
  - Speaker's rule → listen to patient's complaint

## Anticipated Problems Ask AI

Case 6

- Femur → **small canal**
- Options for small canal → shortening? osteotomy?
  - Two-stage procedure?
- Main bone stock; acetabulum viewed from other side
- **Standard cup** vs **augmentation** and preparation?
- False acetabulum → look inside for true walls

## CT Assessment Ask AI

Case 6

- True acetabular walls
- CT **axial cuts**
- Axial cuts → good idea of bone stock type
  - Good area; walls small and very low

## Implants Ask AI

Case 6

- Usual technique → **cemented stem** + standard **cementless cup**
  - Cup → 46 mm, standard
  - Cemented stem → easier with shortening **osteotomy**
  - Insert, good cementation → worked
  - No problem with cement
- Why not **modular stem**?
  - Some reservations; does well
  - Not better than cemented stem
  - Modular stem → acceptable alternative here
- Choice → surgeon's preference

| | Cemented stem | Modular stem |
|---|---|---|
| Use here | Chosen | Acceptable alternative |
| Rationale | Easier with shortening osteotomy; good cementation | Does well |
| Speaker's view | No problem with cement | Some reservations; not better than cemented |

## Second Side Ask AI

Case 6

- After first side → walked month after month → then dislocation (after outside advice)
- Other side → now required
- Time limit noted by Dr. Abu Saud
  - Cup → covered, not uncovered
  - Covering bone stock → lowest, most medial part of previous acetabulum
- Bone at dislocation site → hip had been dislocated, not reduced
  - Patient walking on dislocated hip → abnormal gait
- **Close-up view** → **bone stock** covering cup
  - **Standard cup**; no **bone grafting**
  - Same for other side → size adequate
  - Radiolucent (white) line → surprises some *[C1 · transcript: "white line"]*
- Line closed; not the line below → **dome** fine
- Preoperative image → comparison of both hips intended
  - Outcome → did well

# Conclusions Ask AI

- **Dysplastic cases** → challenging; require special preparation
- Proper **assessment** + proper **planning** → decide needs
  - Most cases → **standard implants;** no special or very large implants
  - Most cases → no **grafting**
  - Both techniques work → if reconstruction done well
- Cemented vs cementless → per preferred technique
  - **Cemented reconstruction** works → case with > 15–16 years follow-up
  - **Cementless techniques** → also work
  - Key → perform chosen technique well

**High Yield**

- Dysplasia → challenging; special preparation, proper assessment, planning
- Most cases → standard implants
- Most cases → no grafting
- Cemented and cementless → both work if done well

## Case Summary Ask AI

| Case | Pattern | Key problems | Reconstruction | Follow-up |
|---|---|---|---|---|
| 1 | Mild dysplasia; woman, 51–52 years | Little subluxation | THR; acetabulum essentially normal | Contralateral THR 1 year earlier; no problems |
| 2 | Subluxated; Shenton's line broken | Bone deficient below; excessive anteversion; narrow canal | Cemented cup + femoral-head graft | 12–14 years; no resorption |
| 3 | High dislocated; false acetabulum | Large anterior dimension (mushroom head) | Femoral-head graft + screws; cementless cup | 14–15 years; doing well |
| 4 | Completely dislocated (Yemen) | Very small true acetabulum; sciatic nerve | Meticulous cemented + shortening osteotomy + bulk autograft | Nine years |
| 5 | Woman, 36 years | Hip problems | Rasp first → subtrochanteric shortening | — |
| 6 | Bilateral (Port Said); Crowe 3–4 | Recurrent contralateral dislocation; small canal | Cemented stem + 46 mm standard cementless cup; no graft | Did well |

# Periprosthetic Fractures

*(Classification and Management of Periprosthetic Fractures)*

**Prof Dr El Sayed Kassem**

**Watch Original Lecture**

*Lecture transcript processed with AI assistance.*

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# AI detected corrections (Mostly bad transcripts)

### Corrected - the text was changed; the transcript said

**C1** (p. 230) “less than 2 diameters” → “at least 2 cortical diameters”

**C2** (p. 232) “6 to 8 months” → “6–8 weeks”

# Overview and Definition Ask AI

## Lecture Scope Ask AI

- Summarized review of **periprosthetic fractures** → essentials only
- Prepared with **Dr. Mahmoud Abdel-Karim**
- Objectives
  - Risk factors and classification after THA
  - Intervention → reducing occurrence
  - Management and take-home messages

## Definition and Epidemiology Ask AI

- Scope includes avoiding **intraoperative periprosthetic fractures**
- Fracture around implant after **total hip arthroplasty (THA)**
  - Sites → acetabulum or femur
- Incidence rising
  - Increasing number of arthroplasties performed
  - Indications extended → younger, more active patients
- Timing → **intraoperative** or postoperative

# General Principles Ask AI

## Risk Factors Ask AI

- Patient-related
  - Female sex (probable)
  - **Advanced age**
  - Reduced bone quality
    - **Osteoporosis**, osteopenia
    - Rheumatoid arthritis, **Paget's disease**
    - DDH, similar conditions
- Implant-related
  - Main → **osteolysis** and loosening
  - Implant type → cemented vs cementless

## Patient Evaluation Ask AI

- History and examination first
  - Pre-fracture symptoms (e.g., pain before injury)
    - Suggest pre-existing **loosening** or osteolysis
- Medical → elderly with comorbidities
  - Identify and optimize before surgery
- Radiological → **AP and lateral X-rays**
  - Fracture classification
  - Implant stability in that region
- **Infection workup** → exclude periprosthetic joint infection
  - Detailed in next lecture (Dr. Mahmoud Abdel-Karim)

- Soft tissue / skin condition (probable)
- **Neurovascular examination**
- Existing implant details → type, company, size
  - Purpose → full preparation

### Aims of Management Ask AI

- **Fracture union**
- Restoration of alignment (probable)
- Preservation of function (probable)
- **Implant stability**

## Intraoperative Periprosthetic Femoral Fractures Ask AI

- Femoral fractures → intraoperative vs postoperative

### Incidence Ask AI

Intraop

- **Cementless** > cemented systems
- Higher in **revisions**

### Risk Factors Ask AI

Intraop

- **Inadequate exposure** → poor visualization
  - See well before **femoral neck cut,** rasping, acetabular reaming
- **Excessive force** → rasping or impaction
  - Especially **osteoporotic** patients
  - Hurried hammering → sudden femoral fracture
- **Press-fit stem larger than medullary canal**
  - Rasped to size `6` → size `6` unavailable
  - Size `7` forced → larger than canal → fracture
- **Long stem mismatching femoral bow**
  - Femur → **anterior bow**
  - Long stem must match bow → else fracture
- Revisions → higher incidence
  - **Compromised bone stock**
  - Cement removal
  - Impaction bone grafting

## Modified Vancouver Classification (Intraoperative) Ask AI

- Most famous → **Modified Vancouver Classification**
- Three main types × three subtypes

| Type | Site |
|---|---|
| A | Proximal metaphyseal |
| B | Diaphyseal, around stem |
| C | Distal to stem |

| Subtype | Lesion |
|---|---|
| 1 | Cortical perforation |
| 2 | Non-displaced linear crack |
| 3 | Displaced / unstable fracture |

### Type A — Metaphyseal Ask AI

Intraop

- **A1** → bone graft
  - Source → femoral head or acetabular reamer reamings
  - Stem stable in metaphysis
- **A2** → cerclage around metaphysis

### Type B — Diaphyseal Ask AI

Intraop

- **B1** → bypass perforation with longer stem
  - Bypass length → **2 cortical diameters**
- **B2**, stable stem → cerclage alone acceptable
  - Fracture fixation preferred
- B2, **unstable stem** → longer stem revision + fixation
- **B3** → revision by longer stem + fixation
  - ± Graft

### Type C — Distal to Stem Ask AI

Intraop

- **C1** → cortical strut graft
  - Bypasses defect (probable)
- **C2** → fixation, cerclage (probable)
- **C3** → fixation (ORIF)

## Prevention of Intraoperative Femoral Fractures Ask AI

- Lecturer → most important point of the lecture
- **Adequate exposure** → no step without good visualization
- Adequate soft tissue release (probable)
  - Difficult exposure, e.g., **protrusio**
- Osteoporotic → no **excessive force** in preparation and impaction
- Proper **templating**
- Check trial / broach size (probable) before final implant
  - Confirms canal fit
- Revision, stem not coming out → **extended trochanteric osteotomy**
  - Controlled osteotomy > uncontrolled fracture
  - "By my hand, not by Amr's hand"
- **Preoperative planning and templating**
  - Patient size known → avoids size 6 / size 7 scenario

**Pitfall**

- Size 6 unavailable → **size 7** forced → femoral fracture
- Long stem ignoring **anterior bow** → fracture
- Hurried impaction in **osteoporotic** femur → sudden fracture

**Clinical Pearl**

- Stuck stem in revision → **extended trochanteric osteotomy**
- Controlled osteotomy > uncontrolled fracture

## Postoperative Periprosthetic Femoral Fractures Ask AI

- Incidence (no intraoperative fracture)
  - Primaries → 1%
  - Revisions → 4%
- Main risk factors → **osteolysis,** loosening
- Mechanism → **low-energy trauma**
- **Swedish National Hip Arthroplasty Register** → mean time to fracture
  - Mean 7.3 years (recording unclear: "7 3.9")
- **Mayo Clinic** study → mean 8.1 years after index operation

## Vancouver Classification (Postoperative) Ask AI

- Most famous for postoperative periprosthetic fractures
- Based on three criteria
  - **Stem stability** → loose vs well fixed
  - Fracture site
  - Bone quality

## Types Ask AI

Postop

| Type | Location | Subtype | Defining feature |
|---|---|---|---|
| A | Trochanteric area | AG | Greater trochanter |
| | | AL | Lesser trochanter |
| B | Around or just distal to stem | B1 | Well-fixed stem |
| | | B2 | Loose stem, adequate bone quality |
| | | B3 | Loose stem, poor bone quality |
| C | Well below stem | — | — |

## Management by Type Ask AI

Postop

- **Vancouver A** → trochanteric region
  - Non-displaced / minimally displaced → **conservative**
  - Displaced (unstable) → fixation, wires and plate (probable)
- **Vancouver B1** → fixation around stable stem
  - Plate or cerclage (probable)
  - i.e., **ORIF**
- **Vancouver B2** → around or just distal to loose stem
  - Stem removal → **longer stem** → fracture fixation
- **Vancouver B3** → loose stem, poor bone quality
  - Longer stem + proximal femur reconstruction, grafts (probable)
  - Reconstruction impossible → **proximal femoral replacement**
- **Vancouver C** → stem untouched → ORIF

### Management Algorithm Ask AI

Postop

| Type | Management |
|---|---|
| A, non-displaced | Conservative |
| A, displaced | ORIF |
| B1 | ORIF |
| B2 | Longer stem + fixation |
| B3 | Longer stem + proximal femur reconstruction, or proximal femoral replacement |
| C | ORIF |

**High Yield**

- Vancouver criteria → stem stability, fracture site, bone quality
- Well-fixed stem (B1, C) → **ORIF**
- Loose stem, adequate bone (B2) → **longer stem** + fixation
- Loose stem, poor bone (B3) → **proximal femoral replacement** if no reconstruction

## Fixation of Periprosthetic Femoral Fractures Ask AI

### Indications Ask AI

- **Well-fixed stem** → B1 and C

### Problems Ask AI

- Stem and cement → prevent **rigid fixation**
- **Poor bone healing** around stem
- **Bone defects**

### Fixation Principles Ask AI

- **Relative stability** → especially osteoporotic bone
- **Long spanning plates**
- **Bicortical screws** when possible
  - Otherwise → unicortical screws + cables
  - Or purpose-designed plates
- **Minimally invasive techniques** → less soft tissue stripping
- Plate–stem overlap → at least 2 cortical diameters (probable) *[C1 · transcript: "less than 2 diameters"]*

### Devices Ask AI

- Very new devices on the market
  - Multi-plane devices
  - **Periprosthetic locking screws** → short
    - Pass beside stem (probable)
  - **Attachment devices**

  - **Positioning pin** for cable system
  - **Proximal femoral hook plates**
  - **Distal femoral locking plates**
  - **Retrograde locked intramedullary nail**

**Exam Pearl**

- Fixation alone → only with **well-fixed stem** (B1, C)
- Osteoporotic bone → **relative stability**
- Long spanning plates, minimally invasive technique

# Periprosthetic Acetabular Fractures Ask AI

## Overview Ask AI

Acetabular

- Uncommon → less than femoral
- When present → **catastrophic**
- Cemented and cementless cups → more with **cementless**
- Intraoperative or postoperative

## Risk Factors Ask AI

Acetabular

- **High impaction force** → cup insertion or reaming
  - Avoid → prevents periprosthetic fracture
- **Revision cases** → higher risk
- **Cup design** → elliptical cups, threaded cup systems
- Acetabular under-reaming (probable)
- **Osteoporosis**, radiation, dysplasia
- Each → ↑ intraoperative or postoperative fracture risk

## Classification Ask AI

Acetabular

- Several classifications; this one → **5 types**

| Type | Description |
|---|---|
| 1 | Intraoperative, during cup insertion |
| 2 | Intraoperative, during cup removal (revision) |
| 3 | Traumatic → patient fall |
| 4 | Spontaneous (probable) |
| 5 | Pelvic discontinuity (probable) |

- Focus → intraoperative and postoperative fractures

# Diagnosis and Management of Acetabular Fractures Ask AI

## Diagnosis Ask AI

Acetabular

- Identify and assess fracture
  - **Fracture stability**
  - **Cup stability**
- Intraoperative suspicion (primary THA)
  - **Stress testing** of cup and pelvis (probable) → cup motion?
  - **Intraoperative imaging**
  - Still suspicious → **remove cup** → inspect acetabulum directly
- Postoperative
  - **AP and lateral hip views**
  - **CT** → fracture assessment
  - **CT angiography** → intrapelvic components
    - Assess vessels before component removal

## Management Principles Ask AI

Acetabular

- Depends on **implant stability**
- Depends on **fracture pattern and displacement**
- Aims
  - Rigid fracture fixation → **bone union**
  - Stable cup integration
  - Restore **acetabular bone stock**
  - Re-establish **center of rotation,** offset, limb length
- Revision cases → full preparation
  - Bone graft, bulk / chip (probable)
  - **Meshes,** rings, augments
  - **Protrusio cups,** jumbo cups

## Intraoperative Fracture Ask AI

Acetabular

- Assess fracture first
- Non-displaced + **stable cup** → nothing major
  - **Additional screws**
  - Close follow-up
  - Protected weight bearing
- Displaced + **unstable cup**
  - Remove cup → **plate** fixation → reinsert cup

## Postoperative Fracture Ask AI

Acetabular

- Stable cup → supplementary fixation (probable) → prevents cup movement
- Non-displaced → **conservative**
  - **Protected weight bearing** → 6–8 weeks *[C2 · transcript: "6 to 8 months"]*
  - Serial follow-up
  - Watch for cup loosening (probable) despite union
- Displaced + **unstable cup** → seen on X-ray and CT

- **Cup revision** after acetabular fixation
- Acetabular **ORIF** first (e.g., posterior column (probable))
- Then cup insertion
- Stable cup unattainable → cup-cage construct (probable)

## Conclusion Ask AI

- Periprosthetic fractures after THA → no longer uncommon
- Correct **classification** + implant stability assessment → essential
- **Prevention** → best, especially intraoperative fractures
- Planning → important
- Full revision preparation → no unexpected surprises
- Main goals → **fracture union** and implant stability

**Must-Know**

- **Adequate exposure** → before every intraoperative step
- Preoperative planning and templating → avoids size mismatch
- Management → driven by **implant stability** and fracture pattern
- Main goals → fracture union + implant stability

# Diagnosis of PJI

*(Criteria, Investigations and Diagnostic Algorithm for Periprosthetic Joint Infection)*

**Prof Dr Mahmoud Abdel Karim**

**Watch Original Lecture**

*Lecture transcript processed with AI assistance.*

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# AI detected corrections (Mostly bad transcripts)

## Queried - kept as spoken, but looks wrong

**Q1** (p. 239) As spoken; ICM 2018 preoperative score: 2–5 inconclusive, 0–1 not infected, not 3–5 and <3.

**Q2** (p. 242) As spoken; leukocyte esterase sensitivity is about 80%; the ~100% figure is its specificity.

**Q3** (p. 244) As spoken; 28 of 65 is 43%, not 61% (61% of 65 is about 40).

# Diagnostic Criteria: Background Ask AI

## Origin of the Criteria Ask AI

- Central question: why does the diagnosis not work? (Dr. Mulham)
  - Criteria first introduced as the American **MSIS** (Musculoskeletal Infection Society) criteria
- Then → **International Consensus Meeting (ICM),** Philadelphia 2013
  - Dr. Wadid Obeid took part
  - Output → the blue book; excellent reference
  - Consensus session design → demanding

## Why Diagnosis Is Difficult Ask AI

- Diagnosis not straightforward; can be wrong
  - Many tests used → almost none **specific** on its own

# 2013 Definition: Major and Minor Criteria Ask AI

## Need for an Algorithm Ask AI 2013

- All available tests described in literature; many commercial kits
  - All tests for every patient → not possible
  - → Prospective **algorithm** needed → infected or not infected

## Criteria Ask AI 2013

- **Major criteria** (either one)
  - Sinus tract **communicating with the joint**
  - **Two positive cultures** → same organism, two separate samples
- **Minor criteria** → three of five
  - Elevated CRP and ESR
  - Elevated synovial fluid WBC count
  - Positive **leukocyte esterase**
  - Elevated synovial **PMN percentage**
  - Positive histological analysis
  - Single positive culture

# 2018 ICM Scoring System Ask AI

## What Changed Ask AI 2018

- 2018 → International Consensus definition changed
- No need to memorize the slide → keep it on a phone
  - Major criteria → essentially unchanged
  - Minor criteria → **weighted scoring system**

## Scored Items Ask AI 2018

- Serum CRP → acute **> 100**; chronic **> 10**
- **D-dimer** → also included

- ESR → not relevant in acute
  - Chronic → **> 30**
- Points per item → summed at the end
- Preoperative + postoperative scores → combined

| Item | Threshold | Points |
|---|---|---|
| Serum CRP | Acute > 100; chronic > 10 | 2 |
| ESR | Chronic > 30; none in acute | 1 |
| Synovial WBC count | — | 3 |
| Leukocyte esterase | — | 3 |

### Score Interpretation Ask AI

2018

- Total score → three outcomes

| Total score | Interpretation | Next step |
|---|---|---|
| ≥ 6 | Infected | — |
| 3–5 | Inconclusive | Molecular techniques (not yet in Egypt) *[Q1 · As spoken; ICM 2018 preoperative score: 2–5 inconclusive, 0–1 not infected, not 3–5 and <3.]* |
| < 3 | Not infected | — |

### Consensus Vote Ask AI

2018

- Debated at Philadelphia → more than 2 hours
- Vote → 68% agreed, 28% disagreed, 4% abstained
- **Proceed with caution** in
  - **Adverse local tissue reaction (ALTR)**
  - **Slowly growing organisms**
  - Inconclusive results → **molecular diagnosis**

**High Yield**

- Major criteria → sinus tract to joint, or two positive cultures of same organism
- 2018 minor criteria → weighted score: CRP 2, ESR 1, synovial WBC 3, LE 3
- ≥ 6 infected; 3–5 inconclusive → molecular; < 3 not infected
- Caution → ALTR, slowly growing organisms

## Preoperative Screening Ask AI

### Clinical Assessment Ask AI

- Main message → **every patient infected until proven otherwise**
- Full history + physical examination

  - Discharge; prior history of infection
  - → **High index of suspicion**

### ESR and CRP Ask AI

Screening

- Universal preoperative screen → **ESR and CRP**; no disagreement
- Both normal ≠ no infection
  - **Negative predictive value** → 95%
  - ~5% → still infected
- Still very useful
- Marker thresholds
  - ESR → chronic 30; no specific value in acute; 1 point
  - CRP → chronic 10; acute 100; 2 points
- Guidelines → both normal, patient may still be infected
- Common belief: normal ESR + CRP → enough; incorrect
- ESR → falling out of use; countless causes of elevation
- CRP → more specific
- Both elevated/abnormal → **aspiration**
  - Early aspiration favoured by some → additional results

**Pitfall**

- Normal ESR and CRP do not exclude infection
- NPV 95% → ~5% still infected
- ESR → many non-specific causes; CRP more specific

## Joint Aspiration Ask AI

### Indication and Culture Yield Ask AI

Aspiration

- Clinical judgment; abnormal ESR or CRP → **aspiration**
- Raising **culture yield**
  - **Off antibiotics 2 weeks** (14 days) before aspiration
  - Request **extended (prolonged) culture**
  - Incubation → 9–14 days
- Ordinary culture container → incorrect
  - Inoculate into **blood culture bottles** → higher yield
  - European Bone and Joint Infection Society (EBJIS) microbiologist → yield increase, possibly up to 9×
- Both points → very important

### Dry Tap Ask AI

Aspiration

- Dry tap → injecting saline **not acceptable**
  - Risk → introducing infection

### Requested Analyses Ask AI

Aspiration

- Request **total and differential leukocyte count incl. PMN**
- Plus culture and sensitivity

- Common error → culture and sensitivity only

### Synovial Thresholds Ask AI

Aspiration

- Synovial WBC → chronic 3,000; acute > 10,000
  - > 10,000 → possible **acute infection**
- **PMN** → subset reflecting infection; reason for differential count

| Synovial marker | Acute | Chronic |
|---|---|---|
| WBC count (cells) | > 10,000 | 3,000 |
| PMN percentage | > 90% | 70% or 80% |

**Clinical Pearl**

- Off antibiotics 14 days before aspiration
- Extended culture → 9–14 days incubation
- Aspirate → blood culture bottles, not ordinary container
- Request total + differential count with PMN, not culture alone
- Dry tap → never inject saline

## Culture-Negative Infection Ask AI

- Culture-negative rate → 5% to 42%
- Negative cultures → possible repeat sample
  - Hold antibiotics 14 days
  - Or **molecular technique** → e.g. **PCR**
- Negative culture ≠ no infection
  - Many returning patients (up to five) → bacteria despite negative cultures
- Molecular techniques → address this gap

## Other Tests Ask AI

### Not Supported Ask AI

- **Gram stain** → no
- Systemic WBC → no
- Intraoperative Gram stain → no supporting evidence
- CT and MRI → no evidence; possibly evolving
  - Requested for other purposes, not to confirm infection

### Nuclear Imaging Ask AI

- **Bone scan** → commonly requested
  - **False positive** 1–2 years after prosthesis placement
  - Positive scan ≠ infection
- Role → when diagnosis not established by other means
  - **Indium scan, sulfur colloid** scan

### D-dimer and Plasma Fibrinogen Ask AI

- D-dimer → threshold **850 ng/mL**
  - Single group only
- Later study (last year) → **plasma fibrinogen** performs better
- D-dimer → not better than conventional serum markers; not superior
- Further tests → leukocyte esterase, alpha-defensin

## Leukocyte Esterase Test Ask AI

### Principle Ask AI LE

- Urine strip (sugar/acetone) → **leukocyte** pad → turns **pink**
- Quite good test
- Enzyme secreted by **activated neutrophils** migrated to infection site
- Likened to a **pregnancy test** → aspirate applied to strip as with urine

### Reading Ask AI LE

- Colour → one-plus (+) or two-plus (++) pink
  - Sensitivity → 80% or 100% *[Q2 · As spoken; leukocyte esterase sensitivity is about 80%; the ~100% figure is its specificity.]*
  - **Two-plus** → very important finding
- Positive and negative predictive values → very high

### Practical Use Ask AI LE

- Blood-contaminated sample (> 3,000 cells) → caution
- Keep a box of strips in the **operating room**
- Sample → **no blood**

## Alpha-Defensin Ask AI

### Properties Ask AI AD

- Covered in detail by Dr. Fouad
- **Not affected by antibiotics**
- Guidelines → good **complement** to existing diagnostic tests
- Not 100% perfect
  - **ALTR** → false positive

### Synovial CRP Combination Ask AI AD

- 2014 key article → add **synovial CRP** to avoid false positives
  - Synovial, not systemic CRP
  - Combined → resolves equivocal results
- Test cost → 500, 5,000 or 4,000 (company-dependent)
  - Speaker → requests synovial CRP

**Exam Pearl**

- LE → enzyme from activated neutrophils; ++ pink most important
- LE sample → must contain no blood
- Alpha-defensin → not affected by antibiotics
- Alpha-defensin false positive → ALTR; add synovial CRP

# Intraoperative Sampling Ask AI

## Number and Sites Ask AI

Sampling

- Workup from first presentation with pain → surgical stage
- **Three to five samples;** preferably **five; odd numbers**
- Different sites → **capsule, acetabulum, femur;** both sides
- → Most accurate yield for infection diagnosis

## Every Revision Ask AI

Sampling

- Sample in every revision → yes
- Prior aspirate cultures → repeat still matters
  - Aspirate organism may differ from intraoperative tissue organism (two different sites)
  - Valid in **polymicrobial infections**
  - Especially **low-virulence organisms**

## Why Odd Numbers Ask AI

Sampling

- Single sample → may be **contaminated**
- Two → one against one
- Three → two against one
- Five → three or more against two or one → higher yield rate

## Bone Culture Ask AI

Sampling

- Two-stage revision → bone culture to raise yield?
  - Consensus → **inconclusive;** can't recommend for or against

**Must-Know**

- 3–5 intraoperative samples, preferably 5, odd numbers
- Sites → capsule, acetabulum, femur, both sides
- Sample in every revision; aspirate organism may differ
- Bone culture in two-stage → no recommendation for or against

# Frozen Section and Sonication Ask AI

## Frozen Section Ask AI

- Useful test → if eminent, well-versed pathologist available
- Threshold → **5–10 PMN per high-power field**
- UK fellowship experience → reliable pathology service
  - Result by phone → within 15–20 minutes

  - Meanwhile → clean and debride
  - Then → proceed, or **staged (two-stage) revision**

### Sonication Ask AI

- Good test → positive = high likelihood of infection
- Cultures commonly negative → sonication adds yield
- Removed **implant** → device → **disrupts biofilm** → fluid extracted
  - Fluid → culture
  - Yield → better and more targeted than ordinary culture
- Use when possible; especially **culture-negative** cases

## Molecular Diagnostics Ask AI

- Last resort → still uncertain after all other tests
  - New technique; the future; already available abroad
- Ordinary cultures → 28 of 65 infected cases positive (61%) *[Q3 · As spoken; 28 of 65 is 43%, not 61% (61% of 65 is about 40).]*
- **Next-generation sequencing (NGS)** → ~90%
  - 61% → 90%
- Detects more infected cases
  - Rarely positive in non-infected cases → not 100% perfect

## ICM Philly Application Ask AI

- **ICM Philly** (Philadelphia) → Android application
- Diagnosis section → enter available data → calculates **score**
- Continues data entry → suggests **next step**
- Strongly advised for everyone

## Take-Home Message Ask AI

- Diagnosis → many elements: aspiration, culture, ESR, CRP
- **Multifactorial process**; no single test enough
  - Not **Synovasure** alone
  - Not ESR alone; not CRP alone
- **Every patient infected until proven otherwise**
  - Painful total hip → assume infection
- Read the guidelines → incorporate and integrate into practice

**Must-Know**

- Every patient infected until proven otherwise
- Painful total hip → assume infection
- No single test sufficient → multifactorial diagnosis

# Management of PJI

*(Surgical Strategies for Periprosthetic Joint Infection)*

**Prof Dr Mohamed Abdel Moneim**

**Watch Original Lecture**

*Lecture transcript processed with AI assistance.*

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# AI detected corrections (Mostly bad transcripts)

### Corrected - the text was changed; the transcript said

**C1** (p. 251) “40 g pack: more than 4 g” → “40 g pack: no more than 4 g”

**C2** (p. 252) “patient has a sinus” → “no sinus”

## Aim of the Lecture Ask AI

- Infected joint → close follow-up → early intervention, before infection establishes
- First presentation: clinic, deteriorating joint → surgical management
- Single goal: **management plan** of periprosthetic joint infection (PJI)
  - Focus: treatment, not diagnosis of infection
- Infected patient presents → surgeon needs a definite plan, not discharge without one
- Aims
  - Concept of PJI management
  - Management techniques
  - Advantages of each technique
  - Disadvantages of each technique
  - When each technique should not be used

## Classification of Infection Ask AI

- Management starts with classification → **acute** vs chronic
- Acute — definition 1: presentation within 4 weeks of **index operation**
- Acute — definition 2: hip replacement present for years
  - Sudden acute symptoms, duration < 3 weeks
  - Example: hip replaced 5 years ago → problematic within one week → **acute hematogenous** infection

## Irrigation and Debridement Ask AI

- First of two options in acute infection: **irrigation and debridement**
- Wound opened within 4 weeks → debridement
  - **Superficial debridement:** skin + subcutaneous tissue
    - Limited superficial infection → may stop, no deeper exploration
  - **Deep debridement:** fascia opened → joint itself
- Prosthesis not removed → debridement around **prosthetic joint**

### Irrigation Methods Ask AI

I&D

| | Pulsatile lavage device | Bulb syringe |
|---|---|---|
| Mechanism | Irrigation + suction simultaneously | Irrigation by squeezing syringe |
| Pressure | High; adjustable level | Low |
| Preference | Not preferred | Preferred |

- High-pressure device problems
  - Drives microorganisms into deeper areas otherwise unreachable
  - Picks up organisms from one area → seeds another
  - Possible damage to surrounding tissue and bone

**Pitfall**

- High-pressure **pulsatile lavage** → organisms driven deeper
- Organisms carried from one area → seeded into another
- Low-pressure syringe preferred

### Determinants of Eradication Ask AI

I&D

- Eradication by I&D depends on
  - Good **immunity** + good general condition
  - Early intervention

### Outcome Ask AI

I&D

- Success rate much lower in this patient than in others: ≈ 65–70%
  - **High-risk** patient → 16–28%

## DAIR Ask AI

- Second option: **DAIR** = Debridement, Antibiotics, and Implant Retention
  - Re-intervention with debridement
  - Antibiotic administration
  - Exchange of **modular components**
  - **Implant retention**
- Joint opened → **femoral head and cup liner** always accessible → easy removal → exchange
  - → reduced **biofilm** load → higher success rate
  - Implant preserved → stable, well fixed

### Success Factors Ask AI

DAIR

- Most important factor determining DAIR success?
- → **interval between index operation and DAIR**
- Modular component exchange
  - No exchange → reportedly ~4× failure risk
  - Exchange → **survival rate** 52% → 79%

**High Yield**

- DAIR = Debridement, Antibiotics, and **Implant Retention**
- Most important success factor → interval from index operation to DAIR
- Femoral head + cup liner exchanged → reduced biofilm
- Modular exchange → survival 52% → 79%

## Chronic Infection Ask AI

- **Chronic infection**
  - Presentation > 4 weeks after surgery
  - Acute hematogenous infection present for some time (e.g., symptoms 2–3 months) → managed as chronic
- Management → **revision** surgery

- One-stage or two-stage

| | Acute | Chronic |
|---|---|---|
| Postoperative | Within 4 weeks of index operation | > 4 weeks after surgery |
| Hematogenous | Long-standing hip; sudden symptoms < 3 weeks | Symptoms for some time (e.g., 2–3 months) |
| Management | Irrigation and debridement; DAIR | Revision: one-stage or two-stage |

## Two-Stage Revision Ask AI

- Classic technique → **gold standard** for chronic PJI

### Indications Ask AI

Two-stage

- Chronic PJI
- Acute PJI or acute hematogenous infection → sometimes used from the start
- Severely **immunocompromised** patient
- Previous failed attempts
  - Repeated operations → ↑ morbidity, ↑ cost, ↑ **arthrofibrosis**, worse final outcome
- Acute PJI → DAIR + other measures all failed → ultimately two-stage revision

### Technique Ask AI

Two-stage

- **First stage**
  - Removal of all implants + all cement
  - Good debridement
  - Spacer of **antibiotic-loaded cement**
- **Intermediate (interim) period** → wait until patient completely free of infection
  - Nutritional optimisation helpful
  - **VAC** (vacuum-assisted closure) — like an implant — stated as contraindicated in PJI
- **Second stage:** spacer removal → definitive reimplantation

**Must-Know**

- Two-stage revision → **gold standard** for chronic PJI
- Stage 1: all implants + cement out → debridement → antibiotic-loaded spacer
- Interim: wait until infection-free
- Stage 2: spacer out → definitive reimplantation

## Three-Stage Exchange Arthroplasty Ask AI

- When is **3-stage exchange arthroplasty** considered?
- Indication: highly resistant/virulent organisms + **severe infection**
- Technique
  - Spacer insertion
  - Repeat debridement after 2–4 weeks, regardless of wound state → eradication
  - Wait for improvement → definitive reimplantation

- Spacer → repeat debridement → reimplantation: three operations

## Antibiotic-Loaded Cement for Spacers Ask AI

- Cement: highly **porous**
- **Hand mixing** preferred over vacuum mixing → more pores → ↑ antibiotic elution
- Antibiotic requirements
  - **Heat-stable** → withstands heat of cement polymerisation
  - **Water-soluble** → elutes into surrounding tissues
- **Combination of antibiotics** better than single antibiotic
  - Second antibiotic → ↑ elution of first (**synergism**)
  - Organism gentamicin-sensitive → not **gentamicin**-loaded cement alone; add second antibiotic
- Dose: antibiotic < 10% of total **cement weight**
  - Example: 40 g cement pack → ≤ 4 g antibiotic *[C1 · transcript: "40 g pack: more than 4 g"]*
  - Absolute maximum 8 g → more weakens cement **mechanically**
  - Recommended maximum 4 g; range 0.5–4 g by antibiotic type

**High Yield**
- **Hand mixing** > vacuum mixing → porosity → elution
- Antibiotic: heat-stable + water-soluble
- Combination > single antibiotic → synergism
- Antibiotic < 10% of cement weight: 40 g → 4 g
- Never > 8 g → cement weakened

## Systemic Antibiotics Ask AI

- **IV antibiotics** → 2 weeks
- Organism identified + general condition improved
  - → switch to **oral antibiotics** → 4 weeks

### No Drug Holiday Ask AI

Antibiotic

- Classic approach: antibiotics for a period → stopped for a period → confirm no persistent infection → reimplantation
- Found to be a poor idea
  - ↑ **drug resistance**
  - ↑ wound complications / reinfection
- → antibiotics not stopped before reimplantation (**no drug holiday**)

**Pitfall**
- Stopping antibiotics before reimplantation → **drug holiday**
- → ↑ drug resistance, ↑ wound complications / reinfection
- Antibiotics continued up to reimplantation

## Spacers Ask AI

- Many antibiotics addable to cement → must be heat-stable + water-soluble
- Antibiotic form → **powder**, not liquid
- Spacer type → static or articulating
- **Static spacer:** cement block inside acetabulum, nothing more
- **Articulating spacer:** several types
  - **PROSTALAC** (Prosthesis of Antibiotic-Loaded Acrylic Cement): very simple total hip → patient moves
  - **Preformed** (ready-made) spacer
  - **Intraoperatively moulded** spacer → shaped by hand during surgery
    - Cement over **Steinmann pin** or wires → Thompson shape

### Advantages Ask AI

Spacer

- Articulating spacer
  - ↑ **joint mobility**
  - Prevents soft-tissue contracture
  - Facilitates reimplantation
  - High local bactericidal antibiotic levels

## One-Stage Revision Ask AI

- **One-stage revision:** single operation, everything revised

### Selection Criteria Ask AI

One-stage

- **Organism** criteria
  - Identified, not unknown
  - Not highly virulent/resistant
  - Sensitive to antibiotic addable to cement
- **Patient** criteria
  - Generally stable condition
  - No sinus *[C2 · transcript: "patient has a sinus"]*
  - Good soft-tissue/bone quality and viability

### Goals and Advantages Ask AI

One-stage

- Goal: operation → **mobilisation** next day → finished
  - ↓ time
  - ↓ cost
  - ↓ hospital stay
  - Rapid return to function
  - Better outcome → no multiple operations, no interim period
- Results comparable, very good → **selected cases** only
- Not every case; "saw someone do it and it worked" → no justification
  - e.g., **Staphylococcus** case → one-stage, went well → not a rule

**Exam Pearl**

- Organism: identified, not highly virulent/resistant, sensitive to cement antibiotic
- Patient: stable, **no sinus**, good soft-tissue/bone quality
- Results comparable → selected cases only

## Resection Arthroplasty Ask AI

- Other management types → very selected cases
- **Resection arthroplasty** (Girdlestone)
- Everything removed → hip left free, no prosthesis

### Indications Ask AI

Girdlestone

- **Non-ambulatory** from the beginning
- Very poor bone stock
- Poor soft-tissue coverage
- **Highly resistant** organism
- Poor general condition → cannot undergo multiple operations, repeated attempts
- Multiple previous failed attempts (e.g., two-stage revision)
- Image: patient after resection arthroplasty

## Chronic Suppressive Antibiotics Ask AI

- **Chronic suppressive antibiotics** → antibiotics only
  - Single indication: patient unfit for surgery → would die if operated on
  - Reason: antibiotics stopped → 80% failure; no eradication of infection

**Clinical Pearl**

- Suppressive antibiotics → **unfit** patient only
- Antibiotics stopped → 80% failure

## Summary Ask AI

- Know management **decision factors**
- Key questions: operation timing, symptom onset → acute vs chronic
- Management **tailored** to each case → good surgical experience
- Antibiotics: major role → careful choice of agents

| Option | Setting / indication |
|---|---|
| Irrigation and debridement | Acute; within 4 weeks; prosthesis retained |
| DAIR | Acute; modular components exchanged; implant retained |
| Two-stage revision | Gold standard for chronic PJI; immunocompromised; failed attempts |

| Option | Setting / indication |
|---|---|
| Three-stage exchange | Highly resistant/virulent organism; severe infection |
| One-stage revision | Selected cases: known, non-virulent, cement-sensitive organism; stable patient; no sinus |
| Resection arthroplasty | Non-ambulatory; poor bone stock/soft tissue; resistant organism; failed attempts |
| Chronic suppressive antibiotics | Unfit for surgery |

# Loosening

*(Osteolysis, Aseptic Loosening and Clinical Work-up)*

**Prof Dr Sherif Khaled**

**Watch Original Lecture**

*Lecture transcript processed with AI assistance.*

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## Overview: From Wear to Loosening Ask AI

- **Aseptic loosening** → mechanism step by step
- Wear → **particulate debris** → inflammatory response
- Radiological expression → **osteolysis** → loosening
- Sequence: wear → debris → osteolysis → loosening on **X-rays**

**High Yield**

- Wear → particulate debris → inflammatory response → osteolysis → loosening

### Learning Outcomes Ask AI

- **Osteolysis** → brief description
- **Clinical picture** of loosening
- Identifying loosening:
  - **Cemented** total hip
  - Cementless total hip

## Osteolysis Ask AI

### Definition Ask AI

- **Time-dependent process**
- Inflammatory reactions → mainly **cell-mediated**
- Activated by **wear debris**

### Debris Sources Ask AI

- Commonest → **polyethylene**
- Others:
  - **Cobalt**
  - Titanium
  - **Methyl methacrylate** cement

### Biologic Response Ask AI

- Mediated by:
  - **Macrophages**
  - Chemical mediators:
    1. Interleukin (likely IL-1, as heard)
    2. **Interleukin-6**
    3. Prostaglandin (likely E2, as heard)
- Mediators = **bone-resorbing cytokines** → periprosthetic resorption → loosening

### Debris Pathway Ask AI

- **Femoral osteolysis** → debris tracks down canal:
  - Bone–cement interface → cemented
  - Bone–prosthesis interface → **press-fit** prosthesis
- Same mechanism → **acetabular** side

  - Acetabular loosening → often not diagnosed on X-rays

### Particle Generation Ask AI

- Wear modes (as heard): adhesive, abrasive, fatigue
- Effects (as heard): local or systemic
- Two particle sources → **wear** and corrosion
- Lecture focus → identifying loosening, not wear or osteolysis

## Clinical Picture of Loosening Ask AI

- Presentation → hip replacement + **pain**
- Loosening → **septic** or aseptic

| Feature | Septic loosening | Aseptic loosening |
|---|---|---|
| Pain pattern | **All the time** | Activity-related |
| Timing | Worse **at night** | **Start-up pain**; worse on movement |
| Fever | Possible; most **afebrile** | — |
| Diagnosis | **Laboratory** investigations + X-rays | — |

- Pain site (as heard) → groin or thigh
- Sometimes anterior pain (as heard) → examination localises true source
- Pain ↑ with **weight bearing**
- Investigations → ↑ **ESR** and CRP, + further work-up

## Investigations and Radiographic Assessment Ask AI

- **CRP** and ESR → important
- X-rays → **AP** + lateral view → look for loosening
- Treatment → **revision**
  - Outside scope → basic total hip course
- Diagnosis easier → **femur** than acetabulum
- Acetabular cases → next session case presentation
- Many cases → loosening easy to diagnose
- **Diagnostic accuracy:**
  - Femoral → 90%
  - Acetabular → 65%
- Greater problem → **cemented** components
  - Cemented → easier to see around femoral component

### Radiolucent Lines Ask AI

Lucency

- Radiolucent lines ≠ necessarily **loosening**
- Common around **cementless** hips → may reflect design
- Key point → lines **not progressive**

> **Pitfall**
> - Radiolucent lines around a cementless stem may be design-related
> - Loosening concern → only if lines are progressive

### Follow-up Ask AI

- **Follow-up** mandatory → patients in Egypt disappear once well
- Schedule → yearly → every 3 years → every 3–5 years
- Practice → rarely seen after 3 years unless **symptomatic**

## Radiographic Zones Ask AI

- Zones needed → **measure** loosening

### Femoral — Gruen Zones Ask AI

- **Gruen zones** → zone 1 to zone 7
  - Zone 1 → lateral
  - Zone 4 → tip
  - Zone 7 → medial

### Acetabular Zones Ask AI

- Zone 1 → lateral (**superior third**)
- Zone 2 → middle third
- Zone 3 → inferior third
- Always → superior, middle, inferior thirds

## Cementless Stem Loosening — Engh Classification Ask AI

- Cementless stem fixation → **Engh classification**

| Engh type | Stem status |
|---|---|
| **Stable bony ingrowth** | Fixed; not loose |
| **Stable fibrous ingrowth** | Stable |
| **Unstable fibrous ingrowth** | Loose |

### Stable Bony Ingrowth Ask AI

Engh

- Used to judge **cementless prosthesis** → loose or not
- Signs of accepted (fixed) stem:
  - **Spot welds** → bone hypertrophying at site
  - **Reactive bone** lines around stem
  - **Remodeling** → reactive bone between stem and cortex
- Example shown → **stable bony ingrowth**
- Minor signs:
  - **Stress shielding**
    1. Calcar → compare bone quality / **cortical density** site to site

    - **Calcar atrophy** labeled
    - **Diaphyseal fitting** → fixed distally → no proximal load
    - Unloaded area → **resorption;** load through diaphysis
    - Normal finding → diaphyseal fitting
  - **Absence of pedestal**
- **Pedestal** → defined below

### Stable Fibrous Ingrowth Ask AI

Engh

- No **spot welds**
- No **migration**
- **Parallel sclerotic lines**
- X-ray → no significant sign of loosening, nor of **fixation**

### Unstable Fibrous Ingrowth Ask AI

Engh

- **Progressive radiolucent lines**
- **Subsidence**
- **Pedestal formation**
- **Subsidence** → unfixed prosthesis moves downward → position change
  - Stem descends → measurable distance
  - Drifts **laterally** → varus
- **Pedestal** → bone hypertrophy at stem tip → attempt to stop subsidence

**High Yield**

- Engh: stable bony / stable fibrous / unstable fibrous ingrowth
- Fixed stem → spot welds, reactive bone, remodeling
- Minor signs → stress shielding, absence of pedestal
- Loose stem → progressive lucencies, subsidence, pedestal

## Cementless Cup Loosening Ask AI

- **Migration** > 2–5 mm
- Radiolucent lines → **all three zones**: superior, middle, inferior
- Lines > 2 mm
- **Progressive** over 2 years
- Pain + very obvious lines → **loose cup**
- Key idea → lines must be **progressive**
- **Change in inclination** (± version, as heard)
- Case → inclination (± version) change after **four years**
  - No other sign present
  - Change in inclination alone → **loose cup**

**Exam Pearl**

- Change in cup inclination alone defines a loose cup

# Cemented Stem Loosening Ask AI

## Classification Ask AI

Cemented

- Most textbooks → **possible / probable / definite** loosening
- **Possible loosening** → radiolucent lines in 50–100% of interface
  - Case → cement–femur lucency, 50–100% of total interface
  - Progression check → followed **5 years**
  - Re-presented → **fracture neck of femur**, other side
  - Index side → no complaints, appearance unchanged at 5 years
  - Cause → poor initial **cementing technique** → radiolucency
- **Probable loosening** → radiolucent line around whole stem (as heard)
  - ± Surrounding changes
- Debris tracks down → **osteolysis** in this area
- **Definite loosening** → four X-ray signs:
  - Stem fracture
  - Cement fracture
  - Radiolucency → **cement–implant interface**
  - Change in stem position
- Each sign → own X-rays

| Grade | Radiological finding |
|---|---|
| **Possible** | Radiolucent lines in 50–100% of interface |
| **Probable** | Radiolucent line around whole stem (as heard) ± surrounding changes |
| **Definite** | Stem fracture / cement fracture / cement–implant lucency / stem position change |

## Definite Signs Ask AI

Cemented

- Sign 1 → **stem fracture**
  - Broken stem → sure diagnosis of loose cemented stem
  - Mechanism → **bending cantilever**
    - Distal stem well fixed in bone → proximal part breaks

**Exam Pearl**

- Mechanism of cemented stem fracture → bending cantilever
- Distal stem well fixed → proximal part breaks

- Sign 2 → **cement fracture** → prosthesis loose
  - Key in **periprosthetic fractures** → loose or not changes classification
  - Stable stem → **fracture fixation** without stem revision
  - Cement fracture + radiolucent lines → loose → **revise stem**
- Sign 3 → **radiolucent lines** between cement and implant

**Clinical Pearl**

- Periprosthetic fracture: loose or fixed stem changes the classification
- Fixed stem → fracture fixation without stem revision
- Cement fracture or lucent lines → revise stem

## Change in Stem Position Ask AI

Cemented

- Sign 4 → **change in stem position;** three types
- **Subsidence** → prosthesis subsides downward
  - Expected: **air-filled centralizer** → stem subsides 1–2 mm (per Dr. Ahmed Hazem)
  - Purpose → fit tightens
  - 1–2 mm → normal in **first year**
  - 2–5 mm → concern
  - > 2 mm → abnormal; centralizer allows no more space
  - Measured: **greater trochanter tip** or head–neck junction → stem tip
  - Two types:
    1. Cement + stem subside together within femur
    2. Stem subsides within cement mantle → **pistoning**
- **Medial midstem pivot**
  - Fixed pivot point → medial, mid-stem
  - Prosthesis rocks around it → moves **laterally**
  - Rocking → radiolucent lines appear
  - Lucency → proximal medial (**superomedial**)
  - Lucency → distal lateral (**inferolateral**)
- **Calcar pivot** (distal toggle)
  - Only calcar fixed → whole distal part loose
  - Distal stem moves right and left → **windshield-wiper effect**
  - **Ballooning** → distal stem
  - Distal area widens
  - Supported proximally, loose distally → keeps wiping back and forth

| Mode | Fixed point | Motion | X-ray sign |
|---|---|---|---|
| **Subsidence** | None | Downward; stem alone in cement = pistoning | GT tip / head–neck → stem tip distance |
| **Medial midstem pivot** | Medial mid-stem | Rocking → lateral | Superomedial + inferolateral lucency |
| **Calcar pivot** | Calcar | Distal windshield-wiper | Distal ballooning |

**Must-Know**

- Definite cemented stem loosening → four signs
- Stem fracture, cement fracture
- Cement–implant radiolucency
- Change in stem position: subsidence, medial midstem pivot, calcar pivot
- Subsidence 1–2 mm in first year → normal

## Cemented Cup Loosening Ask AI

- Example → cemented cup **postoperatively** vs 7 years
- Sign 1 → **radiolucency**
  - Normal **cement mantle** → lucency / bone resorption all around cup
  - Lines > 2 mm, extensive, **progressive**
  - Over years → **all three zones:** superior, middle, inferior
- Sign 2 → **migration** or subsidence
  - Direction → superior or medial
  - Medial → entering **protrusio** → loose cup
  - ± **Medial wall fracture**
- Sign 3 → **change in inclination** → loose cup
- Other signs:
  - **Cup wear**
  - Cup or cement fracture → rare

| Sign | Cementless cup | Cemented cup |
|---|---|---|
| **Migration** | > 2–5 mm | Superior or medial → protrusio ± medial wall fracture |
| **Radiolucent lines** | All three zones; > 2 mm; progressive over 2 years | All around cup; > 2 mm; extensive; progressive; all three zones |
| **Inclination** | Change (± version, as heard) → loose | Change → loose |
| **Other** | — | Cup wear; cup or cement fracture (rare) |

## Take-Home Message Ask AI

- **Osteolysis** → definition
- Clinical picture:
  - **Aseptic vs septic** loosening
  - Start-up pain
  - Related laboratory investigations
- **Radiological diagnosis** → loosening of cemented
  - And cementless components; management

# Instability

*(Causes, Diagnosis and Management of Dislocation After THA)*

**Prof Dr Ahmad Azzam**

**Watch Original Lecture**

*Lecture transcript processed with AI assistance.*

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# AI detected corrections (Mostly bad transcripts)

## Queried - kept as spoken, but looks wrong

**Q1** (p. 268) as spoken; 70% of the first-year 75% is ~52%, not 90% — two conflicting figures

# Instability After THA — Overview Ask AI

## Learning Objectives Ask AI

- Instability → many contributing factors
- Objectives
  - Magnitude of the problem
  - **Risk factors** for instability
  - **Multifactorial** nature of the problem
  - **Management algorithm** for instability after total hip arthroplasty (THA)
- **Javad Parvizi:** despite decades of progress in the profession
  - No answer yet to instability after hip arthroplasty
  - Advances in implants and surgical technique → problem not yet behind us

## Magnitude of the Problem Ask AI

- **Incidence:** ~3% of primary THA in most series
  - Higher after revision THA
- **Timing:** early vs late dislocation
  - Late → after ~5 years from surgery
  - Early → before that
- 75% of instability → **first year** (12 months)
  - Of these, ~70% → first 3 months
  - Also quoted: ~90% of dislocations within first 3 months *[Q1 · as spoken; 70% of the first-year 75% is ~52%, not 90% — two conflicting figures]*
- **Cumulative risk** persists after first year
  - ~1% per year → first 2–3 years
  - Then ~1% every 5 years
- **Late dislocation** (> 5 years)
  - More than half recurrent
  - Most → revision surgery

**High Yield**

- Incidence ~3% after primary THA
- 75% within first year; ~70% of these within first 3 months
- Cumulative risk ~1%/year for 2–3 years, then ~1% per 5 years
- Late (> 5 years) → mostly recurrent, mostly revision

## Diagnosis Ask AI

- Same as native hip dislocation
  - Sudden **pain**
  - **Loss of function**
  - **Rotational deformity** → apparent immediately
  - **Limb shortening**

# Risk Factors Ask AI

## Categories Ask AI

- Two axes → patient-related vs surgical; modifiable vs non-modifiable

| Category | Modifiable? | Note |
|---|---|---|
| Patient-related | Essentially non-modifiable | Can still be optimised |
| Surgical | Modifiable | Totally avoidable |

## Patient-Related Factors Ask AI

- **Spinopelvic pathology** / lumbar spinal fusion → ↑ instability
  - Alters hip mechanics and stability
  - Intraoperative effect → **component positioning**
  - Postoperative effect → hip **dynamics**
    - Sitting and standing → ↑ chance of instability
- **DDH** → higher incidence of instability (see Hip Dysplasia)
- Anatomic factors
  - **Abductor insufficiency**
  - **Trochanteric malunion / non-union** — esp. with migration
    - **Fibrous non-union** → better prognosis than non-union with migration
  - Higher preoperative **range of motion**
  - **Spinopelvic dysfunction**

## Surgical Factors Ask AI

- Modifiable, totally avoidable
  - **Component malposition** → most important, highest incidence, commonest cause of dislocation
  - **Primary stability** of implant (primary fixation)
  - **Surgical approach** — posterior vs anterior
  - **Soft tissue tensioning** and capsular repair
  - **Larger femoral head** use
  - **Surgical expertise**
    - Low annual THA volume → ~50% higher dislocation rate
    - **Learning curve** → very important

# Clinical Assessment and Work-up Ask AI

## History Ask AI

- Beyond diagnosing the dislocation, establish
  - **Time** from surgery to dislocation
  - **Surgical approach** used
  - **Incident position** — position in which hip dislocated
  - Clinical signs of **infection**

### Laboratory Ask AI

- **Infection markers** → if infection suspected
  - **Aspiration** → positive markers or clinical judgement

### Imaging Ask AI

- Standard AP and **cross-table lateral** radiographs
- **Standing and sitting views** → spinopelvic mechanism, possible missed problem
- **CT** → accurate component position
  - Esp. **version** of acetabular and femoral components
- **MRI with MARS** (metal artifact reduction sequence) → late instability
  - Suspected **osteolysis** and wear

## Classification of Instability Ask AI

### Dorr Classification Ask AI

- Type 1 → **Positional**
- Type 2 → **Soft tissue imbalance**
- Type 3 → **Component malpositioning**

### Wera Classification Ask AI

- Based on 75 revisions of total hips for dislocation → six types

| Type | Cause |
|---|---|
| 1 | Acetabular malposition |
| 2 | Femoral malposition |
| 3 | Abductor insufficiency |
| 4 | Impingement |
| 5 | Late wear |
| 6 | Unresolved (unknown) etiology — no diagnosis or main factor after full work-up |

- Highest recurrent dislocation after revision → **abductor insufficiency** group
  - Lecturer's view: revision protocol only partially addresses abductor insufficiency

**Exam Pearl**

- Wera: six types, from 75 revisions for dislocation
- Worst recurrence after revision → abductor insufficiency
- Dorr type 3 → component malpositioning

# Spinopelvic Dynamics Ask AI

## Normal Mechanism Ask AI

- Standing → sitting → accommodative change in pelvic position
  - Pelvis tilts posteriorly ~20–35°
- **Radiological indices** of spinopelvic dynamics
  - **Pelvic incidence**
  - **Sacral slope**
  - **Pelvic tilt**

## Dynamic Component — Fusion or Stiff Spine Ask AI

- **Spinopelvic fusion**
- Limited spinal **range of motion**
- Either → **accommodative mechanism** fails
- Sitting normally → pelvis tilts backwards
  - ↑ **acetabular anteversion**
  - Anterior acetabular rim moves away from femoral neck → no **impingement** in sitting
- Mechanism lost → ↑ instability in sitting

## Static Component — Fixed Sagittal Deformity Ask AI

- **Fixed sagittal spinal deformity** with loss of lumbar lordosis (flat back) → altered mechanics
- Compensation for upright posture → pelvis tilts backwards
  - → ↑ instability in standing

| | Dynamic | Static |
|---|---|---|
| Cause | Spinopelvic fusion / limited spinal ROM | Fixed sagittal deformity (flat back) |
| Pelvic behaviour | No posterior tilt on sitting | Compensatory posterior tilt |
| Unstable position | Sitting | Standing |

## Outcome and Planning Ask AI

- Instability after **spinal fusion** ~8%
  - Despite correct technique and component position
- → address more than one factor
- Preoperative plan
  - Assess **deformity**
  - Assess spinal **mobility** vs stiffness
  - Plan **implant design**

# Hip Dysplasia Ask AI

- **DDH** → higher instability rate; more than one factor
- **Component positioning** → highly relevant
  - **Shallow acetabulum** → more medialized cup

  - **Deficient walls** → exaggerated anteversion
    - Both acetabular and femoral components

## Component Positioning Ask AI

- **Component malposition** → most important, most modifiable surgical factor

### Lewinnek Safe Zone Ask AI

- Target zone for cup position
  - **Anteversion** 15 ± 10°
  - **Inclination** 40 ± 10°
- Within zone ≠ guaranteed stability
- Later studies questioned safe zone → other targets proposed

### Combined Anteversion Ask AI

- **Combined anteversion** = acetabular + femoral anteversion
  - One component compensates for the other
  - Optimal → 30–50°

**Must-Know**

- Lewinnek: anteversion 15 ± 10°, inclination 40 ± 10°
- Safe zone ≠ guaranteed stability
- Combined anteversion target → 30–50°

### Pitfalls in Achieving the Safe Zone Ask AI

- Narrow safe zone → risk in more than one direction → precise positioning needed
- Patient factors (**obesity**, female pelvis) → ↑ error
- **Lateral decubitus** → misjudged cup position
- **Gender** differences in pelvis
- Pelvic position change → lateral decubitus vs supine
- **Residual osteophytes**, esp. anterior wall → intraoperative misjudgement of cup version
- **Wall deficiencies**
- **Distorted femoral version** (table)

| Femoral version | Causes |
|---|---|
| Increased anteversion (native) | DDH, juvenile rheumatoid arthritis |
| Less anteverted / retroverted (native) | Previous trauma, Legg–Calvé–Perthes disease |
| Neutral / less anteverted neck | Lower neck cut |

### Ensuring Correct Position Ask AI

- Intraoperative reference aids → cup oriented to fixed anatomical landmark
  - e.g., **transverse acetabular ligament**

### Surgical Approach Ask AI

- Dislocation rate → anterior approach lower than **posterior approach**
- More epidemiological and **anatomical risk factors** → lower-risk approach preferred

## Soft Tissue Tension Ask AI

- **Soft tissue tension** → maintains coaptation of femur and acetabulum
  - Influenced by many factors
- Restore offset and leg length
  - Offset in vertical and anteversion planes
  - Unqualified "offset" → usually **medial offset**
  - Medial offset + leg length restoration → key to correct tension
- Not a discrete surgical step → depends on **preoperative planning**
  - Component size
  - Component position
  - Offset and leg length restoration
- **Posterior soft tissue repair** → better tensioning
- **Trochanteric non-union** with migration → more instability than fibrous non-union
  - **Trochanteric advancement** → may improve stability in some cases

## Patient Education and Support Ask AI

- Patient repeats **instructions** before discharge
  - Restates them at follow-up visits
- Maintain patient **independence**
  - **Assistive tools** for getting up from / sitting on bed
  - → no violation of instruction safe zone when getting out of bed
- **Caregivers** → major help
  - Dementia, alcoholism, cognitive impairment

## Femoral Head Size and Capsular Repair Ask AI

### Femoral Head Size Ask AI

- **Larger femoral heads** → among best measures against instability
- Biomechanics of larger head
  - Greater **jump distance**
  - Greater range of motion before **impingement**
  - → lower dislocation risk
- Wear concern
  - Conventional polyethylene → higher wear
  - **Highly cross-linked polyethylene** → improved
  - Concern persists → wear / trunnion problems with metal heads

### Capsular Repair Ask AI

- **Capsulectomy** without capsular repair → major cause of instability
  - Unrepaired capsule → early postoperative dislocation, ↑ instability

## Management Ask AI

### First Step Ask AI

- Rule out **infection** first

### Treatment by Type Ask AI

- Dorr types 1–2 (positional, soft tissue imbalance) → best treated without component revision
- Options
  - **Liner exchange**
  - **Elevated-rim liner**
  - Larger head
  - **Dual mobility** or constrained liner
  - **Trochanteric advancement**

### Closed Reduction Ask AI

- **Closed reduction** → most successful in early, first-time dislocations
  - **Sedation and analgesia** usually enough; general anesthesia rarely needed
  - Longitudinal traction under **image intensifier** → abduction → usually reduces
  - Traditional hip reduction methods also usable
- Polyethylene liner may **dissociate** during dislocation, before patient arrives

**Pitfall**

- Liner may dissociate during the dislocation episode
- Capsulectomy without repair → early postoperative dislocation
- Always rule out infection first

### Post-Reduction Mobilization Ask AI

- Recurrent dislocation → **hip abduction brace**
  - ~20° abduction
  - Flexion limited to 60°
  - Duration → 6 weeks to 3 months

### Isolated Head/Liner Exchange Ask AI

- Literature: **isolated head/liner exchange** with larger head → well-positioned components
- Lecturer: selective use
  - Even 10° malposition → component revision required
  - Shell not changed → larger head preferred

### Increased Constraint Ask AI

- Methods to increase constraint → **dual mobility**, constrained liners
  - Useful in **multifactorial** instability
  - Many patients → multifactorial or **idiopathic** (cause unknown)

### Soft Tissue Reconstruction Ask AI

- Preferred in proven **abductor deficiency**
  - **Achilles tendon allograft** or synthetic graft

### Resection Arthroplasty Ask AI

- **Last resort**
  - Multiple dislocations
  - Severely debilitated
  - Failed appropriate management
  - Non-ambulatory
  - Severe cognitive impairment

**Clinical Pearl**

- Closed reduction → best in early, first-time dislocation
- Traction under image intensifier, then abduction
- Recurrent → brace: ~20° abduction, flexion ≤ 60°, 6 weeks–3 months

## Take-Home Message Ask AI

- Instability → important, common, highly **multifactorial**
- **Prevention** better than treatment → do the primary THA correctly from the start
- Search for the **true cause** → apply appropriate management
- **More than one treatment modality** → liner and soft tissue improvement best options
- **Component malposition** and loosening
  - → best treated by **revision**

| Cause | Treatment |
|---|---|
| Positional / soft tissue imbalance | Liner exchange, elevated rim, larger head, dual mobility / constrained liner, trochanteric advancement |
| Multifactorial / idiopathic | Dual mobility / constrained liner |
| Abductor deficiency | Soft tissue reconstruction (Achilles allograft, synthetic graft) |
| Component malposition / loosening | Revision |
| Failed management, debilitated, non-ambulatory | Resection arthroplasty |

# LLD Nerve Palsy

*(Neurovascular Injury, Leg-Length Discrepancy and Nerve Palsy)*

**Prof Dr Khaled Fawzy**

**Watch Original Lecture**

*Lecture transcript processed with AI assistance.*

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# AI detected corrections (Mostly bad transcripts)

## Corrected - the text was changed; the transcript said

- **C1** (p. 283) “shortening less than 1 cm, 8%” → “shortening more than 1 cm, 8%”
- **C2** (p. 286) “shortening less than 1 cm, 8%” → “shortening more than 1 cm, 8%”

## Vascular Injury — Anatomy, Incidence and Mechanisms Ask AI

- Session: two talks → **neurovascular injury**, then leg length discrepancy (LLD)
- Hip muscles stripped → complex regional neurovascular anatomy exposed
  - Not limited to **sciatic nerve**, femoral and external iliac vessels
  - Numerous other vessels present
  - **Superior gluteal vessels** → not to be forgotten
- **Vascular injury** incidence: 0.2%–0.3%
- Commonly injured vessels
  - **Iliac**
  - **Femoral**
  - **Superior gluteal**
  - **Obturator**
- Mechanisms of injury
  - **Sharp instruments** → knife, osteotomes
  - **Blunt instruments** → retractors → compress, stretch or tear vessel
  - **Screw placement** → through acetabular shell of cementless cup

## Acetabular Quadrant System Ask AI

- Acetabulum → four quadrants → **safe zone** for screws: posterior-superior
- Safe zone — structures still at risk
  - **Sciatic nerve**
  - **Superior gluteal nerve and vessels**
  - → Screws within the lines, not at their periphery
- Construction
  - Line 1: **ASIS** → center of acetabulum
  - Line 2: perpendicular to line 1
- Four zones

| Zone | Quadrant | Structures at risk |
|---|---|---|
| Safe | Posterior-superior | Safest → screws placed here |
| Death | Anterior-superior | Iliac vessels → perforation → death on table or later |
| Cautious | Posterior-inferior | Screws up to 20 mm; over-penetration → sciatic nerve, inferior gluteal vessels, pudendal nerve and vessels |
| Danger | Anterior-inferior | Obturator vessels |

- **Posterior-superior quadrant** → safest → screw placement site
- Pelvis viewed from inside → screw perforations and structures affected
  - **Superior gluteal** vessels
  - **Iliac** vessels

  - **Femoral** vessels
  - **Obturator** nerve
  - **Pudendal** nerve
  - → Screws placed with great care

**High Yield**

- Quadrant lines → ASIS to acetabular center + perpendicular
- Safe zone → posterior-superior; sciatic and superior gluteal still at risk
- Death zone → anterior-superior → iliac vessels
- Danger zone → anterior-inferior → obturator vessels
- Cautious zone → posterior-inferior → screws up to 20 mm

# Vascular Injury — Risk Factors, Presentation and Management Ask AI

## Predisposing Factors Ask AI

Vascular

- **Revision surgery**
- **Atherosclerosis** → vessel tears easily with malpositioned retractor
- **Previous vascular surgery** or vascular injury
- **Abnormal anatomy** → e.g. dysplasia

## Presentation Ask AI

Vascular

- Intraoperative → **bleeding**; never underestimate
- Postoperative
  - **Occult bleeding** → nothing visible externally unless in drains
    - Blood pressure falls, heart rate rises
  - **Ischemia**
  - → Postoperative vigilance; regular examination

## Cement-Related Injury Ask AI

Vascular

- Over-reaming → **acetabular floor** breached → cement extrudes into pelvis
  - **Heat of polymerization** → vessel injury → thrombosis
  - Postoperative erosion into iliac vessels → **pseudoaneurysm**
- Case: marked cement extrusion → vessels spared → **obturator nerve** irritated

## Management Ask AI

Vascular

- Intraoperative bleeding
  - Control, transfuse, pack field
  - Call **vascular surgeon** or interventional radiologist without hesitation
  - Never underestimate
- Postoperative
  - **Angiography**
  - Repeated clinical examination
  - Vascular surgeon as soon as possible

## Nerve Injury in THA Ask AI

- **Nerve injury** incidence: 1% — somewhat higher than vascular
- Nerves at risk
  - **Sciatic** → most commonly injured, especially posterior approaches
  - Then femoral, superior gluteal, obturator
  - Anterior approach → **lateral cutaneous nerve of thigh**
- Mechanisms
  - **Sharp laceration** of sciatic nerve by knife
    - Rare
  - Commoner: **retractor pressure** → traction, stretching, contusion
  - Postoperative **hematoma** → sciatic nerve compression
  - Postoperative **dislocation** → nerve injury
- Risk factors
  - **Female sex**
  - **Revision**
  - **Dysplasia**
- Approach-specific risk

| Approach | Nerve at risk |
|---|---|
| Posterior | Sciatic — slightly more at risk |
| Lateral | Superior gluteal |
| Anterior | Lateral cutaneous nerve of thigh |

## Sciatic Nerve Injury Ask AI

### Pattern Ask AI

Sciatic

- **Lateral popliteal (common peroneal) division** → commonly injured
  - Less fat between fasciculi → less protection
  - Lateral, not medial, position
  - Relatively **tethered at sciatic notch**
- → **Foot drop**: common presentation

**Exam Pearl**

- Peroneal division injured → less interfascicular fat, lateral, tethered at sciatic notch
- Result → foot drop

### Outcome Ask AI

Sciatic

- Recovery categories

| Outcome | Proportion |
|---|---|
| Good recovery | 40% |
| Mild residual motor or sensory symptoms | 45% |
| Severe motor or sensory deficit | 15% |

- Recovery possible up to 2–3 years → long regeneration distance
- **Axonotmesis** → regeneration 1 mm per day

### Diagnosis Ask AI

Sciatic

- Clinical examination
- **EMG** and nerve conduction studies → level of lesion, signs of recovery

### Management Ask AI

Sciatic

- Any foot drop → **foot drop splint**
- Acute postoperative nerve injury
  - Compressive hematoma → **evacuate**
  - Dislocation → **reduce**
  - Persistent palsy → **explore nerve**
  - Excessive lengthening → reoperate → **shorten leg** (see LLD)

### Prevention Ask AI

Sciatic

- Identify nerve intraoperatively where necessary
- **Short external rotators** reflected laterally → sciatic nerve protected
- **Knee flexion** → relaxes sciatic nerve
- Lateral approach → no gluteal split > 5 cm above **tip of greater trochanter**
- Careful **retractor** placement

## Neurovascular Injury — Summary Ask AI

- Neurovascular injuries → uncommon
- Know anatomy; care with **knife and osteotomes**
- Retractors → appropriate position
- **Sciatic nerve** → protect in posterior approach
- No screws in acetabular **death and danger zones**
- **Bleeding** → never underestimate

## Leg Length Discrepancy — Normal Variation and Incidence Ask AI

- Second talk: **leg length discrepancy (LLD)**
  - Audience poll → majority have seen LLD after hip replacement
  - Test: **femur protrudes forwards** → leg possibly lengthened
- Normal variation
  - Many normal individuals → 5–15 mm difference between legs

  - Not everyone perfectly symmetric
  - Majority unaware if < 1 cm
- LLD after **total hip replacement (THR)**

| Finding | Incidence |
|---|---|
| Shortening or lengthening within 10 mm → unimportant, patient unaware | ~72% |
| Over-lengthening > 1 cm | 22% |
| Shortening > 1 cm | ~8% *[C1 · transcript: "shortening less than 1 cm, 8%"]* |
| Noticeable or consequential difference overall | ~30% |

### Consequences Ask AI

LLD

- Surgeon forced to **revise** hip
- Patient may **sue** surgeon
  - USA: winning case → compensation + **punitive damages**
    - e.g. $10,000 compensation + $50,000–$100,000 penalty for lack of care
  - → Care: risk of **litigation**

## Preoperative LLD and Effects of LLD Ask AI

### Preoperative LLD Ask AI

- Common; causes
  - **Acetabular erosion**
  - **Femoral head collapse**
  - **Subluxation**
  - **Fixed deformities**
- Example: acetabular articular cartilage erosion + damaged femoral head

### Effects of LLD Ask AI

- Especially postoperative
  - **Limp** → dips on shorter side
  - Large **shoe raise** forced → very distressing
- Normal spine + LLD
  - **Pelvic obliquity**
  - **Compensatory scoliosis** or list
  - **Tiptoeing** on shorter side
  - **Knee flexion** on longer side, especially standing
- Other effects
  - **Abductor strain** → pain or ache
  - Decreased **abductor lever arm**
    - → Abductors generate more force → increased **joint reaction force**
      - → Long-term loosening and wear

# Causes and Types of LLD Ask AI

## Shortening Ask AI

LLD

- Shortening → may lead to **dislocation**
- Causes
  - **Short head**
  - **Acetabular cup** placed too high
  - Deeper **stem insertion**
  - Most important: **failure to restore offset** — horizontal and vertical
    - → Decreased myofascial tension

## Lengthening Ask AI

LLD

- Lengthening > 2.5 cm → **nerve palsy**
  - Sciatic, femoral or both → sciatic commonest

## Types of LLD after THR Ask AI

- **Type 1 — component-related**
  - Femoral component **too proud** → lengthening
  - Cup too high or too low
  - **Neck cut** too long or too short
- **Type 2** → components malpositioned in retroversion or anteversion
  - → Hip unstable → **long neck** deliberately used for stability

**High Yield**

- Shortening → dislocation
- Lengthening > 2.5 cm → nerve palsy; sciatic commonest
- Key cause of shortening → failure to restore horizontal and vertical offset
- Type 2 LLD → long neck to stabilize malpositioned components

# Acceptable LLD and Preoperative Planning Ask AI

- Acceptable LLD
  - Mean in most studies: ~3–9 mm (figure unclear in recording)
  - **Within 1 cm** → acceptable; reinforced by Charnley
  - More → **unhappy patient**
  - Reported range: 22 mm shortening → 3.5 cm lengthening
- Minimizing LLD
  - **Preoperative examination**
  - **Preoperative planning**
  - **Intraoperative measurement and adjustment**
    - Postoperative → too late; only shoe raise left

## Preoperative Examination Ask AI

LLD

- Check **fixed deformities**
- Measure **apparent and true leg length**

### Preoperative Planning Ask AI

LLD

- Radiographs → **preoperative planning**
  - **Standard- vs high-offset** prosthesis
  - Stem insertion depth
  - Cup position
  - **Digital planning** → software on market
    - → Restore center of rotation
- Acetabulum → size, coverage, **level of insertion**
  - No medialization or lateralization of cup
- Stem insertion and **neck cut** → planned properly
- Femoral component
  - Restore **center of rotation of head**
  - Check component size and neck cut

## Intraoperative Assessment of Leg Length Ask AI

- **Supine position** (some surgeons) → easy
  - Square pelvis → palpate medial malleoli → almost same level
- **Lateral position** → difficult, tricky
  - **Longitudinal traction** → joint surfaces separate a few mm; depends on
    - Force applied
    - Muscle relaxation by anesthetist
    - Extent of capsular release
    - e.g. lateral approach → one third glutei detached → traction against two thirds → more separation
  - **Soft-tissue tension** → longer leg → tight hip
  - **Knee-flexion test:** lateral, hip extended, knee flexed → leg kicks back on release → too long
- Measurement devices
  - Dedicated or simple: **Steinmann pin** in ilium → distance to tip of greater trochanter
    - Especially after trial reduction
  - More complex devices
  - **Computer navigation** → used in the West
    - Not every operation; not available in every hospital
  - Simple alternative: **intraoperative X-ray** on trial reduction

### Restoring Offset Ask AI

LLD

  - Very important: restore **center of rotation** + horizontal and vertical offset
  - Not done → lax soft tissues
    - e.g. short neck cut + lax tissues → tendency to lengthen leg → stabilize joint, minimize dislocation
  - Longer leg → **unhappy patient**

**Pitfall**

- Short neck cut + lax soft tissues → leg lengthened to stabilize hip
- Longer leg → unhappy patient

## Postoperative LLD and Management Ask AI

- Majority of postoperative inequality → **functional**
  - Resolving flexion soft-tissue contractures
  - Flexible pelvic obliquity
- **True LLD within 1 cm** → asymptomatic
- Symptomatic LLD > 1.5 cm → most probably **shoe lift** or shoe insert
- **Bilateral hip disease** → further chance to equalize at second operation

### Revision for LLD Ask AI

- Avoid if possible → shortening risks **destabilizing joint**
- Indications
  - Significant shortening → dislocation → revise to **longer head**
  - **Severe nerve palsy** from significant lengthening → reoperate, shorten leg
    - → Nerve recovery hoped for
  - Patient intolerant of large **shoe raise**
- Shortening leg → great care not to **destabilize hip**

## Leg Length Discrepancy — Summary Ask AI

- **Lengthening** commoner than shortening after THR
  - Majority within 1 cm; many patients unaware
- **Shortening** tolerated better than lengthening
  - > 1 cm not tolerated
- LLD after THR → cannot be eliminated, **can be minimized**
- **Revision** → avoid where possible

**Must-Know**

- Normal individuals → 5–15 mm difference; < 1 cm unnoticed
- After THR → over-lengthening > 1 cm 22%, shortening > 1 cm ~8% *[C2 · transcript: "shortening less than 1 cm, 8%"]*
- Acceptable LLD → within 1 cm (Charnley)
- Symptomatic > 1.5 cm → shoe lift or insert

# PJI

*(Case-Based Discussion of Periprosthetic Joint Infection)*

**Prof Dr Mohamed Abdel Moneim**

**Watch Original Lecture**

*Lecture transcript processed with AI assistance.*

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# AI detected corrections (Mostly bad transcripts)

## Queried - kept as spoken, but looks wrong

**Q1** (p. 296) As spoken; colistin is the nephrotoxic drug, tigecycline needs no renal adjustment - reasoning looks reversed.

# Case 1 — Chronic PJI After Implant Removal Ask AI

## Presentation Ask AI

Case 1

- Not yet on dialysis; potassium reportedly acceptable
- General condition reasonable for 1 year
- Previous surgeon → **hip joint removed** ~6 months ago
- Current status: **wound closed**, patient bedridden
- Difficult case: wound closed + **no prior work-up** available

## Pre-operative Imaging Ask AI

Case 1

- Pre-operative images at clinic presentation
- Unusual finding: prosthesis removed but **cement left in femur**
  - Retained cement → no blood supply → avascular foreign material

## Pre-operative Planning Ask AI

Case 1

- Direct re-implantation → no
- Planned **second stage** → what to request before theatre?
- **Aspiration** first → establish and confirm diagnosis before surgery
- Investigations
  - **ESR** 80 (figure 125 also quoted — unclear)
  - **CRP** 48 (normal 6)
- Hip **aspiration** — reasons
  - Wound closed → no sinus
  - No documentation of previous **cultures**
  - Without it → **organism** and antibiotic unknown
- Aspiration done → culture results obtained
- Management: **two-stage** approach → starting with first stage

**Clinical Pearl**

- Closed wound, no sinus, no prior cultures → **aspirate** before surgery
- Unknown organism → no targeted antibiotic

# First Stage — Cement Removal and Spacer Ask AI

## Cement Removal Options Ask AI

- Instruments: **burr** or others
- Two options
  - **Intramedullary removal** → through femoral canal, femur not osteotomized
  - **Femoral osteotomy** → cement removal

## Case 1 Procedure Ask AI

Case 1

- Retained cement removed → thorough **debridement**
- Antibiotic-loaded cement spacer → **PROSTALAC**
- Cup: standard cemented cup, **snap-fit** type

  - Snap-fit → head held firmly → dislocates less than standard cup
  - Slightly different from a **constrained** cup

### Spacer Stem Selection Ask AI

PROSTALAC

- Always choose **extra-small (thin) stem** → maximum cement around it
  - Goal: whole prosthesis **covered with cement**
  - ↑ antibiotic **elution,** ↑ antibiotic surface area
  - Allows **immediate mobilization**
- Image after first stage

## Interval Antibiotic Protocol Ask AI

- Question: how are antibiotics given after first stage?
- Case 1 operated 2 years ago → older protocol (2013–2016 practice)
  - Antibiotics per **culture** results
  - Targeted by **sensitivity**
  - 6 weeks: 2 weeks IV → 4 weeks oral
  - End of 6 weeks → antibiotics **stopped**
  - ESR/CRP repeated → confirm return to normal
  - After stopping → ESR/CRP repeated again
- Antibiotic-free interval → **drug holiday**
- Current practice: **no drug holiday** → proceed to revision
  - 2 weeks IV + 4 weeks oral antibiotics
  - Then check **ESR/CRP** → usually proceed to revision
  - Option: **extend antibiotic course**, even beyond 4 weeks
  - Never an antibiotic-free rest period
- Case 1 → **revision** chosen; pre-op image re-shown

| | 2013–2016 protocol | Current protocol |
|---|---|---|
| Antibiotic choice | Culture + sensitivity | Culture + sensitivity |
| Course | 2 weeks IV → 4 weeks oral | 2 weeks IV → 4 weeks oral, may extend |
| After course | Stopped → drug holiday | No drug holiday |
| ESR/CRP | Repeated at end and after stopping | Checked → usually revision |

**High Yield**

- Interval antibiotics: 2 weeks IV → 4 weeks oral
- **Drug holiday** → no longer recommended
- Course may be extended beyond 4 weeks
- ESR/CRP → then usually proceed to revision

# PROSTALAC Spacer — Principles and Technique Ask AI

## Why the Spacer Is Placed Loose Ask AI

- Q: why does a spacer become **well fixed** → hard removal → more bone removed?
- Answer: inserted at first stage, when ESR/CRP not yet good
- **Lucency** all around the cup on first-stage image
  - Cup not pressurized → simply placed → comes out easily
  - Nothing fixing it initially
  - First stage → cement only used to place it
- Construct differs from **G-stem** shape
- Cement spacer shapes familiar; hemi (head-only) form also referred to (unclear)

## Rationale Ask AI

PROSTALAC

- Functions in the **interim** period; simple principle
- Stem, cup, all components completely **loose** → easy removal
- Aim: allow **weight bearing**
- **Dislocation** incidence lower than G-stem
- **Acetabular erosion** lower than hemi spacer
  - Hemi spacer moves in acetabulum → erodes it if left longer

| | PROSTALAC | Comparator |
|---|---|---|
| Dislocation | Lower | G-stem |
| Acetabular erosion | Lower | Hemi spacer → moves in acetabulum, erodes if left longer |

## Spacer as Foreign Body Ask AI

- Antibiotic-loaded **G-stem** → elution lasts 4–6 weeks
- After 4–6 weeks → cement = **foreign body**
  - Both spacer types → foreign body
- ↓ elution → uncleared bacteria adhere → **biofilm**
- Accept G-stem → no major objection to **PROSTALAC**
- Second point (any implanted joint)
  - Patient **cured** → single revision possible → remove all, place joint, mobilize
  - Patient **still infected**
    1. → must remove a **fixed** prosthesis
    2. Bone stock loss, **osteotomies**, revision implants
    3. Unneeded procedures at both first and second stages

## Loose Insertion Technique Ask AI

PROSTALAC

- Construct deliberately **loose**
  - Cup: hold in hand → apply cement by hand
  - Place in acetabular position without **pressurization**
  - Not cement first then pressurize
  - Stem: hold in hand → coat with cement → insert into femur when cement near setting

  - Result: no **interdigitation** with bone → easy removal
- Allows movement for a limited period
  - e.g. 2–3 months
  - Cannot be left long → loosening, **periprosthetic fracture**

**Pitfall**

- Pressurized cement → interdigitation → **fixed** spacer
- Fixed spacer + persistent infection → bone stock loss, osteotomies, revision implants
- Spacer left too long → loosening, periprosthetic fracture

## Case 1 — Second Stage Ask AI

- Patient **renally impaired** → creatinine 7, about to start dialysis
- → **whole cemented revision**
- End of Case 1

## Case 2 — Early Post-operative Infection Ask AI

### Presentation Ask AI

Case 2

- Age 30 years; no comorbidities
- **AVN** → total hip arthroplasty 3 weeks earlier
- Pre-operative image
- Post-operative image
- Post-op joint appears good
- Presented 3 weeks after surgery
  - **Tenderness**
  - **Pain on weight bearing** (± at rest — unclear)
  - **Hotness**
  - **Serous discharge** from sutures
- At 3 weeks → **DAIR**
- Q: straight to theatre, or culture first?
- Participant answer → judged correct

### Management — DAIR Ask AI

Case 2

- **Do not wait**
  - Theatre → **debridement**
  - Change **modular components** → retain implant
  - Intra-op **cultures** → targeted antibiotics post-op
- Must not delay; review whether index operation performed properly
- Signs evident: high temperature, discharge
- Delay (a week, 5 days, 9 days) → infection established → **chronic** infection
- Catch early; otherwise situation lost
- Outcome: DAIR → patient did well

**Exam Pearl**

- Early infection at 3 weeks → **DAIR**
- Debridement + modular exchange + implant retention + cultures
- Delay → **chronic** infection

# Case 3 — Chronic PJI With a Sinus Ask AI

## Presentation Ask AI

Case 3

- Male, 36 years; **multiple operations** for total hip replacement
- Last procedure 5 years ago (image shown)
- 1 year later → **sinus**
- Presents with image + sinus → management?
- Q: what first before theatre?
- Participant proposal: first stage

## Pre-operative Work-up Ask AI

Case 3

- **No aspiration** needed → sinus present
  - **Culture** from sinus
  - Point of disagreement with participant ("aspiration")
- **ESR/CRP** drawn
  - Not to prove infection → sinus itself confirms it
  - Purpose: post-operative **follow-up**, as in other case
- ESR/CRP = **baseline**, even if very high with obvious clinical picture
  - Post-op repeat improved → reassuring
  - Not improved → another risk not eliminated
- Done: ESR/CRP + **culture & sensitivity**

| | Closed wound, no sinus (Case ) | Sinus present (Case ) |
|---|---|---|
| Aspiration | Required | Not needed |
| Culture source | Hip aspirate | Sinus |
| Infection proof | Aspiration confirms diagnosis | Sinus confirms infection |
| ESR/CRP | Requested pre-op; repeated in interval | Baseline for post-op follow-up |

**Clinical Pearl**

- Sinus present → **no aspiration**; culture from sinus
- ESR/CRP → baseline, not proof of infection

## First Stage Ask AI

Case 3

- Culture & sensitivity obtained
- **2-stage revision** → spacer
  - **Articulating** spacer preferable to static spacer

  - Options: G-stem, custom-made, PROSTALAC → surgeon preference
- **PROSTALAC** used
- Large **acetabular defect** → all filled with cement
  - Unlimited cement possible
  - Much cement around stem → avoid **fixed** prosthesis
- Image after first stage → proceed to second stage
- Q: imaging needed before second stage?

## Case 3 — Second-stage Planning and Reconstruction Ask AI

### Planning Ask AI

Case 3

- **CT** → plan second stage
- Prepare **revision implants**
- Patient image
- CT findings
  - Coronal cut → **cavitary defect** + superior defect
  - Axial → anterior and posterior walls; mostly **cavitary**, small part possibly segmental

### Reconstruction Ask AI

Case 3

- **Allograft**
- **Reinforcement ring** + cemented cup
  - Ring → cup **must be cemented**
  - No metal on metal
- **Cemented stem** preferred → possibly for extensive antibiotic loading
- Cement separates ring from cup
- **Press-fit** cup cannot sit directly on ring

**Exam Pearl**

- Reinforcement ring → cup **must be cemented**
- No press-fit cup on ring → no metal on metal

## Case 4 — Infected Hemiarthroplasty Ask AI

### Presentation Ask AI

Case 4

- **Hemiarthroplasty**, not THA → concept similar
- Female, 65 years; **renal** and cardiac disease (abdominal disease also mentioned — unclear)
- **Thompson** hemiarthroplasty 10 months ago
- 3 months of deterioration, possibly discharge
- Significant **renal** disease
- Image: Thompson prosthesis at presentation
- Previously mobile for simple activities (e.g. visiting her children)

## Work-up and Antibiotics Ask AI

Case 4

- **Culture** first
- ESR/CRP high
- Culture → highly resistant **Enterobacter**
  - Sensitive only to colistin and tigecycline → the two traditionally available
  - **Tigecycline** excluded → renal concerns → colistin given *[Q1 · As spoken; colistin is the nephrotoxic drug, tigecycline needs no renal adjustment - reasoning looks reversed.]*
- Options: second stage, or **Girdlestone** (participant) → two-stage attempted

## First Stage — Implant Removal Ask AI

Case 4

- Removal **without osteotomy** (same principle)
  - Ream cement gradually → **tibial locking-nail reamers**
  - Remove prosthesis gently → **preserve bone stock**
- Prosthesis removed without problems
- **PROSTALAC** → first stage
- **IV antibiotics** 2 months in hospital → wound healed, well
- Then **partial wound dehiscence** → readmitted 2 weeks, then another week → large wound
- Q: decision if she were 25 years old?
- Three weeks on → decision differs from a 25-year-old?
- Participant suggestion → **another stage**

## Decision — Third Stage vs Excision Arthroplasty Ask AI

Case 4

- Options: **three-stage** procedure or Girdlestone
- **Counsel** patient and family
  - High-risk → each surgery carries specific risk
  - Third stage (change spacer, restart) vs **excision arthroplasty**
- Family decision → excision arthroplasty
  - Reason: afraid of multiple further operations
- Joint removed → **excision arthroplasty**

> **Clinical Pearl**
> - High-risk patient, repeated failure → **counsel** family
> - Options: third stage vs excision arthroplasty (Girdlestone)

# Periprosthetic Fractures

*(Case-Based Discussion of Periprosthetic Fractures)*

**Prof Dr Mahmoud Abdel Karim**

**Watch Original Lecture**

*Lecture transcript processed with AI assistance.*

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## Case Presentation and History Ask AI

### The Case Ask AI

- **Hip arthroplasty** in place → 7 years
- Group exercise: classify the fracture — B, C, D, or other?
- Age: 75 years
- Full history gathered before the answer

### Pre-Fracture History Ask AI

- **Ambulatory status** before the injury
- **Pre-fracture pain**
  - Prior pain → loosening or **osteolytic process** already started
- **Anticoagulants** → relevant to both surgery and anaesthesia
- Trauma case → surgery not postponed a week
  - Exception: unfit for both general and **spinal anaesthesia** → wait
- **Neurological disorder** — very important
  - → drives implant choice: large head / tripolar / **dual mobility**
- Diabetes
- Hypertension
- **Diabetes** in this setting
  - Uncontrolled (glucose ~400) → postpone surgery
  - Internist or endocrinologist + anaesthesia team
  - Up to 300 sometimes accepted by anaesthetists → off guidelines
  - **Acute adjustment** of glucose sometimes performed
- Key point, above the case itself
  - Pain + **limited range of movement** since end of surgery
  - Pain since index operation → suspect **infection**
  - → ask about implant type and whether it is the relevant type

**Must-Know**

- Pain since the index operation → suspect infection
- Prior pain → loosening or osteolysis already started
- Neurological disorder → large head, tripolar, or dual mobility

## Imaging Work-up Ask AI

### Plain Radiographs Ask AI

- Imaging mandatory → X-rays before CT
- No planning on CT alone
- Required views
  - **Long lateral**
  - **Long AP**
  - Most important besides the long films

    - **AP pelvis** (both hips) → basic requirement
        - Both sides visible → contralateral comparison in **templating**
        - → gives the **offset**
- AP pelvis alone insufficient → **distal end** of prosthesis not visible
- **Long film** of entire prosthesis → AP + lateral

### Other Investigations Ask AI

- **Bone scan** → loose or not; infected or not
    - False positive here (fracture) → not needed
- Limited range of movement → reason for further studies
- **CT scan** — important
    - **Bone stock**
    - Fracture **extension** not seen on X-ray
    - Displacement
    - **Implant seating**
    - **Wear process**
    - Finer bone detail → fragments, comminuted fragments
    - Extent of **comminution**

### Before Calling B1 vs B2 Ask AI

| Item | Purpose |
|---|---|
| Pre-fracture status | Walking? Pain? |
| Radiography of entire femur | AP + lateral |
| CT | Prosthesis stability; wear of prosthetic components |
| Intraoperative test | Implant stability |

**Pitfall**

- Bone scan after periprosthetic fracture → false positive
- AP pelvis alone → distal end of prosthesis missed
- CT alone → never the basis for planning

## Vancouver Classification Ask AI

### Classification Systems Ask AI

- **Vancouver classification** → presented by Dr. Sayed; the more important
- **UCS** (Unified Classification System) → relevant for exams
- Two systems differ considerably → both must be known
- **Vancouver** → A, B, C only; no D
- This case → type B; B1 vs B2 to be justified

## Applying the Subtypes Ask AI

- **B** → fracture around the stem
- Audience proposals, challenged by the lecturer
  - Comminution → B2
  - Displaced → B3
- Treatment based on classification → brainstorming matters
- This case → **B2:** around the stem, loose implant, good bone
- B3 proposed because "unstable" → revisited below
- Unstable prosthesis → B2, not B3
- B1 excluded → is the stem loose?
- Stem not loose → **B1**

## Three Elements Ask AI

- **Location** → A / B / C
  - Proximal → greater or lesser trochanter
  - Around the stem
  - Distal
- **Prosthesis fixation** — the most important
  - Well fixed + trochanteric → A
  - Well fixed + around the stem → B1
  - Unstable / not well fixed / loose → **B2**
- **Bone quality** → loose stem may be **B3**
- Algorithm: **location** → **fixation** → **host bone**

| Type | Location | Stem | Bone |
|---|---|---|---|
| A — AG | Greater trochanter | Well fixed | — |
| A — AL | Lesser trochanter | Well fixed | — |
| B1 | Around or just distal to stem | Stable | — |
| B2 | Around or just distal to stem | Unstable (loose) | Good |
| B3 | Around or just distal to stem | Loose | Poor |
| C | Well distal to stem | — | — |

**High Yield**

- Vancouver → A, B, C; no type D
- Always: location → fixation → host bone
- Fixation is the most important element
- Loose stem + good bone → B2; poor bone → B3

## Ruling Out Infection Ask AI

- Poor bone possibly present from the start
- **Infection** → must be ruled out, per the infection lecture
- Preoperatively → look for active inflammation; check **inflammatory markers**
- **Examination** + history

| Test | Note |
|---|---|
| Alpha-defensin | Confirmation if still needed; expensive → not routine; acceptable if affordable and in dilemma |
| Leukocyte esterase | Confirmation if still needed |
| Frozen section | Needs in-hospital expertise → very difficult in Egypt |
| Cultures | Always taken, even if findings normal |

- **Any revision** → rule out infection
- Suspicion → **spacer** on standby, or know how to make one

**Must-Know**
- Any revision → rule out infection first
- Cultures sent even when the field looks normal
- Suspicion → spacer on standby

## Preoperative Preparation Ask AI

### Multidisciplinary Team Ask AI

- **Multidisciplinary team** → mandatory; patient can die easily
  - **Neurology** → neurological disorder; age 75–80
  - **Cardiology**
  - **Anaesthesia** → assessment; possible **ICU** need
- All must be involved
- Any history of illness or **comorbidity**

### Informed Consent Ask AI

- Family told about everything
- Lecturer's consent form → sent via WhatsApp to all hospitals
- One form per operation type (4–5 types) → all known **complications** listed
- Serves as a reminder → stated to patient and family

### Planning Checklist Ask AI

- **Planning** → most important step before theatre
- Never enter the operating room without knowing the plan
- Personal **checklist** → what to bring, do, have available
- Start a plan checklist → add to it with every patient

## Prevention of Periprosthetic Fracture Ask AI

- Best treatment → **prevention** (augmenting Dr. Sayed's lecture)
- Better than the most successful treatment → exam answer
- During arthroplasty
  - Avoid **cortical defects**
  - Avoid **malalignment**
- Regular **radiological follow-up**
- "Catch the patient before he catches you" → detect **osteolysis** and loosening before fracture
- **Loosening** seen (Dr. Sherif Khaled) → fracture prevented → best treatment

**Exam Pearl**

- Best treatment of periprosthetic fracture → prevention
- Avoid cortical defects and malalignment
- Regular radiological follow-up → catch osteolysis and loosening early

## Treatment of the B2 Fracture Ask AI

### Revision Plan Ask AI

- Agreed type → **B2**
  - Loose implant + **bone stock** present
- Algorithm (Dr. Sayed): fix or revise → **revision ± fixation ± augmentation**
- Steps
  - Remove the **implant**
  - Remove the **cement** → may or may not be easy
  - Fractured femur → **extended trochanteric osteotomy** possible
  - **Long stem** bypassing the fracture
    - By twice the femoral diameter, or 4–6 cm (yesterday's figure)
    - Lecturer uses 6 cm → **more cortical contact**
    - Large gap → less stable construct; consensus → more cortical contact

### Fixation Ask AI

- Long stem alone → fracture union?
- → **fixation** needed in addition
- Options → **plate, cable,** or cable-plate
  - All three → added expense
  - Evidence of superiority → questioned
- Cables alone may suffice in such fractures vs added plate
- Intraoperative principle
  - Place **cables** or a couple of wires
  - Assess construct **stability**
  - Add fixation only if needed

**High Yield**

- B2 → revision ± fixation ± augmentation
- Long stem bypasses fracture → lecturer uses 6 cm
- More cortical contact → more stable construct
- Cables first → add plate only if unstable

# Instruments and Implants Ask AI

## Prepared Kit Ask AI

- **Instruments** → very important; any revision fully prepared
- Plan included
  - **Dual mobility cup** → size unknown in advance
  - **Cerclage**, wire passers, Biznoska bands
- **Cables** → unavailable at that hospital
  - Alternative resembling a cable → **Biznoska bands**
- **Wagner stem** → shape shown
- Simple **plate** included → not intended for use
- Standby only; plan already known
  - **Long-stem cementless** revision (e.g., Zimmer)
  - **Long-stem cemented** stem
  - **Cemented snap-fit cups** → if dual mobility unavailable
  - **NCB periprosthetic femoral plate**
- No reinforcement **ring** → cup had to be removed in this case
- Everything may turn out loose
- Company representative → always puts a **ring** or similar items in the box
- Work in your own setting with everything available
- Cleaning and **cement removal** → may be easy in this case

## Additional Equipment Ask AI

- **Flexible drill** → very important; fancy foreign instruments not available to all in Egypt
  - Drill → makes holes
- Power saw and bone graft (probable) → stored even if not intended
  - Unknown intraoperative findings, e.g.
  - Very poor **bone quality**
- **Screws** → 3.5 and 4.5 mm; all sizes and lengths
- Company representative (Hussein, Zimmer) → always present; must hold your profile or reliably bring items
- Surgical drains (probable)

**Clinical Pearl**

- Any revision → full instrument preparation
- No cables → Biznoska bands as alternative
- Flexible drill → very important
- Screws 3.5 and 4.5 mm, all sizes and lengths

## Operative Principles and Steps Ask AI

### Surgical Goals Ask AI

- Fracture + implant → addressed equally
  - **Stable fixation**
  - Preserve **fragments** and their soft-tissue attachments
  - Restore **length, offset, orientation** → as templated

### Plate Options Ask AI

- Plate shown → used in another case, not this one
  - **Screws from above** + anterior and posterior screws (as Dr. Sayed showed)
  - Alternative → **Locking Attachment Plates** (Synthes)

### Operative Steps Ask AI

- First step, before **reaming** → **distal cerclage**
  - Most important step; 75-year-old woman → caution
  - **Shattering** risk → prophylactic wiring very important
- Bypass the fracture by 6 cm
- End of case, rotating the leg → **greater trochanter** fracture
  - Managed with screws and cerclage available

### Follow-up Ask AI

- Patient returned last week by coincidence → follow-up
- Now about 80 years (was 77; now 78 or 79)

**Clinical Pearl**

- Distal cerclage before reaming → prevents shattering
- Greater trochanter fracture possible while rotating the leg

## Take-Home Message Ask AI

- Careful handling of **bone fragments**
- Preservation of **soft-tissue attachments** (probable)
- **Stable fixation** of fragments
- Restoration of **length, offset, rotation**
- Follow the **treatment algorithm**; prepare all instruments and assistance
- **Attention to detail**

# Loosening

*(Case-Based Discussion of Aseptic Loosening)*

**Prof Dr Sherif Khaled**

**Watch Original Lecture**

*Lecture transcript processed with AI assistance.*

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# Cemented Femoral Stem Loosening Ask AI

## Reading the Radiograph Ask AI

- Case series → prosthetic **loosening**
- Per case → loose or not?
- **Full X-ray** → examine each zone around implant
  - Note extent of involved zones → **zones 1 and 7**

## Case 1 — Cement Fracture and Radiolucent Lines Ask AI

Cemented

- Key finding → **cement fracture**
  - Site → **lateral cement mantle**
- Plus **radiolucent lines** almost all around implant
- Verdict → **definitely loose**
- Diagnosed because **symptomatic**
  - Asymptomatic → would have been **followed up**
- No symptoms → prosthesis still **definitely loose**

## Case 2 — Improper Cementing, Then a Fall Ask AI

Cemented

- Presentation → **periprosthetic fracture**
  - Same topic as Mahmoud's preceding lecture
- Timing → **one year after the surgery**
- Problem → **cement mantle**
  - Cementation not as it should be → **improper cementing**
- Old X-rays unavailable
- Only definite event → **fall**
- Fall → prosthesis failed

## Case 3 — Subsidence, Then Pistoning Ask AI

Cemented

- Two similar mechanisms
- **Subsidence** → cement and stem migrate downward together
- **Pistoning** → stem moves within the cement
- Distal opacity → **cement plug**, not distal stem
- Sequence:
  - Whole construct subsided
  - → stem pistoned within cement
  - → **cement fracture** → further sinking
- Summary of findings:
  - **Cement fracture**
  - **Subsidence of cement and stem**
  - **Radiolucent lines all around**

**Pitfall**

- Stem here → cemented
- **Pedestal** sign → cementless stems only
- Distal opacity → cement plug, not distal stem

**High Yield**
- Cement fracture, lateral cement mantle → loosening
- Radiolucent lines all around implant
- Subsidence of cement and stem together
- Stem pistoning within cement

## Cementless Acetabular Loosening Ask AI

### Case 4 — Broken Hardware Ask AI

Cup

- **Cementless** acetabular component
- Finding → **broken hardware**
- Broken hardware → **subsidence**
- Loosening in some zones around cup

### Signs of Cementless Cup Loosening Ask AI

- **Breakage of screws**
- **Medialization**
- **Protrusio**
- **Proximal migration**
- **Radiolucent zones** all around
- This cup:
  - **Broken screws**
  - Radiolucent lines > 2 to 5 mm
  - Involvement of **all three zones** of acetabulum
  - Cup in **protrusio**
  - Cup **migrated upward**

**High Yield**
- Broken screws
- Medialization / protrusio
- Proximal migration
- Radiolucent lines > 2 to 5 mm, all three zones

## Femoral Loosening Patterns Ask AI

- Femoral failure modes include **cantilever** (bending)
- Confusing patterns → e.g. **windshield wiper effect**
- Distinguish **medial midstem pivot** from windshield wiper effect

| | Windshield wiper effect | Medial midstem pivot |
|---|---|---|
| Fixed part | Proximal — held firmly above | Midstem pivot point |
| Loose part | Distal | Above and below the pivot |

| | Windshield wiper effect | Medial midstem pivot |
|---|---|---|
| Motion | **Distal toggle** | Swings back and forth |
| Bone effect | — | Progressive erosion of bone on one side |

- Medial midstem pivot → stem swings → **erodes** bone on one side

## Femoral Revision Options Ask AI

### Options Discussed Ask AI

80 years

- Patient aged 80 years
- Option → **long stem Wagner**
- Cup → **dual mobility**, cemented or cementless?

| Option | Note |
|---|---|
| Long stem Wagner | Suggested by audience |
| **Long stem locked** | Modular |
| **Long stem cemented** | Surgeon's choice |
| **Short stem** | Depends on its actual length |

- **Implant sets differ** → stem length varies by system
- Some systems → length changes from size to size
  - **Size 1** → length 13.3
  - **Size 6** → length 17.9
- **Revision stem** vs standard → depends on actual stem length
- Adequate length → standard stem can be done

### Procedure Performed Ask AI

- Surgeon's choice → **long stem**
- **Long stem cemented**
- **Tripolar** bearing
- **Cup cemented**
- Age 80 years → less demanding construct acceptable

## Composite-Beam Stem with Lost Cement Ask AI

### Findings Ask AI

75 years

- Friend's father, aged 75 years
- Finding → **subsidence**
- Known **cemented stem** → cement visible only in **zone 1**
- At removal → cement found inside
- Original surgery ~16 years earlier → **cement disappeared**

- Stem type → **composite beam**, not load taper
  - Recognized from **design of the stem**
- Plan → revision

### Revision Choice Ask AI

- **Wagner** → proposed
- Must be **long stem**
  - Reason → **original stem already long**
- Performed → **long stem cemented**
- Cup → **dual mobility cup**

## Ceramic-on-Ceramic Failure in a Young Patient Ask AI

### Presentation Ask AI

39 years

- **Police officer**, 39 years old
- Injury (likely fall) on 12 December 2016
- Associated **head trauma**
  - → cognitive impairment, like Alzheimer's
  - Retired by police; not doing well now
- Cognitive functions → no longer as high as before

### Primary Arthroplasty Ask AI

- **CT** → evaluation
- Workup → **rule out infection**
- Plan → **cementless**
- Performed → **total hip, cementless**
- Assess **inclination** and anteversion
- Cup slightly high → compare with **lesser trochanter**
- Relative to lesser trochanter → **cup set well**
- Surgeon's inclination → 40, not 30
- Most cups → 40 degrees

### Failure at 7 Months Ask AI

- 7 months later → failure
- Bearing → probably **ceramic**
- Principle → always blame the **surgeon**
  - Never the patient, circumstances, life or nature
  - Error suspected; unclear if in the operation itself
- Fine for seven months → not an **early failure** (within three months)
- Plan → **revise**
- Revision problem → the **ceramic**
- Bearing → **ceramic on ceramic**, not surgeon's choice

### Bearing Choice in a 39-Year-Old Ask AI

| Preferred: hard on soft | Avoid |
|---|---|
| Metal on poly | Ceramic on ceramic |
| Ceramic on poly | |

- **Not ceramic on ceramic** → see what happened here

### Revision to Dual Mobility Ask AI

- Intra-operative finding → **ceramic head** failure
- **Cognitively impaired** → strange behaviour → **tripolar** construct
- Performed → **dual mobility** construct
- **Leg lengthened slightly** → compare level around lesser trochanter

**Clinical Pearl**
- Young (39-year-old) patient → hard on soft bearing
- Metal on poly or ceramic on poly
- Not ceramic on ceramic
- Cognitively impaired → tripolar / dual mobility for stability

## Asymptomatic Acetabular Loosening on Follow-up Ask AI

- Longer construct → for **stability**
- **Stem not changed** → stem was good
- Follow-up → August 2018
- July 2019 → **lucency** around acetabular component
- **Zone 1** → superior lucency
- **Zone 3** → obscured on imaging, probably loose
- Revision offered → patient had no pain
- **Acetabular loosening** → often asymptomatic
  - → **follow up with X-rays**
- 7/2019 → no pain → no revision needed

**Clinical Pearl**
- Acetabular loosening → often asymptomatic
- → mandatory follow-up with X-rays

## Failure of the Dual Mobility Construct Ask AI

### Presentation in 2020 Ask AI

- 2020 → construct failure
- Approach → "wait until he gets fed up", not "wait until it unites"
- Previous: **dual mobility** cup placed well
- First operation → **cementless cup**

- First operation → **cementless cup**
- Current appearance → failed

### Dual Mobility vs Bipolar Ask AI

- **Dual mobility ≠ bipolar**
  - **Large cup** in acetabulum → very large head inside
  - A **total hip** construct
  - Articulation → large metal cup ↔ large polyethylene head
    - Inside → **small metal head**

| | Dual mobility | Bipolar |
|---|---|---|
| Outer head | **Plastic** (large polyethylene head) | Metal |
| Articulates with | Large metal cup in acetabulum | — |
| Inner head | Small metal head | — |
| Construct | Total hip | — |

- **Tripolar** = dual mobility
- Course → **dislocation,** progressive displacement → everything disrupted
- Classroom exchange → student presents the case

**Exam Pearl**

- Tripolar = dual mobility
- Dual mobility ≠ bipolar
- Outer head plastic, inside large metal cup
- Small metal head inside polyethylene head

## Infection Workup and Antibiotic Policy Ask AI

### Inflammatory Markers Ask AI

Infection

- Student who knows the case → brief exchange
- **ESR** and **CRP** performed
  - CRP 30
  - ESR 100

### Surgical Management Ask AI

- Outcome → not yet known
- **Removed everything**
- **Cultures** taken and sent
- Treatment → starts after culture results
  - No antibiotics yet

### Antibiotic Policy Ask AI

- **Antibiotics are withheld until the culture results return** → identify organism
  - Exception → dose pre-operatively + dose post-operatively

  - This → quite sufficient

### Tissue Cultures Ask AI

- **Tissue cultures** → organism search:
  - **Gram stain** → Gram-positive / Gram-negative
  - **Aerobic cultures**
  - **Anaerobic cultures**
- **TB** → sometimes investigated
  - Especially **chronic cases** referred from outside
- All investigations requested
- High ESR and CRP → **serial readings**

**Must-Know**

- Antibiotics withheld until culture results return
- Exception → one dose pre-operatively, one post-operatively
- Gram stain, aerobic and anaerobic cultures
- TB → chronic cases referred from outside
- High ESR / CRP → serial readings

## Loose Cementless Cup with Cavitary Defect Ask AI

- Clearly loose
- Signs:
  - **Migration** → change in position, movement of acetabular cup
  - **Radiolucent zones**
- **Cavitary defect**
- Another **cementless cup** → all signs visible
- Presented in the first lecture
- **CT** → localize the defect:
  - **Medial wall** broken or not?
  - Cavitary defect?

## Septic Loosening Ask AI

### Characteristic Picture Ask AI

Septic

- Diagnosis → **septic loosening**
- This appearance → **no differential diagnosis**
- All signs together → **very rare**
- Signs:
  - **Periostitis**
  - **Reactive bone formation**
  - **Endosteal scalloping**
  - **Large radiolucent zones**
    1. **Expand very rapidly** → week by week
    2. Not every 6 months or a year

- → **characteristic septic loosening picture**
- Included because **rare**
- Dr **Mohamed Abou Saoud** → sees such cases often
  - Lecturer has never seen it himself
- Diagnosis → **septic loosening**

**Must-Know**

- Periostitis
- Reactive bone formation
- Endosteal scalloping
- Large radiolucent zones expanding week by week
- This picture → no differential diagnosis

## Take-Home Messages Ask AI

- Loosening → **septic** or **aseptic**
- Clinical → **pain which increases with weight bearing**
- Covered → **radiological diagnosis of loosening**
- **Beware of septic loosening**
  - Must **exclude it** even if case looks aseptic

# Instability

*(Case-Based Discussion of Instability After THA)*

**Prof Dr Ahmad Azzam**

**Watch Original Lecture**

*Lecture transcript processed with AI assistance.*

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# AI detected corrections (Mostly bad transcripts)

## Corrected - the text was changed; the transcript said

**C1** (p. 319) “5-year-old girl” → “55-year-old female”

## Case 1 — Early Dislocation After THA Ask AI

### Presentation Ask AI

Case 1

- 55-year-old female, **AVN** → total hip arthroplasty *[C1 · transcript: "5-year-old girl"]*
- 2 days post-op → **posterior dislocation**
  - Limb in internal rotation
  - Dislocated three times
- Managed early
- Radiograph → comment on implant position systematically

## Radiographic Assessment of Implant Position Ask AI

### Cup Ask AI

- **Inclination**
- **Version**
- **Cup position**
  - Mediolateral
  - Superior/inferior
- **Head eccentric within cup** → not clearly seen in Case 1

### Stem Ask AI

- **Medial (femoral) offset** → compare with contralateral side
  - Case 1: no major problem
- **Vertical offset** → likewise assessed
- Case 1: **lateral view** also obtained

### CT Ask AI

Case 1

- Radiographs → then **CT**
- CT → component malposition, not apparent on plain films
- **Component malposition** > 10 degrees → management must include revision of malpositioned component

**High Yield**

- Component malposition > 10 degrees → revise the malpositioned component
- CT may show malposition not apparent on plain films

## Case 2 — Late Dislocation with Loose Stem Ask AI

### Presentation Ask AI

Case 2

- Female, hip arthroplasty → stable hip initially
- **Recurrent dislocation** 3 years post-op
- Radiograph → sign of **loosening**

## Work-up and Revision Ask AI

Case 2

- **Infection markers** → negative
- **Aspiration** → negative
- Stem → **loose**
- Revised with **long stem**
  - Fixation cemented vs uncemented — not specified

# Case 3 — Post-traumatic Dislocation Ask AI

## History Ask AI

Case 3

- **Trauma** (accident) → hip dislocated
- Several months post-injury → hip unstable
- Patient in fact **arthritic**
- Primary fixation → then **total hip**

## Cup Assessment Ask AI

Case 3

- Assess point by point
- **Inclination** ≈ 60 degrees → above 40 degrees
- **Version**
  - Some degree of **anteversion** → does not by itself require revision
  - Anteversion ≈ 15–20 degrees → acceptable
- Difficult situation → possibly no adequate acetabular bone stock
- **Hip center** slightly higher vs contralateral side

## Femoral Component Ask AI

Case 3

- Joint in this view → appears reduced
- **Long neck,** high offset
  - → possible minor **impingement** → levering of joint

## Medial Offset and Rotation Ask AI

- **Medial offset** → first compare with **contralateral side**
- Comparison requires images **rotationally identical** on both sides → axes comparable
- Slightly rotated → cannot judge
- Later view obtained
- Pelvic rotation difference → check:
  - Shape of **obturator foramen** on each side
  - Configuration of **pelvis**

**Pitfall**

- Offset and version comparison invalid unless both views rotationally identical
- Rotation clue → obturator foramen shape and pelvic configuration

## Comparison Between Views Ask AI

Case 3

- **Rotation** differs between images
  - Cup inclination → approximately the same
  - Cup version → anteversion appears greater on one view

    - Rotational difference → cannot be certain
- **Lucency** between **cement and bone**?
- Change in **metalwork** shape/position at **screws**?
- Change in lucency?
- These findings may be present in any cause of failure

## Constrained Liner Ask AI

### Indications Ask AI

- Preferable → address instability by treating its cause
- Obvious problem (e.g., **component malposition**) → exclude → must be **revised**
- Cause **multifactorial** or **unclear** → **constrained liner** may be used
  - Not with > 10 degrees of component malposition
- Case 3 → revised with constrained liner

### Design Ask AI

- Liner extends beyond **circumference of the head**
  - → constrains head within cup

### Failure at One Year Ask AI

Case 3

- Post-op view → **screws** fix the liner
- Did well ≈ 1 year → then re-presented
- **Hardware failure** beginning
- **Cup position** changed
- Compare with previous view → new change in this part
- **Hardware broke**
- Defect in **superior wall**

### Lessons Ask AI

Constrained

- Constrained liner → not to be relied on heavily
  - Did well 1 year → still not sufficient
- Not for liberal use in multifactorial or unsuitable situation
- Reserved for **salvage**

**Pitfall**

- Constrained liner not a substitute for revising malposition > 10 degrees
- Case 3: good for 1 year → hardware broke, superior wall defect
- Reserve for salvage, not liberal use in multifactorial instability

## Revision After Constrained Liner Failure Ask AI

### Options Ask AI

- **Dual mobility construct** planned → useful to convert to it
- Any problem with **component position** → revise the components

### Revision Ask AI

Case 3

- Case 3 → revised
- All components removed
- New reconstruction performed
- Dedicated acetabular construct used
- Follow-up time not yet long
- Follow-up view

## Prevention and Management Algorithm Ask AI

### Prevention Ask AI

- Reference: **2011**
- Very important → in **initial primary total hip**, surgeon must:
  - Choose patient and implant carefully
  - Technically **restore the anatomy**
    - **Soft-tissue tensioning** across hip joint
- Treatment always far more costly than **prevention**

### Management by Risk Factor Ask AI

- Identify **risk factors** → first exclude:
  - **Component malpositioning** → revision
  - **Loosening** → revision
- Absent → other adjuncts:
  1) Increase **head size**
  2) Change the **liner**
  3) **Soft-tissue procedure** (e.g., trochanteric advancement)
  4) **Constrained liner**
- Choice approximately according to the problem

| Finding | Management |
|---|---|
| Component malposition > 10 degrees | Revise malpositioned component |
| Loosening | Revision |
| Multifactorial / unclear cause | Adjuncts: larger head, liner change, soft-tissue procedure, constrained liner |
| Constrained liner failure | Revise components; consider dual mobility |

**Must-Know**

- Prevention in primary THA → patient and implant selection, anatomy restoration, soft-tissue tensioning
- Exclude malposition and loosening first → both need revision
- Multifactorial cause → more than one method

# Case 4 — Clear Mechanical Cause Ask AI

Case 4

## Presentation Ask AI

- 65-year-old, **fracture neck of femur**
- Operated 3 weeks after trauma
- **Dislocation** ≈ 3 months post-op
- Simple → very clear **mechanical** problem

## Principle Ask AI

- Other cases → no **clear cause** known
  - As **Dr. Sherif** discussed in the **loosening** lecture
- Cause not apparent or **multifactorial** → use **more than one method** in treatment
- Case 4 → extensive revision
- Excellent result after revision